Milestone 6

Design Implementation

Bring ROAR to Life - Team 1

Joseph Dale (Mechanical)
Joshua Shoenfelt (Mechanical)
Kevin A. Rivera (Mechanical)
Maria Rodems (Mechanical)
Mark Shenouda (Mechanical)

Due: April 17th, 2025

Faculty Advisor: Kurt Stresau
Technical Advisor: Gabriel Vazquez

EML 4502 – Senior Design II
Tuesday/Thursday 3:00 PM-4:15 PM

Executive Summary

Bring ROAR to Life is a mechanical engineering senior design project. The goal of this project was to create an animatronic stegosaurus skeleton with the capability to walk. This project was largely inspired by the film *Night at the Museum*, which features a T-Rex skeleton that comes to life.

The project was designed with a museum application in mind, to be an additional exhibit that guests can interact with. The animatronic consists of a mechanical system, electrical system, and control system. The main goals of the project were that the animatronic must display a high degree of anatomical accuracy and a smooth gait, which the team achieved.

The team utilized engineering principles to complete the project. The structure was analyzed for stress and torque results. The team selected the best material to handle the stress and torque, as well as fit within budgetary constraints. The power system was substantial enough to actuate all joints, while compact enough to fit within the skeletal structure. The control system was able to operate the joint actuators. The team analyzed the project to identify any failure modes and effects before iterating and testing each function of the animatronic.

The progress of the project has been documented throughout this report, and the final results will be displayed in a promotional video.

Table of Contents

List of Figures

Revision History

Version	Date	Reason for Changes
-	03/10	Initial document.
A	04/11	Main Setup
B	04/20	Completion

I. Introduction

This project originated from Dr. Kurt Stresau, who many years ago purchased for his daughter a stuffed toy of a stegosaurus from a museum gift shop [48]. Several years later Dr. Stresau was inspired by his daughter and “Rex-y” from *Night at the Museum*, who was modeled after a complete fossil specimen, AMNH 5027, from the American Museum of Natural History in New York City [47]. He developed a senior design project around the concept. The focus of the project is creating a stegosaurus skeleton with a biologically accurate appearance, anatomically correct movement, and ease of control.

Figure 1: `Image of ROAR Stresau for Reference [48].

As an unsponsored project, the team tailored the animatronic to a museum that would use it as an educational tool/aid within an exhibit or showcase. The primary user is museum staff and will exclude guests or visitors. The secondary user is the companies purchasing the animatronic. The tertiary user is the audience; this is important as the design also needs to reflect and consider the experience from their perspective. The design is not intended to be used for military or industrial purposes and furthermore should not by any means be used to endanger injury or hurt any of the three forms of users. The team has met with and interviewed the Lawrence Hall of

Science to better understand the wants and needs of the user. The team is in active communication with them and will continue to do so throughout the progression of the project.

The focus of this report will be to document the steps the team took to design and bring to life the project. It will include the requirements, concept generation, and analysis of the animatronic. The report will also include the successes and failures of the final design, as well as future work that other teams could take on.

II. Project Objectives & Scope

The planned objectives of our Bring ROAR to Life project is to…

- Complete a full structural and electrical analysis of the total robot, to ensure that parts will not break during operation and that our motors and servos will be able to move the robot.
- All components used in the final prototype have been chosen and tested for efficiency and integration verification.
- All manufacturing in order to construct a fully functioning prototype will be prioritized.
- By the end of the semester we will have a fully functioning, tested, and validated robot along with all complete and relevant documentation, ready for the Senior Design Showcase.
- It will be able to walk with a realistic gait, take and respond to commands smoothly, and function optimally for at least half an hour.
- A user manual will be compiled and provided with the finished prototype.

The long-term objective of our Bring ROAR to Life project is to…

- Develop a lifelike animatronic stegosaurus that can operate reliably in a museum environment.
- Maintaining a realistic anatomical accuracy and realistic motion.
- Beyond the senior design scope, the system is intended to be maintainable and adaptable for long term use or future upgrades.
- The payoff for stakeholders includes a unique, interactive exhibit that enhances visitor engagement and STEM education, offering a visually compelling way to demonstrate robotics, and the engineering design process.
- For museum staff, the project emphasizes safe operation, and ease of maintenance to minimize downtime and operating costs, while faculty and students benefit from a strong engineering platform and real-world design experience.

III. Assessment of Relevant Existing Technologies and Standards

In order to better understand where to take the final design, the team first needs to understand and research industry standards to inform design decisions and to develop a final prototype. Further, as the model is based on a previously-living creature, care should be taken to understand how a stegosaurus lived in its environment. Research should also cover the current animatronic industry, along with more specific research into joints and control systems, that way we can model the project as needed to get to a near real product.

I. Anatomy and Diet of the Stegosaurus

The stegosaurus is a dinosaur found mostly in the mid to late Jurassic period, although specimens have been found dating from the Cretaceous period [5]. They were a member of the genus *Stegosauria*, which was a family of quadrupedal herbivores found across the globe, including China, Siberia, Morocco, and Australia. Fossil remains of the stegosaurus, however, have been discovered primarily in North America and Europe [6]. The stegosaurus is one of the larger of the stegosaurus, with fossils reaching up to 9 meters in length..

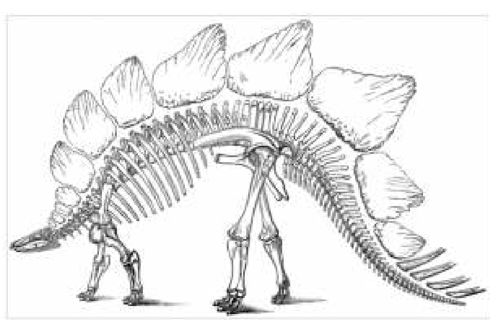


Figure 2: Stegosaurus Skeleton [69].

Diet

Due to the lack of soft tissue preservation, there's little evidence for what the stegosaurus ate. Many of the tooth specimens lack substantial wear, making it difficult to predict the diet of the stegosaur. The common consensus is that they fed close to the ground, near the 1 m level. The one dissenting opinion suggests that the stegosaur stood on its hand limbs, using its tail to balance, similar to a kangaroo [6]. Jaw and dental analysis implies that the stegosaurus had a bite force ranging from 166 to 410 N [5]. This bite force would have allowed the dinosaur to bite through branches, though the main function of their teeth appears to be centered on grinding down leaves and other plant matter.

Anatomy

Estimated masses of the stegosaurus range from 1082 kg to 2256 kg [3]. The mass of the animal depends heavily on the spread of the legs and descent of the body cavity. This information is lost to time without the presence of soft tissue or more complete skeletal specimens. Further, it is predicted that the stegosaurus grew to its full size slowly, implying it did not need to reach its final dimensions quickly for protection [4].

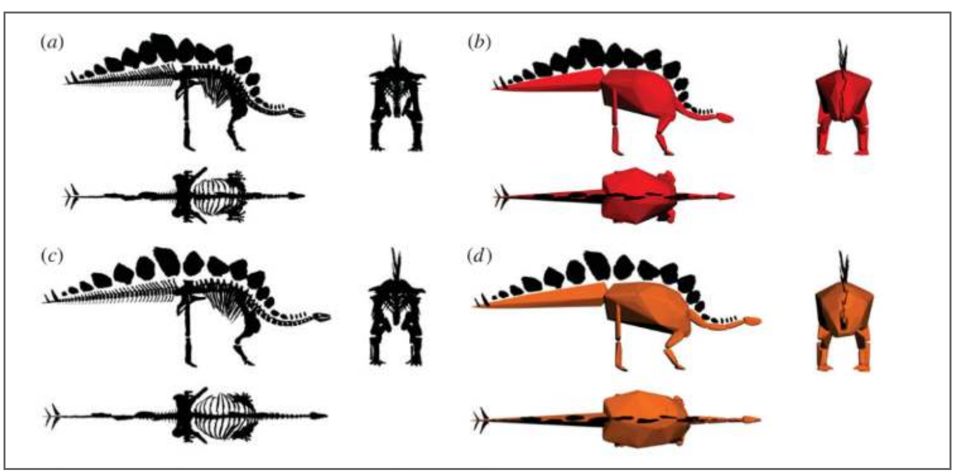


Figure 3: Reconstructions of Stegosaurus stenops (NHMUK R36730) and associated convex hulls [3].

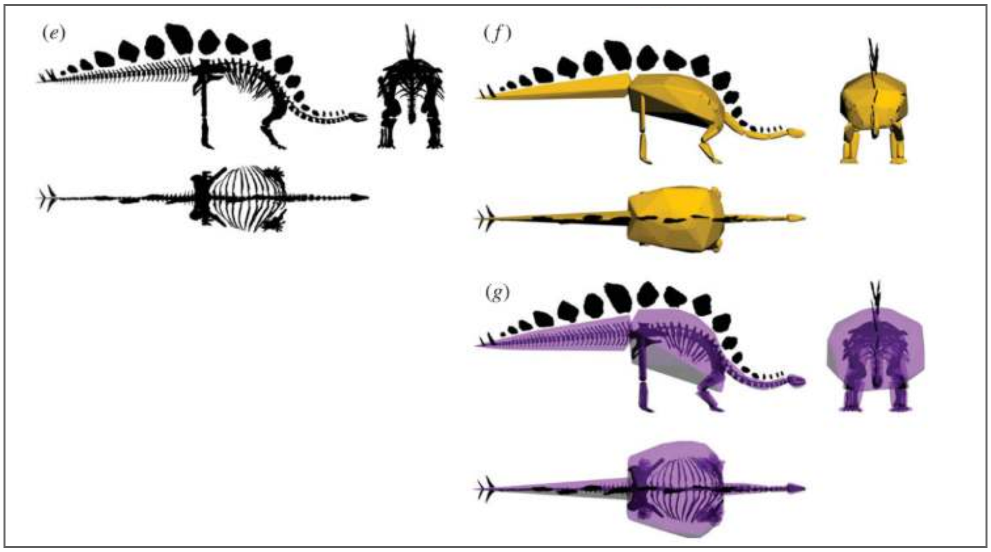


Figure 04.1: Reconstructions of Stegosaurus stenops (NHMUK R36730) and associated convex hulls [3], cont.

Head

Notable features of the stegosaurus anatomy include a proportionally small head compared to the body. The top of the skull was flattened, and the cranial cavity was small. It is likely the animal had a small brain as a result of its diminished cranium [6].

Body and Legs

Fossil records demonstrate that the beginning of the rib cage was narrow, widening when approaching the pelvis. Additionally, the forelimbs were short in proportion to the hind limbs, which introduced a dramatic curve down in the spine. This results in a visual center of mass targeted over the hind legs.

The ribs of the stegosaurus are T-shaped in cross section. Females have an additional sacral rib. The neural passageway through the spine is narrow, which would limit the thickness of blood vessels and nerves able to pass through [6].

Tail

One of the most notable features of the stegosaurus were the dorsal spines running down the vertebrae. They began after the skull and extended down the tail. The plates alternated angles at 50° along the body, while falling more in line when proximal to the tail. The tail finished with four large spikes.

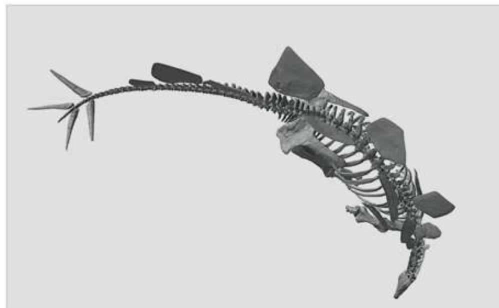


Figure 4: 3-D Scan and Model of Stegosaurus Triebold Paleontology, Inc., Denver Museum if Nature and Science.

The function of the dorsal fins is disputed. A possible purpose could have been for defense, making the animal too large and boney for predators to easily eat. Additionally, it has been proposed that its plate could have performed thermoregulatory functions. The fins contain vertical grooves, which may indicate the presence of blood vessels. Another explanation may be that the fins distinguished each animal, allowing for the recognition of individuals. Its spikes are bony and presumed to be defensive measures [4].

Summary

The anatomy of the stegosaurus resembles many modern quadrupedal herbivores. Stegosauruses have a low center of mass. Their wide ribcage and sturdy limbs resemble animals such as elephants or rhinoceroses. In contrast to modern herbivores, stegosaurus fossils display small heads, spiked tails, and dorsal fins. Additionally, their long necks and tails result in a wider splay of the animal in comparison to the aforementioned animals.

This anatomy will create an interesting design challenge. The short forelimbs will create dramatic stability issues in the animatronic. Another potential struggle will be providing enough support and power to animate the far-reaching head and tail. While the solutions to these problems are not yet clear, the team will be better equipped to develop an accurate animatronic when backed by the fossil data.

II. Animatronic Development in Industry

This project at its core is a project dealing with animatronics, specifically animatronics and robotics in entertainment. Creating a dinosaur that is lifelike in its movements and actions, as well as its interactions with the environment is an integral point of success for this project. In order to better achieve this goal, it is good to be familiar with the problems that engineers before this group encountered when attempting to take on similar projects. Looking at the specifics of their problem statements and solutions, gives the group a more detailed guide when it comes to figuring out what functionality and design specifications will likely be most important. Additionally, studying examples from past attempts in industry may illuminate potential pitfalls the group is likely to encounter during the design and prototyping process. Knowing where those pitfalls are likely to occur, and how other groups went about solving them, gives the group a roadmap to success that doesn't need to be built from the ground up, saving valuable time and resources.

Bringing Art to Life

Companies like Disney and Universal are at the forefront in the development of robotics and animatronics in entertainment. From Walt Disney's very first bird animatronic which he found in an antique store up to today's highly advanced models with face projection, fully electric precision servos, speakers, and expertly designed costumes.

Figure 5: Disney Imagineer demonstrates features of Louis animatronic [9].

Animatronics in this field tend to be a highly coordinated effort between interdisciplinary teams of engineers and highly skilled digital artists and animators. Robotics experts work with costume designers [9] back and forth to bring every detail of a fictional character to life from their attitudes and mannerisms to their voices. Software interfaces allow for direct and precise control of every part of the animatronic, coming together to form smoothly coordinated, natural movements. Disney's Audio-Animatronics [10] are famed precisely for their ability to unite movement and sound, bringing together songs and gestures perfectly coordinated and choreographed by teams of animators. For the purposes of the project, nailing down as small a level of precision as possible will be paramount to achieving the smoothest and most natural movement possible.

Disney's Droids

Within the past few years, Disney especially has had to ramp up their development of new animatronic solutions to accommodate their new and expanding repertoire of themed parks based upon their newer intellectual properties. Most interestingly for the purposes is an animatronic robot based upon the droids from Star Wars. These new robots incorporate elements of engineering design and artistic creativity to deliver a final product that, when paired with a skilled puppeteer, behaves in an incredibly lifelike manner.

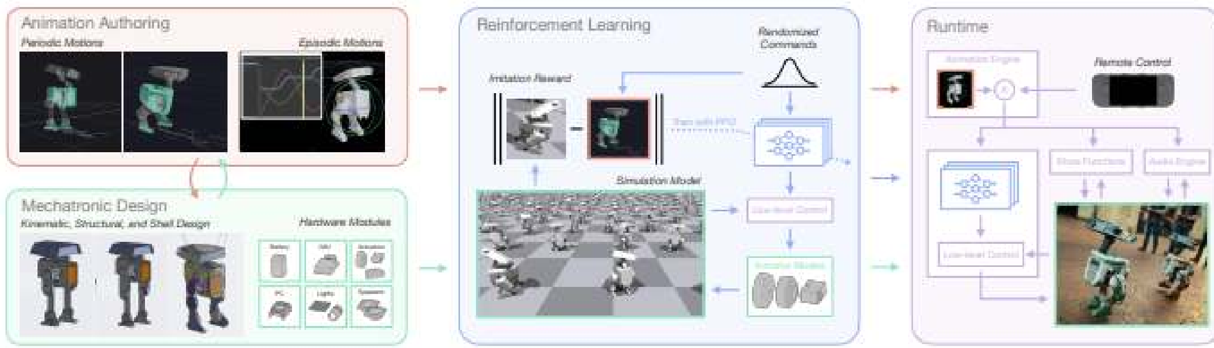


Figure 6: Summary of the Development Process [7].

Through a unification of artistic animation and computer simulation based upon the initial character design, animators are able to use kinematic simulations of the actual robot to test realistic walking patterns within the physical bounds of the robot's weight and size, as well as range and speed of motion. Using reinforcement learning [7] the robot can be trained to move within the bounds of the intended animation as closely as possible while also maintaining proper balance across a broad range of possible environmental variables. Special attention was given to the use of both 3D printing and off the shelf parts which allowed for much quicker development time and a reduction in cost from having to develop a multitude of systems from the ground up. Puppeteers use the already existing Steam Deck [7] in order to move the robot and issue commands, which also makes training puppeteers easier. The use of a video game style controller was deliberate due to widespread familiarity with the platform, making it easy to pick up on the basics of the controls. Using an existing video game platform also goes back to allowing off the shelf parts to reduce development time and cost.

This project also focuses very heavily on having creative design guide the functionality of the robot, especially in the realm of Human-Robot Interaction. Every aspect from the use of puppeteers to guide spontaneous behavior to allowing artists and animators to define pre-set animations results in a robot which can display a wide variety of emotions as well as dynamically responding to a variety of environmental factors. Robots can appear to inspect their environment, interact with objects, respond to questions, voice commands, and touch [7] in much the same way a household pet or a very young child would.[8]

Figure 7: (top) A robot interacts with an object in a set scene performing a routine of emotional controls, (bottom) A robot interacts with a human, showing curiosity and receiving affection [7].

Simulation and Reinforcement Learning allow much of the heavy lifting in terms of calculation to be taken off of engineers and computer scientists, further reducing development time and allowing for more effort to be expended towards the creative goals of the project.

Summary

This dinosaur project has a very broad scope that largely focuses upon a lot of creativity driven solutions rather than pure functionality. Examining this case in industry sheds light on foundational conclusions including the use of off the shelf parts, 3D printing, simulation, and machine learning as tactics to greatly reduce development time and cost. It also helps open the door to functions which the team may want to consider such as necessary range and types of motion, predetermined animations, and Human-Robot Interaction. Considering aspects of who and where the robot will be interacting with, gives insight as to what behaviors it may need to express.

III. Robotic Joint Types and Movement Systems

One important aspect of creating lifelike animatronics is having joints that are able to mimic fluid movement. For this project, this will mean the use of high torque, lightweight drive systems. Another important factor will be a consideration on size and packaging. There are many different drive systems already used in robotics, the challenge of this will be inventing a way to hide these systems in order to preserve biological accuracy to the model.

Figure 8: Capstan Drive Example [11].

Looking to the industry today, one drive method is a capstan drive, these systems trade out the use of gears for synthetic rope. This allows for the system to transmit high torque, while decreasing noise and slop. These systems will typically have a reduction ratio less than 10:1 [11]. however there will need to be a careful look into packaging, as the primary drum is quite large when compared to the rest of the joint.

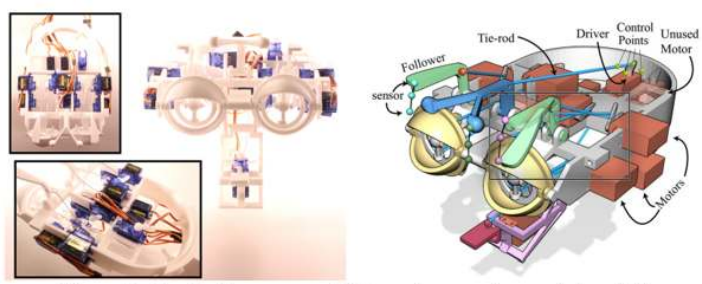


Figure 9: Head with motors and linkages for an animatronic face [12].

Another consideration could be through the use of tie rods & cables. These can allow for drive motors to be relocated away from a joint. In doing so, it allows for the combination of electronics, drive motors, and mechanical linkages to drive separated movement. [12] This could possibly be used to actuate finer movement, such as the head, neck, and tail. Along with relocating motors for the legs in order to preserve biological accuracy.

Figure 10: Shoulder with linkages and range of motion [12].

Along with using linkages and separated motors for facial features, another example used was for a shoulder. This uses the same basic principles however includes a larger focus on bone structure and lifelike movement. [12] Which will become a large focus of this project.

For a more direct drive example, The team can take a look at cycloidal drives. This system uses a set of eccentric gears which results in a very space efficient gear reduction. The basic working principle of a cycloidal drive is that of a disk rolling about an outer ring of larger radius. As it rolls, the disk rotates at a velocity that is proportional to the relative radii, which provides the gearing ratio of the gearbox. [13] These are able to be low-cost, 3D-printable, and use off-the-shelf bearings and steel components. [13]

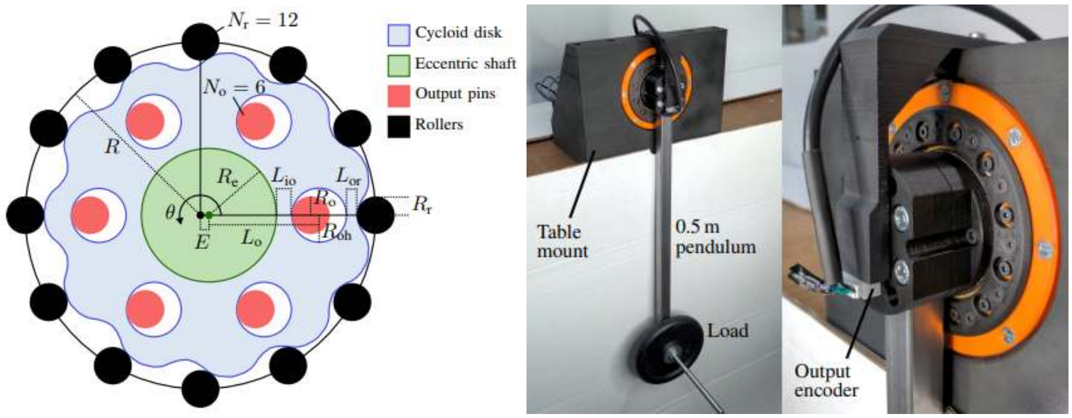


Figure 11: Cycloidal Drive Geometry [13]

When beginning any robotics project, many design considerations will need to take place. Aspects include robot weight, required torque, speed, and cost. Most of these joint types aim to increase the torque output of most electric motors which is needed on most legged robots. In order to determine these factors, The team will have to estimate the dinosaur's weight and expected forces on each leg. This will then allow the team to calculate the required mechanical advantage needed from these joints.

Another creative technique is through the use of elastic or springs along with rigid linkages. These can be used to provide a static mechanical advantage to joints, here the relationship between the applied torque and the joint torsion is identified [14]. In turn allowing for the movement mechanisms to become smaller which can aid in overall packaging. Additionally these systems can be used to help prevent backlash and reduce slop [14].

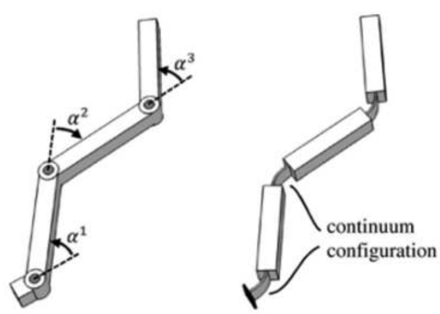


Figure 12: Comparison of Standard and Flexure Joints [15].

Moving on from joint movement, the team can take a look into types of physical joints., which will need to mimic biologically accurate joints. Here, industry standard joints may be less desirable, however required depending on specific requirements. One alternative is to use a Flexure joint, these trade out pins and bearings for a flexible material. Advantages include the lack of sliding parts, which reduce points of backlash or slop. Along with the ability to use a variety of different material types that can possibly be 3d printed at a much lower cost to traditional joints [15]. These may be worth looking at as these joint types could be hidden into a biologically accurate appearing joint, which can increase realism. One drawback to flexures as robot joints is the complex mechanical behavior that they exhibit compared with pin joints [15]. In order to determine accurate joint movement, beam bending models may be needed, along with a higher level of research in material selection and manufacturing.

Although more research is to follow, the previous joint types allow the team to gain exposure to a few unique methods used outside of traditional robotics. Which opens up conversations about material selection, packaging, maintaining biological accuracy, and many other important aspects of animatronics. And will likely lend useful as the team progresses further into this project.

IV. Simplified Linkage Types to Mimic a Realistic Gait

Problem statement: The success of the walking dinosaur depends heavily on its ability to walk in a stable manner, while allowing movement for special effects, including standing on its 2 back legs, and using its tail for a 3rd leg. One other special effect the team agreed on implementing is having the dinosaur walk, trot, canter, gallop, etc. Unlike wheeled systems, legged locomotion requires precise coordination.

Jensen Linkage

One mechanism is the Jansen linkage. The Jansen linkage is quite simple, it is an eleven-bar mechanism, and its motion mimics a walking animals' motion. I have designed this mechanism in my internship at Siemens and have familiarity with its function. The mechanism has 13 different length numbers that can be edited. The genetic code of the numbers are called the "13 holy numbers" [16] and these numbers are a ratio, and provide the best walking motion, ensuring there are always two legs touching the ground simultaneously. Each two legs would only use one motor, unless The team has a motor for all of the legs.

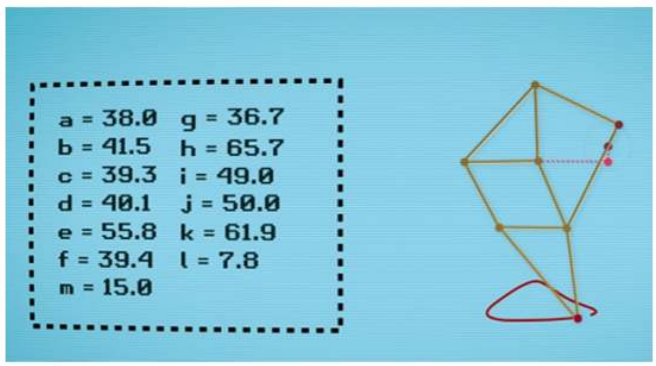


Figure 13: Jansen Linkage [16].

This walking mechanism has pros and cons. The cons are that if the team has one motor for each side, it would be hard to make the robot stand up on its hind legs. One way to fix this is to have 1 motor for each leg, that way, each leg can behave independently, and the robot can stand on its hind legs, and balance using its tail. The pros are that it is a simple mechanism to work with and only requires 2 or 4 motors. Another pro to this mechanism is that the team can add one motor on each side to not only have one degree of motion (front and back), but side to side. This would enable the robot to move its legs from side to side.

AlienGo Robot

In this article, The AlienGo robot shows how complex locomotion can be controlled and trained. Unlike fixed linkage mechanisms, advanced robots use multiple actuators and control systems to replicate walking, trotting, and galloping with high accuracy. For the dinosaur, understanding these systems can guide the team in how to design both stability and special effects (such as standing on hind legs or using the tail for balance).

[17] AlienGo is a quadrupedal robot weighing approximately 23 kg and designed to replicate agile animal-like movement. [17] It has 12 actuated joints: 3 per leg (hip, thigh, and calf). This design allows it to mimic a wide variety of gaits, including walking, trotting, galloping, and even acrobatic maneuvers like backflips. The key advantage of AlienGo [17] is its ability to learn gait motions from videos of real animals, which makes its locomotion both efficient and realistic.

The pros of the AlienGo approach include precise independent control of each leg, making complex behaviors like standing on hind legs or balancing with a tail much more feasible. [17] With 12 motors, each joint can contribute to stability and dynamic movement. Another strength is flexibility; new gait patterns can be programmed without redesigning the hardware. This design has the most flexibility and would be great especially if The team only made the skeleton dinosaur.

The main con is the complexity: [17] 12 motors means higher cost, more power requirements, and greater control system complexity compared to simpler mechanisms like the Jansen linkage. For the project, this approach may be too resource-intensive for a prototype, but the principles behind AlienGo's motion studies are valuable for informing gait control in the dinosaur.

Figure 14: [17] Liu Zhao (results section)

Development of quadruped walking robots - article

This article [18] has many different types of quadrupedal motion. The purpose of this article is to summarize key findings from the review by Biswal & Mohanty (2020) that are particularly relevant to the walking dinosaur project. In particular, this includes how modern quadruped robots are designed in terms of actuation (motors/joints), gait patterns, structural trade-offs. Many quadruped robots in the literature use two rotary joints per leg; each leg often has a hip and a knee that are actuated, this enables flexion/extension motions which are essential in the robot in order to perform the trot, canter, gallop motions.

Some more advanced designs have three actuated joints per leg (hip, thigh, knee / ankle), this is for dynamic gait (trotting, running) and complex terrain adaptation. While this approach includes precise independent control of each leg, it would require higher costs, and more power requirements.

(a) (b)

Figure 15: [18] Complex actuated joints robot.

Summary

These articles collectively show the challenge of designing effective quadrupedal locomotion for the walking dinosaur: balancing stability, complexity, and realism. The Jansen linkage [16] offers a mechanically simple and motor-efficient solution, but its limited degrees of freedom restrict more advanced movements like standing on hind legs. In contrast, the AlienGo [17] quadruped demonstrates the power of multi-motor actuation and advanced control, using 12 independent joints to achieve dynamic gaits and agile maneuvers, but the tradeoff is the cost of higher complexity and budget. Finally, the review by Biswal & Mohanty [18] situates these approaches within the broader field, showing how two to three joint legs per limb are standard in quadrupeds and highlighting tradeoffs between actuator count, gait diversity, and adaptability. Together, these insights guide the project toward a design that balances the simplicity of linkage mechanisms with the flexibility of the gait to achieve both realistic walking and special effects.

V. Digital and Control Systems

The team has been tasked with constructing a free standing quadrupedal robot able to achieve a walking motion with dinosaur anatomy. To accomplish this the quadrupedal robot will require the implementation of digital and control systems that will drive its ability to walk as well as any additional functions/features that have been approved and discussed by the team such as for example opening and closing the mouth of the robot etc. The Tech memo is about the selection and a catalog of options for digital and control systems. Below the following will be discussed: a summary of control theory with a theoretical analysis as pertains to the project, discussion of the Hardware such as microcontrollers, microcomputers, and microprocessors, the programming languages and operation systems with regards to use in robotics, a discussion of references for algorithmic and mathematical methods for use in controlled and autonomous robotics systems. Achieving these objectives will allow the team to make more informed decisions when creating design requirements as well as create a reference document for future design and development.

Control Theory and Systems

Control theory "is a mathematical framework that gives the team the tools to develop autonomous systems" [20]. A feed forward controller or open loop controller: takes in the desired output/reference and generates the control inputs without ever needing to measure the state. A feed forward controller does not rely on feedback from the state of the system which is why it is an open loop, it never loops back. Feedback control/closed loop control relies on both the reference and the current state of the system to generate the appropriate control inputs. This allows for if the state of the system begins to deviate due to errors or external disturbances the output is fed back into the feedback control so it can adjust the input accordingly. Refer to *Figure*

17. for types of feedback controllers. State Estimation is another critical part of control theory. This is because the state of the system is read using sensors, so measurement noise becomes a problem.

Type of Controller	Key Characteristics
Linear	Based on linear system assumptions; easy to analyze and design (e.g., PID, state feedback)
Nonlinear	Handles nonlinear dynamics; can deal with real-world complexities
Robust	Maintains performance under model uncertainties and disturbances
Adaptive	Adjusts its parameters in real-time based on system changes
Optimal	Minimizes a defined cost function (e.g., LQR, MPC)
Predictive	Uses a model to predict future behavior and optimize control actions (e.g., Model Predictive Control - MPC)
Intelligent	Uses AI techniques (e.g., fuzzy logic, neural networks, reinforcement learning)

Figure 16: Types of feedback controllers

The team must also be able to observe every state of the system even if not every state is directly measured. So the goal is to “reduce measurement noise” and “manipulate measurements in such a way that allows the team to accurately estimate the state of the system.” [20] It is also vital to test the control system via analysis, simulation, and test to ensure it works properly and gather all data from testing for final conclusions and edits. The primary requirement of the project is to achieve a walking motion, implementing control theory will aid in the design process. In order to achieve motion control, control theory can be used to create a model to regulate the position, speed, and torque of the robot. For example a feedback controller or closed loop controller would adjust the motor input based on the desired state and the actual measured state. Another example would be for trajectory tracking, the control system can in real time adjust the input based on disturbances.

Hardware

The purpose of this section is to define and create an understanding of the hardware and aims to give the reader understanding when selecting Hardware for project use. Proper selection is crucial for planning and implementation of the project. Determining the proper hardware for the project is crucial to achieve the locomotion goal, factors like processor architecture, clockspeed, RAM, Flash memory, I/O ports, power consumption, operating voltage, and the development environment, must be taken into consideration.

The architecture of a processor is a direct determinant of the performance of the system as "microcontrollers are generally classified according to processor characteristics such as processor architecture, processor word processing length, processor clock operating frequency, and so on." This determines how well it handles data. For example, "RISC architectural microprocessors are generally preferred in electronic applications such as control and automation, [while] CISC architectures are used in microcomputer applications such as signal processing and embedded systems" [21]. Therefore when choosing the architecture it should account for computational power and the complexity of its intended application. The Clock speed, measured in MHz or GHz, showcases how fast a processor can carry out instructions, thus "attention should be paid to technical features such as word processing length, working clock frequency, memory amount, bus width and internal additional hardware" [21]. If it has a higher clock frequency then the general consciousness is that there is also an increase in power consumption and heat generation. Another key factor to take into account is memory. Random access memory or RAM "stores the information that the processor needs during its operation, while ROM, PROM, EPROM, and EEPROM memories…store information such as command set, program and program data"[21]. Enough Ram is important for dynamic operations while

Flash memory is generally used to store firmware or any permanent code. Having enough Flash storage is a necessity so it can adequately hold all the required data and code. Just as important as the memory capacity, the I/O ports are equally as crucial to consider. It is recommended that the designer should list the hardware needed for different " communication interfaces (USB, SPI, UART etc.)" and list the "port connections… for the input/output elements such as sensor, LCD display, driver and relay" [21]. How much power the system will consume is also important to consider especially if it is battery-powered or energy-sensitive. It is important to check the " energy consumption, operating frequency and other variables" to assure you are making the correct choice in the microcontroller [21]. When looking at operating voltage it is important to look at voltage compatibility since it can directly impact the ability and safety of a system. Finally it is important to check the "check the development environment and programming languages provided for the hardware…[and] search the programming interfaces and tools to be used" [21]. Together all these factors are crucial to understand when trying to make the proper choice for which microcontroller to be used for the project.

Software/Programming

The purpose of this section is to create an understanding of the different types of software tools and programming languages. It aims to give the reader understanding when selecting software and programming language for project use. Selecting the appropriate software tools are crucial to insure performance, compatibility and development efficiency. Robotics has a diverse range of programming languages that it relies on. Each language has its own strengths in terms of control abstraction and usability. C/C++ are one of the most commonly used languages due to their low-level efficiency and hardware access in industrial and research robotics, which offered extensions for controlling sensory, actuators, and planning activities [22]. In comparison, Python is versatile for high level logic. It is popularly used for scripting and AI [22]. Robot-specific extensions for Java, like motion description languages have been developed emphasizing object-specific abstraction with portability [22]. Another programming language is MatLab, a specialized environment design for data analysis, algorithm development, and simulation [23]. ROS supports multiple languages, providing tools and libraries for robot software development making it flexible to work with [24]. Micropython is another lightweight and "efficient implementation of the Python 3 programming language that includes a small subset of the Python standard library and is optimised to run on microcontrollers" [25]. Lua is one of the lightweight scripting languages used for rapid prototyping and embedded scripting, it was "designed to be simple, small, portable, fast and easily embedded into applications" [26].

When selecting the programming language there are multiple factors to consider such as: the execution speed and real-time capabilities, hardware compatibility and low-level access, library and API support, community support, and the integration with existing robotics platforms, as stated in FIGURE B. It is important to evaluate the programming language in relation to the intended task, performance constraints, hardware choices etc.

Methods for Algorithmic and Mathematical Techniques

The purpose of this section is to identify and summarize key methods and programming techniques commonly used in robotics applications. Below is a table (*FIGURE A*) summarizing the technique, purpose, area of application and a source with the method being used. The table in Figure A can be applicable to the making of the project for reference of possible techniques to use.

Summary

Further research into this subject is recommended once design requirements have been designated. The background knowledge as well as all the research provided in this document should be valuable reference for multiple of the development and design phases. I believe that the key concepts provided here should allow the team to do the following: select the hardware, software, design the control system structure and architecture, and implement the techniques mentioned or variations as needed for the needs. It is recommended that further study also looks into the cost and difficulty of implementation and use. This Technology memo fails to take these factors into account and is primarily of the purpose of familiarizing the team with the methods, models, key concepts, and systems used in digital systems and control systems in a generalized matter.

IV. Professional & Societal Considerations

The target customer would be a museum that would utilize the animatronic as an educational tool/aid within an exhibit or showcase. The primary user will consist of an employee of said museum and will exclude guests or visitors. Our robot will be used in the future as a teaching tool to get students and children interested in engineering and STEM fields in general. Our robot is also additively manufactured; most engineering prototype projects in the industry can be additively manufactured, whether they are plastic, metal, or some type of alloy or composite. It is important to keep them in mind in terms of cost of maintenance and functionality, as their needs will be different from the needs of the operator. The audience; this is important as the design also needs to reflect and consider the experience from their perspective. The design is not intended to be used for military or industrial purposes and furthermore should not by any means be used to endanger injury or hurt any of the three forms of users.

V. System Requirements and Design Constraints

The requirements for this project were developed in the previous semester to guide the design. A difficulty the team encountered was the lack of defined stakeholders. Due to the open nature of the project description, the team pursued the Lawrence Hall of Science at UC Berkeley to assist in defining some key requirements for the project. This semester the team transitioned from concept generation to design actuation, leading to the requirements transitioning to fall more in line with the direction of the project.

The team's goals and the museum's constraints were expanded upon so individual functions, components, target values, and testing methods could be determined. These are split up into functional and component decompositions, which are included in Appendix B and C. The requirements provided direction for the team in concept generation and gave the team a metric of success for the final system evaluation. Unmet requirements indicate areas of future study for up-in-coming teams.

Functional Requirements

The base functional requirements focus on what the team wants the animatronic to achieve. Primary areas of focus include Structure, Material, Power, Articulation, and Safety requirements. These are the "shall's" that the team wanted the final robot to meet. The requirements guided the team in design; target values gave the team a way to measure their success in these guidelines.

The key functional requirements of this project included an animatronic that is visually anatomically accurate to a stegosaurus skeleton. The animatronic must be able to walk, and therefore balanced dynamically. Additionally, the motion of the animatronic must reflect a plausible natural motion of the living stegosaurus.

In terms of material functional requirements, the stegosaurus must be durable, and economical. Further, the material should be easily repairable.

Looking at the power requirements, the animatronic must have a reasonable runtime, and a reasonable recharge time. The control system must be able to coordinate between the mechanical and electrical components. The animatronic must be able to respond to user input.

All these requirements have key specific target data, as well as validation methods, included in Appendix B and C.

Component Requirements

Component requirements focused on what should be included as a physical part in the final animatronic. The component decomposition outlines each individual component and subsystem the team intends on using to create the robot. The components are categorized into Control Systems, Electrical Systems, Main Structure and Joints. Each individual component is listed in Appendix B, and the expected requirement the component must achieve to be viable. Target values for validation were set with their associated units which guided individual design choices for each component and were carefully considered when choosing between various component specifications for a single integrated system.

When reviewing the component requirements, the team determined that the main structure should not exceed 3 feet. Further, the team wanted each limb to have 2 degrees of freedom.

The motors/actuators must provide sufficient torque to move the limbs to the specified position. The control system will be operated from a remote control with a minimum range of 50 feet. It will contain a computer that is capable of controlling the joint actuators so that they reach the appropriate position. Further component specifications are included in Appendix C, along with their validation method.

Design Constraints

Design constraints are restrictions that typically fall outside of the team's control, either due to customer requirements or limited resources. The majority of constraints on this project were the result of resource limitations. As is typical for senior design, the team had a narrow timeline to finish the project, with 3 months to complete system designs, order material, manufacture parts, assemble, and test. Further, this project consisted of only mechanical engineering students, but its scope reached into computer science and electrical engineering. The team had access to two 3-D printers, but the combined print-time of all the components reached over 180 hours. The print beds were small enough that the parts needed to be split and glued after the fact.

Customer design constraints were much fewer, but the staff at UC Berkeley had relevant input. The staff wanted the electrical parts to be off the shelf, not custom-made. Additionally, they wanted it to be easily operable by museum staff and able to be maintained . All these constraints were taken into consideration when developing the animatronic.

VI. System Concept Development

The majority of concept generation was conducted during the previous semester; final decisions were made early in this semester to allow for timely delivery and assembly. The team's process for concept generation began with reviewing the requirements and constraints. The team then provided numerous solutions to each requirement. These solutions were analyzed, and the team selected the best to use in the final design.

3D Modeling:

Figure 18 represents the final 3D model the prototype was based from, all of the joint types remain consistent with previous reports just the digital representation of them has been finalized. Below is a condensed outline on joint types and selection, followed by a discussion on the physical prototype.

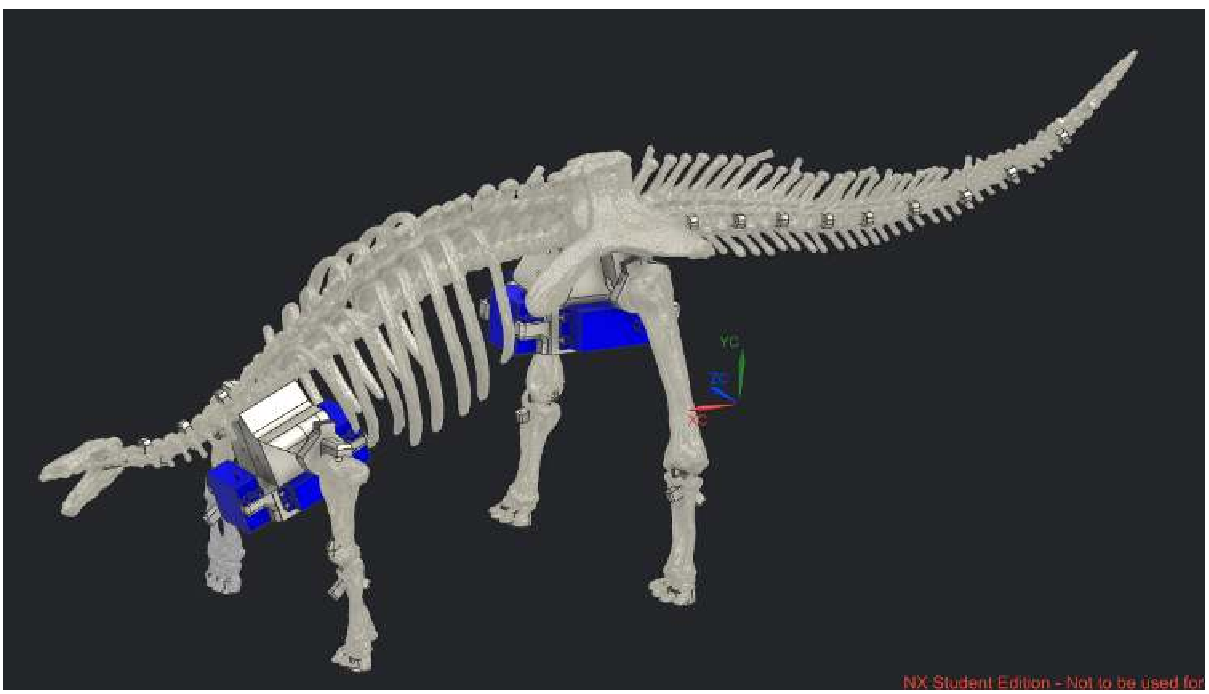


Figure 17: Finalized CAD Model

Legs:

Final leg design remains consistent with previous reports. These use capstan drives for hips/shoulders and will utilize internal cable routings for the knees/elbows which can be seen below.

Another main note will later be discussed in the manufacturing section of this report, but the use of 3D printing for this project makes the internal cable routing and a large portion of the geometry possible as it is not feasible with other manufacturing methods.

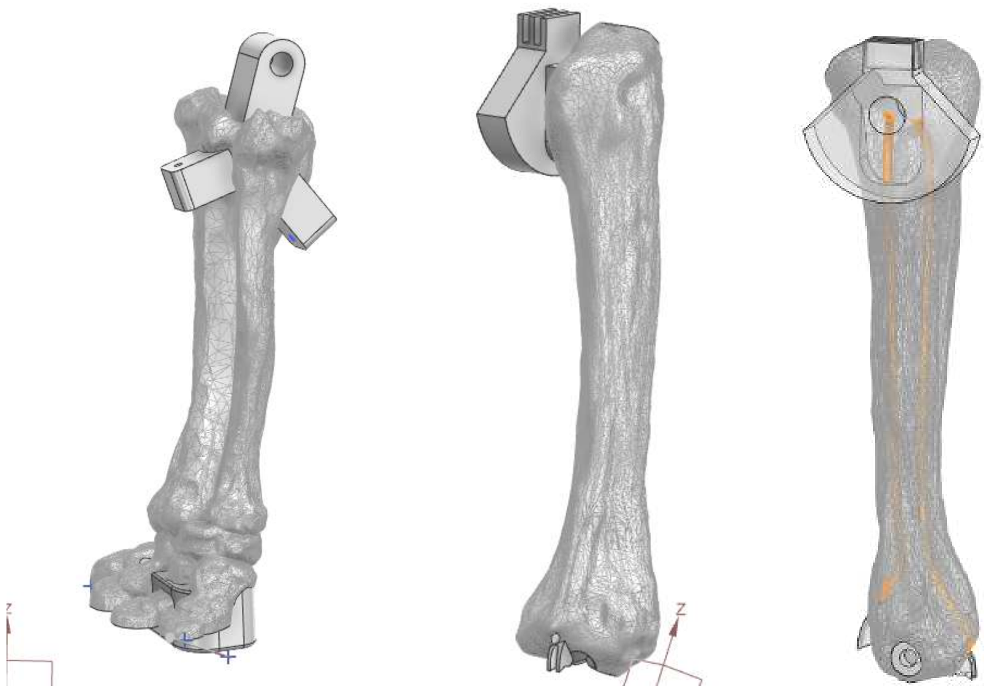


Figure 18: Rear Leg Design

Knee / Elbow Joints:

The knees and elbows shall utilize the internal cable routings seen in the femurs and upper-arms. These cables are then routed through the hips and shoulder mounts to remotely mounted servo motors with pulleys for movement. The initial concept drawing can be seen in figure 18 and the joints themselves can be seen below.

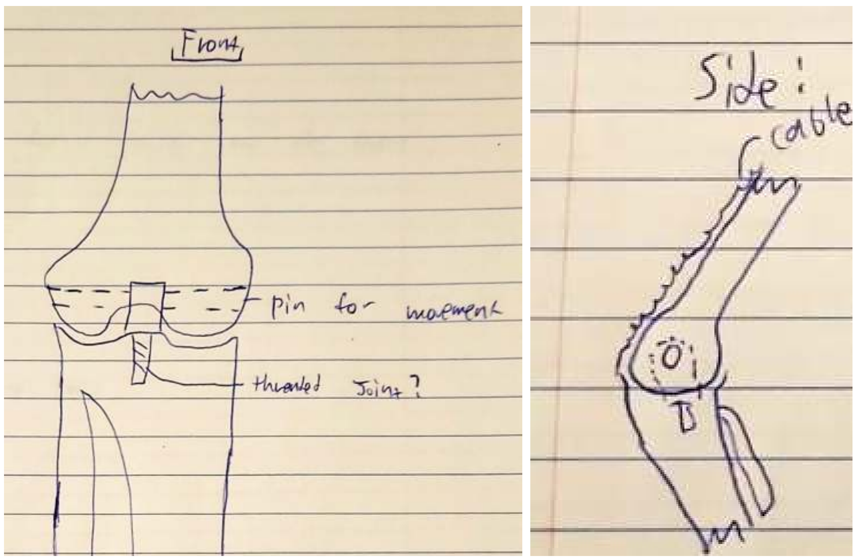


Figure 19: Knee base joint

On the topic of motors The team did consider the use of linear actuators, which would allow for the actuator to essentially act as a muscle with cables for tendons. However, their relatively slow speeds will likely not allow the team to meet the final walking speed goal set by the functional requirements, so it is replaced by servo motors instead for better speed and availability. However, it is curious to take inspiration from bio-mechanics through how muscles and tendons would have been set up if it were a living creature.

With the control motor being separated from the physical joint does present a few challenges, the more severe of which being that The team need a way to route the cables through another moving part, while attempting to minimise added slop or movement when the hips move separate to the knee. It may also be possible to counteract some of this on the software side of things. however an effort should be made to minimise this to simplify the finished software.

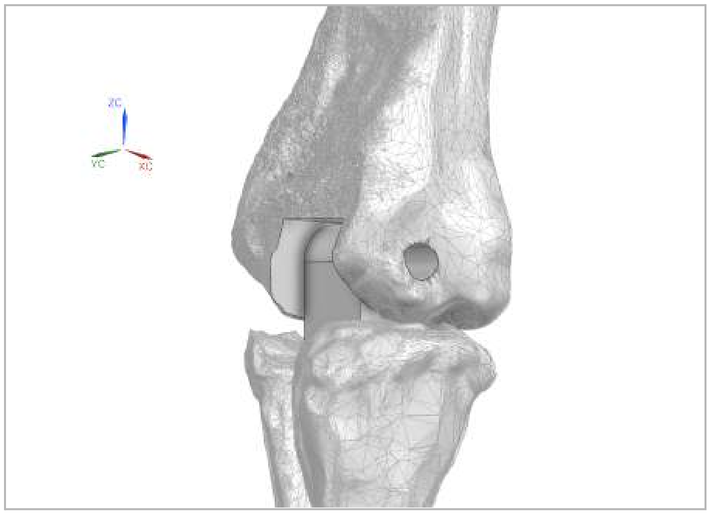


Figure 20: Base model for Knee Hinge.

To begin the knee design, an initial hinge is needed to provide structural support and a rotation point. When looking at the knees of living creatures, they are not fixed at a pin, but provide a sliding motion. This can be mimicked by moving the rotation point of the hinge upwards into the femur, here the top joint on the tibia will appear to slide over the femur instead or rotating at a hard point.

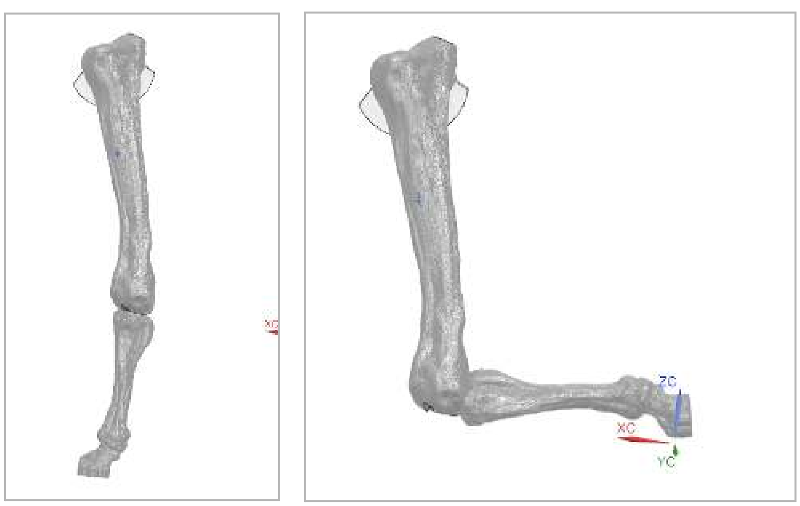


Figure 21: Knee Range of Motion.

Hips:

After the leg joints have been selected, they need to be mounted so they can rotate and support the full weight of the final prototype. To accomplish this, bearings can be mounted into some integrated supports for the hips. A nearly identical design will be used for the shoulder so only the rear legs will be shown in *figure 22* below. Servo motor mounts will also be mounted to this point and this is where the capstan drive wheel will be connected to the hips themselves. For the knees, a takeoff point can also be placed into the central part of this mount where the cables will exit, and connect to additional motors that will provide movement to the knees.

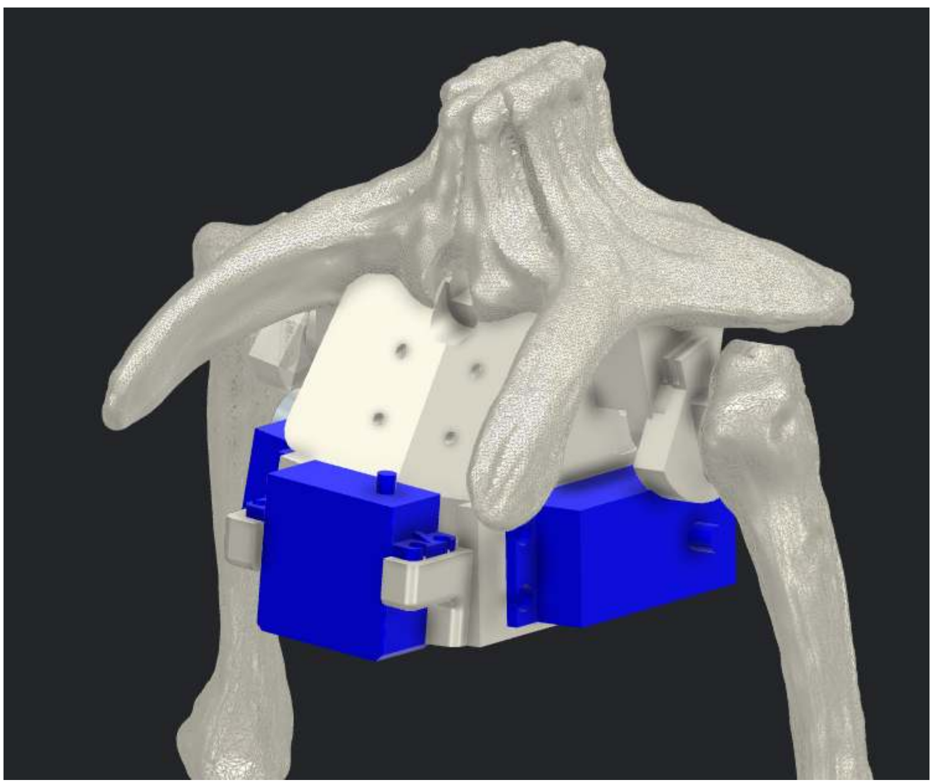


Figure 22: Hips with Base Legs Attached

Tail:

The tail of a Stegosaurus makes up nearly 50 percent of its entire body length. And although it will not directly impact the walking ability of the robot. It will be required to move for both balance and emotion. Per the current design requirements, the tail will need at minimum 8 separate vertebrae sections that are able to articulate a total of 45 degrees side to side across the length of the tail. In order to meet this, the final tail will be made up of a single flexible section. Splitting the tail into individual sections with joints was considered, however a single piece ultimately simplifies the design and greatly lowers part count.

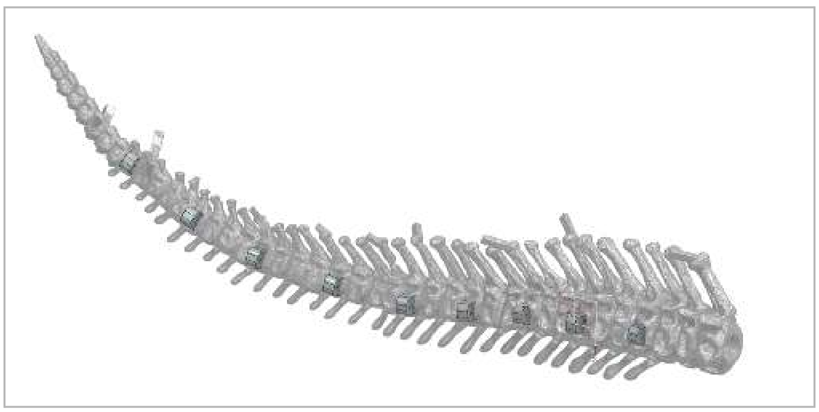


Figure 23: Tail with pull blocks for cables.

Neck:

Similar to the tail, the neck will contain multiple vertebrae, in this case it will be made from a single flexible element. And will need to be able to support the neck, move in 2 axes (up/down and left/right). A preliminary model of the neck is shown below in *figure 24*, per the pugh matrix this will end up being a fully flexible section. Another observation can be made in the side cable routing. The initial scan was first modified by sculpting the mesh in both blender, and thenMeshMaker in order to add a solid structure in between the forward vertebrae structure and the vertebrae to mimic bone structure. And although it is not biologically accurate, it provides a clever way to minimize protrusions from the exterior boundaries of the neck itself.

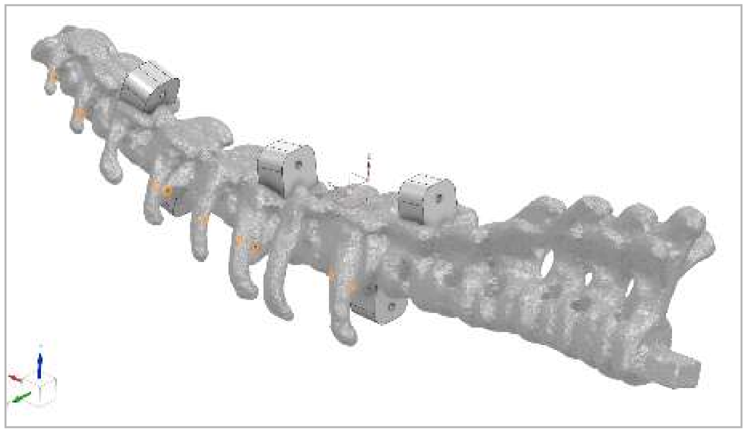


Figure 24: Base Neck Design

Skull/ Mouth:

The skull and mouth will need to provide non-structural movement, and will not be critical to the final operation of the robot. however it will be necessary to provide an emotional element to the dinosaur. A simple hinged design will be sufficient to produce the necessary 20 degrees of movement outlined in the functional requirements. Movement will be provided from a micro linear servo that can fit within the boundary of the skull itself. Torque requirements will be minimal as it will only need to provide aesthetic movement and will not be interfacing with a solid surface.

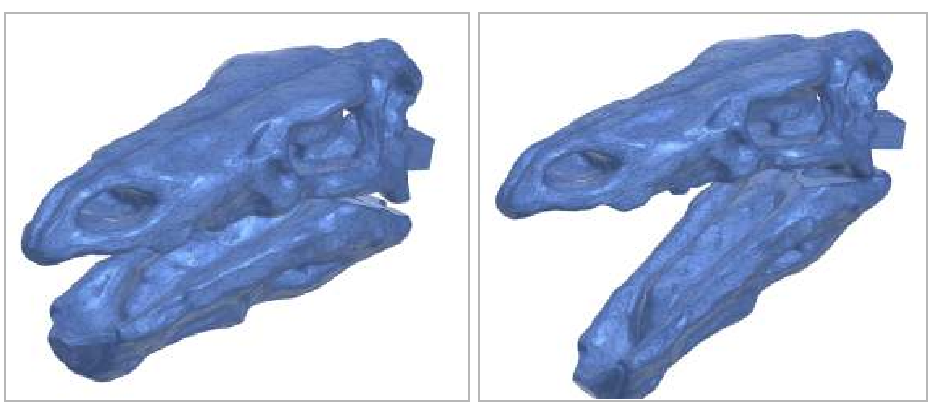


Figure 25: Skull with mouth closed vs open.

Hardware and Manufacturing

Most of the parts in this robot will be 3D printed. If a prospective product were to come to market, then injection molding, machining, and possibly resin printing may be used in order to increase the surface finish and overall quality of parts.

For some of the leg members, 3D printing will be the only way we will be able to create features such as internal routing for cables in a relatively small part. The leg joints will also need to be supported. In this case we could use anything from plastic on plastic joints, to bushings, bearings, and more. The top joint on each leg will be supported by needle bearings to create smooth motion, the knees and elbows will first use a plastic on plastic joint with a screw acting as a pin. This may need to move to a steel or metal rod end to better support the lower leg members.

Electronics Packaging

The housing for the electronics is approximately 5.2in. x 4.5in. x 3in. Strategic placement and orientation of the components in the housing is essential. Many components are fitted precisely into a tight space. This housing will fit under the ribcage; the cover will be connected to the dinosaur’s spine, and the body of the housing will be connected to the cover via heated inserts and bolts. Some electronic components have slots and will be bolted into the housing via heated inserts, and some of the electronic components do not have slots, and will be free floating in their designated space. To avoid inconsistencies and damage, all cables must have a minimum bend radius of five times the overall cable diameter, which is a safe standard rule of thumb when working with wires. Appropriate space was left for the wires, and cutouts were made so that wires can go out of the housing and into the motors. The housing will be 3D printed upright and will use supports where needed. Tolerancing and spacing is accounted for when it comes to cutouts and holes for heated inserts. Fillets were placed to strengthen the housing structure.

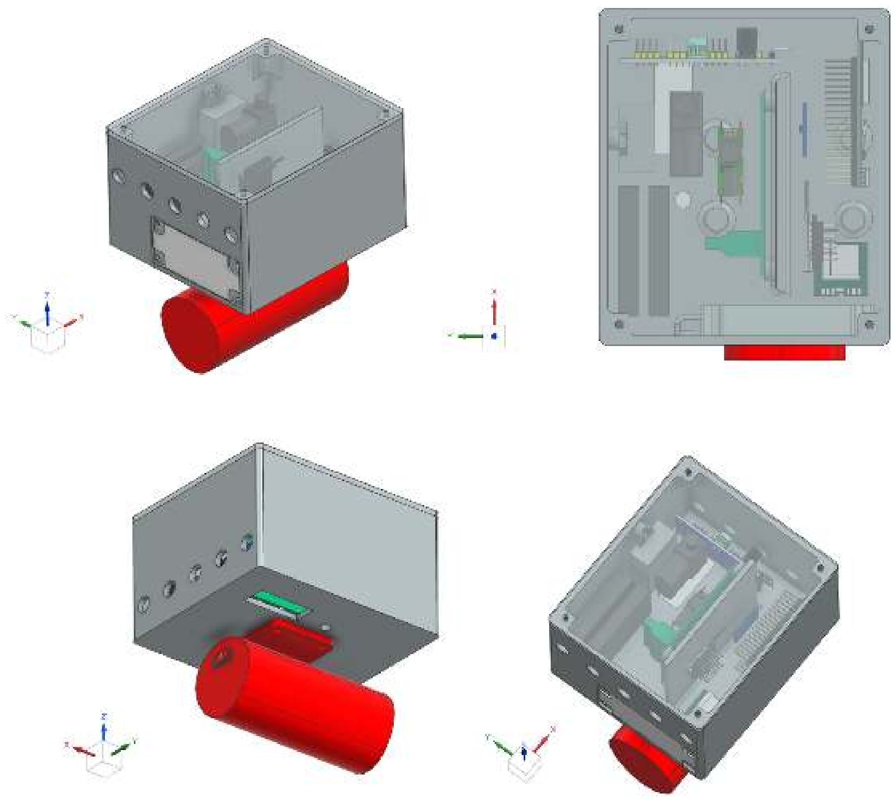


Figure 26: Electronics Housing

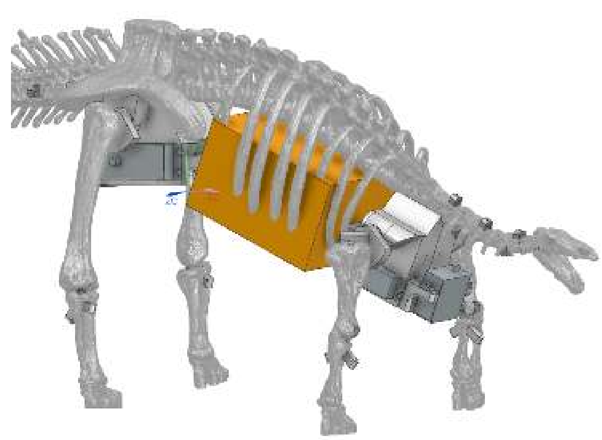


Figure 27: Electronics Housing (orange) Realestate

Materials

The team 3D printed the majority of the structure of the stegosaurus with ASA filament. The figure below compares the properties of many materials [54]

	ABS	Flexible	PLA	HIPS	PETG	Nylon	Carbon Fiber Filled	ASA	Polycarbonate	Polypropylene	Metal Filled	Wood Filled	PVA
Ultimate Strength	40 MPa	26 - 43 MPa	65 MPa	32 MPa	53 MPa	40 - 85 MPa	45 - 48 MPa	55 MPa	72 MPa	32 MPa	20 - 30 MPa	46 MPa	78 MPa
Stiffness	5/10	1/10	7.5/10	10/10	5/10	5/10	10/10	5/10	6/10	4/10	10/10	8/10	3/10
Durability	8/10	9/10	4/10	7/10	8/10	10/10	3/10	10/10	10/10	9/10	4/10	3/10	7/10
Maximum Service Temperature	98 °C	60 - 74 °C	52 °C	100 °C	73 °C	80 - 95 °C	52 °C	95 °C	121 °C	100 °C	52 °C	52 °C	75 °C
Coefficient of Thermal Expansion	90 µm/m-°C	157 µm/m-°C	68 µm/m-°C	80 µm/m-°C	60 µm/m-°C	95 µm/m-°C	57.5 µm/m-°C	98 µm/m-°C	69 µm/m-°C	150 µm/m-°C	33.75 µm/m-°C	30.5 µm/m-°C	85 µm/m-°C
Density	1.04 g/cm³	1.19 - 1.23 g/cm³	1.24 g/cm³	1.03 - 1.04 g/cm³	1.23 g/cm³	1.06 - 1.14 g/cm³	1.3 g/cm³	1.07 g/cm³	1.2 g/cm³	0.9 g/cm³	2 - 4 g/cm³	1.15 - 1.25 g/cm³	1.23 g/cm³
Price (per kg)	$10 - $40	$30 - $70	$10 - $40	$24 - $32	$20 - $60	$25 - $65	$30 - $80	$38 - $40	$40 - $75	$60 - $120	$50 - $120	$25 - $55	$40 - $110
Printability	8/10	6/10	9/10	6/10	9/10	8/10	8/10	7/10	6/10	4/10	7/10	8/10	5/10
Extruder Temperature	220 - 250 °C	225 - 245 °C	190 - 220 °C	230 - 245 °C	230 - 250 °C	220 - 270 °C	200 - 230 °C	235 - 255 °C	260 - 310 °C	220 - 250 °C	190 - 220 °C	190 - 220 °C	185 - 200 °C
Bed temperature	95 - 110 °C	45 - 60 °C	45 - 60 °C	100 - 115 °C	75 - 90 °C	70 - 90 °C	45 - 60 °C	90 - 110 °C	80 - 120 °C	85 - 100 °C	45 - 60 °C	45 - 60 °C	45 - 60 °C
Heated Bed	Required	Optional	Optional	Required	Required	Required	Optional	Required	Required	Required	Optional	Optional	Required
Recommended Build Surfaces	Kapton Tape, ABS Slurry	PEI, Painter's Tape	Painter's Tape, Glue Stick, Glass Plate, PEI	Glass Plate, Glue Stick, Kapton Tape	Glue Stick, Painter's Tape	Glue Stick, PEI	Painter's Tape, Glue Stick, Glass Plate, PEI	Glue Stick, PEI	PEI, Commercial Adhesive, Glue Stick	Packing Tape, Polypropylene Sheet	Painter's Tape, Glue Stick, PEI	Painter's Tape, Glue Stick, PEI	PEI, Painter's Tape
Other Hardware Requirements	Heated Bed, Enclosure Recommended	Part Cooling Fan	Part Cooling Fan	Heated Bed, Enclosure Recommended	Heated Bed, Part Cooling Fan	Heated Bed, Enclosure Recommended, May Require All Metal Hotend	Part Cooling Fan	Heated Bed	Heated Bed, Enclosure Recommended, All Metal Hotend	Heated Bed, Enclosure Recommended, Part Cooling Fan	Wear Resistant or Stainless Steel Nozzle, Part Cooling Fan	Part Cooling Fan	Heated Bed, Part Cooling Fan

	ABS	Flexible	PLA	HIPS	PETG	Nylon	Carbon Fiber Filled	ASA	Polycarbonate	Polypropylene	Metal Filled	Wood Filled	PVA
Flexible	–	✔	–	–	–	✔	–	–	–	✔	–	–	✔
Elastic	–	✔	–	–	–	–	–	–	–	–	–	–	–
Impact Resistant	✔	–	–	✔	–	✔	–	✔	✔	–	–	–	–
Soft	–	✔	–	–	–	–	–	–	–	✔	–	–	✔
Composite	–	–	–	–	–	–	✔	–	–	–	✔	✔	–
UV Resistant	–	–	–	–	–	–	–	✔	–	–	–	–	–
Water Resistant	–	–	–	–	✔	–	–	–	–	✔	–	–	–
Dissolvable	–	–	–	✔	–	–	–	–	–	–	–	–	✔
Heat Resistant	✔	–	–	✔	–	✔	–	✔	✔	✔	–	–	–
Chemically Resistant	–	–	–	–	✔	–	–	–	–	–	–	–	–
Fatigue Resistant	–	✔	–	–	✔	✔	–	–	✔	✔	–	–	✔
Heated Bed Not Required	–	✔	✔	–	–	–	✔	–	–	–	✔	✔	–

Figure 28: Materials and property comparison [54].

Materials that were under consideration include: PLA, PETG, ABS, ASA, NYLON, TPU, and Carbon fiber reinforced (nylon, onyx,). When it comes to the bulk of the material, ASA will be used because of its availability and properties. ASA is strong and has deflection properties similar to ABS, and it can handle heat better than PLA and ABS. When prototyping, it is beneficial to see deflection; the team used ASA for the final product., and for the tail and neck areas, the team used TPU. TPU was chosen for these areas because when the robot walks, the neck and tail areas will oscillate side to side, giving it that lifelike motion, and realism.

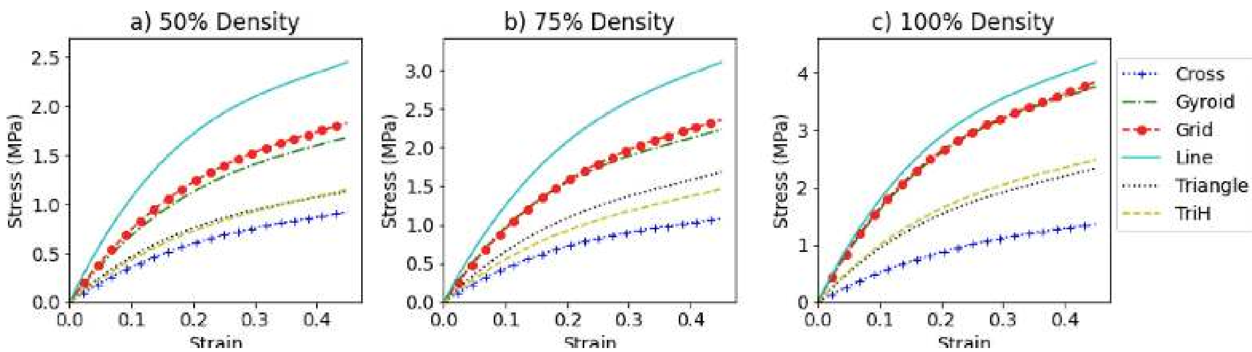
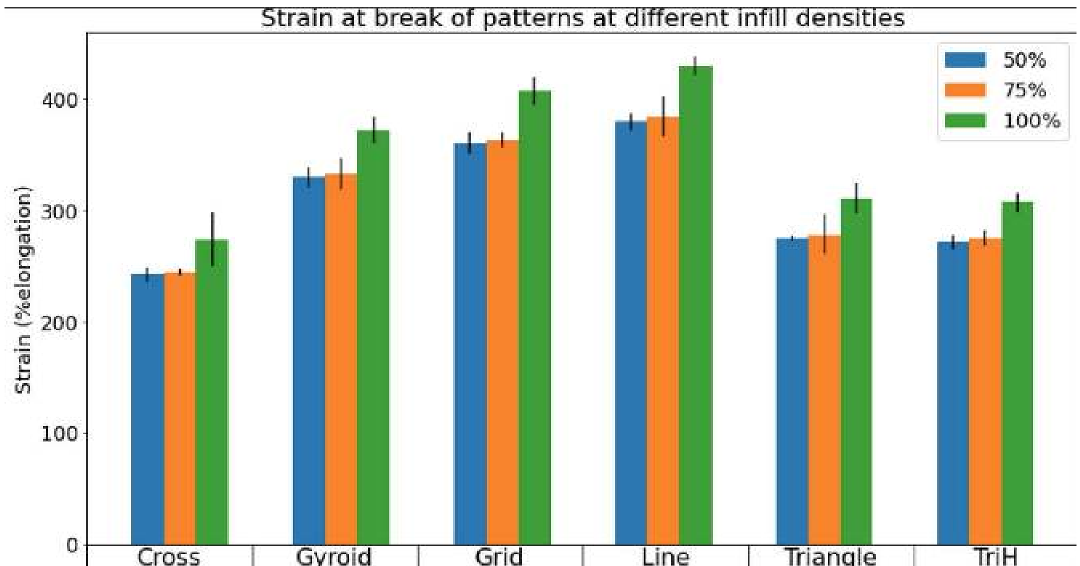


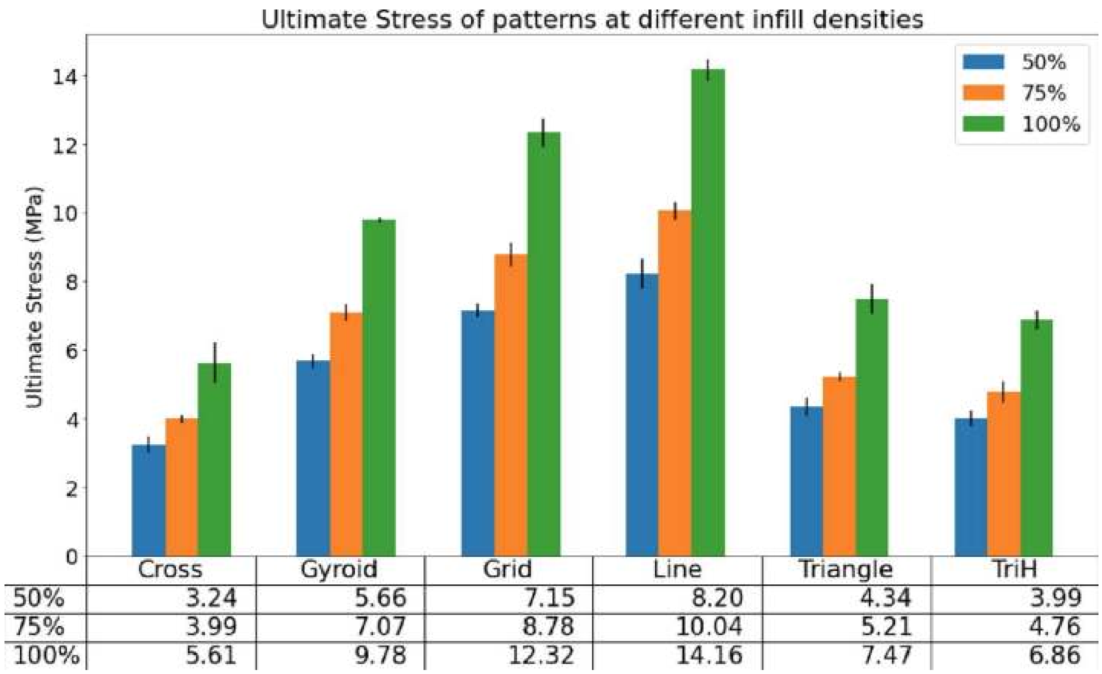
Figure 29: Infill Patterns at Different Densities [55].

When it comes to infill patterns, hex, or honeycomb, has the highest strength in all directions [55]. If a part is experiencing load in one direction, line and grid are the strongest option. Gyroid is also another alternative to hex/honeycomb. In regards to infill density, anything above 40% is adding more weight than strength, and it is not worth it to spend more money to get a slightly stronger bone [55]. Taking into consideration the motors strength, having low density bones should be considered when calculating torque required when deciding on which motors will be chosen. Further, having a lower infill density is better for materials like ASA because it will bend before breaking, so it acts like a shock absorber if any load is applied in a rapid manner. All these materials do not have a problem with clogging when printing. TPU may be used for damping and vibration. TPU can be challenging to print, but the team should be able to overcome this. In our final design, the team used 12 walls, and 50% infill, and for the pattern, the team used rectilinear. For the hips, and the rest of the printed parts, the team went with 6 walls, and 25% with most of it being gyroid infill pattern.



	Cross	Gyroid	Grid	Line	Triangle	TriH
50%	241.8%	329.4%	359.9%	379.7%	274.7%	271.4%
75%	244.4%	332.6%	363.0%	383.8%	278.1%	274.2%
100%	273.8%	371.5%	406.6%	429.2%	310.5%	306.8%

Figure 30: Compares different infill patterns with density and shows the strain % elongation [55].



	Cross	Gyroid	Grid	Line	Triangle	TriH
50%	3.24	5.66	7.15	8.20	4.34	3.99
75%	3.99	7.07	8.78	10.04	5.21	4.76
100%	5.61	9.78	12.32	14.16	7.47	6.86

Figure 31: Compares different infill patterns with density and shows the ultimate stress in MPa [55].

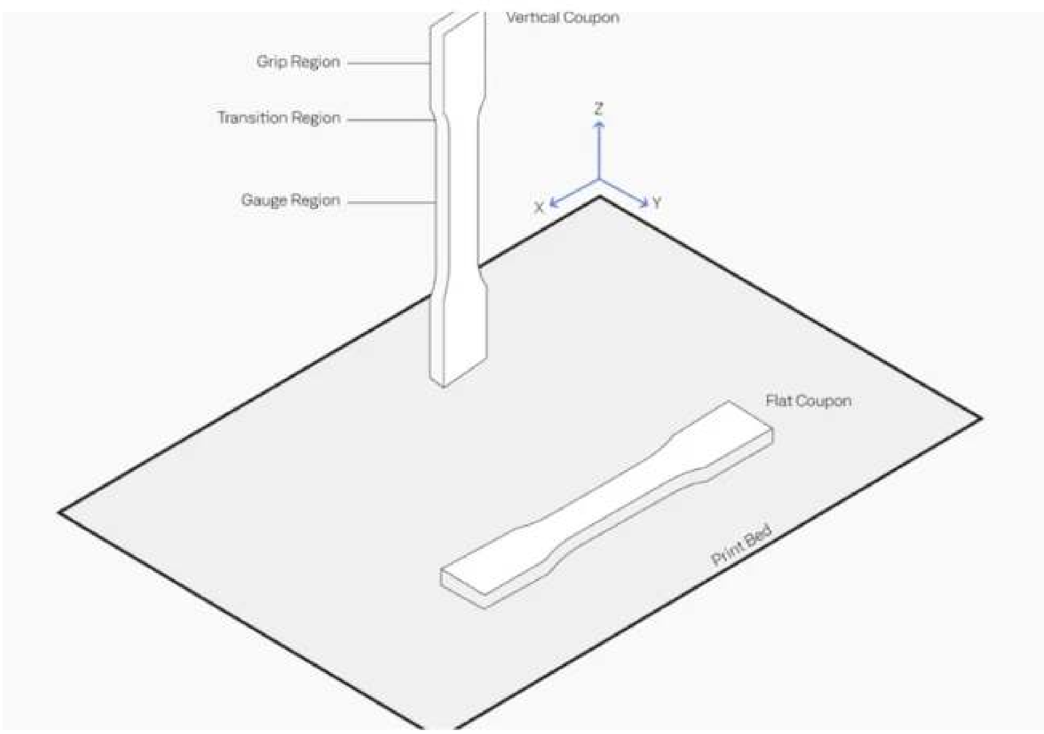


Figure 32: Importance in printing orientation [56].

Printing orientation is extremely important when 3D printing; simply changing the orientation can make a big difference when it comes to the success of a 3D printed object [56]. For example, in figure 70, when testing the print in tension (pulling the grip regions), the vertical coupon will easily snap as the layer lines are laid horizontally which create many small notches which set up the part for failure. On the other hand, the flat coupon (printed horizontally) will take much more tensional force as the layer lines run along the part.

Another thing to take into consideration is print quality. Errors in prints such as "u" and "v" notches can cause a 3D printed part to fail when undergoing tension forces. Having a notch as seen in figure 6 in a 3D printed part puts the structural integrity in danger [57]. This is because notches create a place where cracking begins. This is why quality is important when it comes to these 3D printed bones.

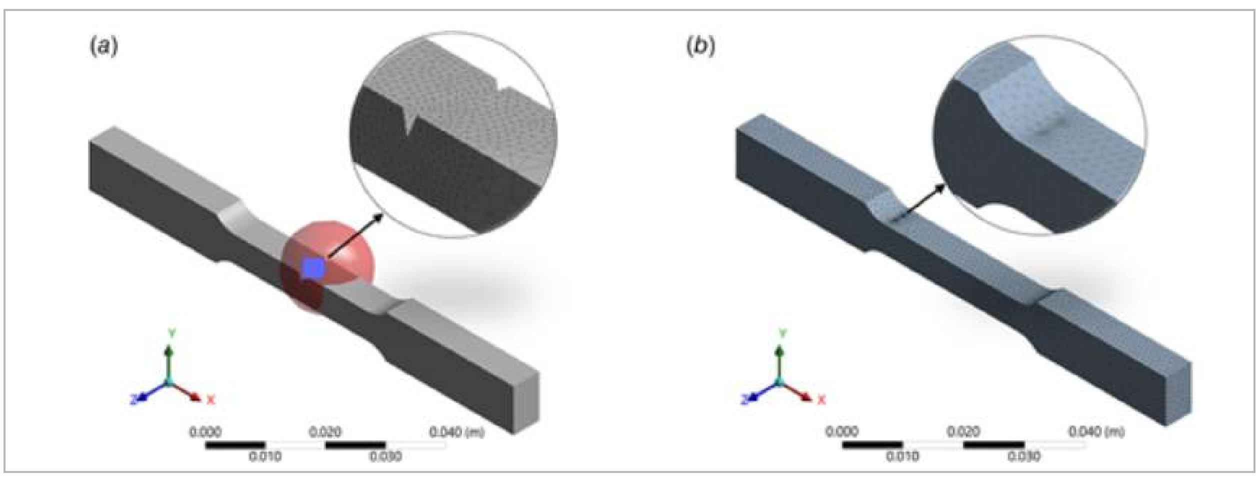


Figure 33: [57] Importance of print quality

Controls:

In regards to the programming section of the project, the following section defines the generalized plan and functional requirements of the control code. It will go in depth on how software will interact with the physical parts of the robot and execute physical movements.

This section presents the original aim of the code and electricals. It outlines the envisioned ideal approach of what the code would have looked like given more development time and experience with the current electrical components. In later sections, a discussion on what was actually achieved will explore any constraints and adjustments.

Code Structure Overview:

Once powered on, the selected microcontroller will run a startup routine. It will initialize all hardware components, set initial parameters, and enter State 1 (Idle State). The initialization should confirm all systems are operational before proceeding. For example it will confirm that the remote control is connected and all motors are responsive. If any issues arise during start up, the user will be prompted with an audio cue, alerting that something has gone wrong. If no issues occur and the start up is successful, a different audio cue will play showing the user that it is safe to proceed with standard operation of the robot.

Once time to power off, the robot needs to shut down safely thus it will have a shutdown procedure to ensure all hardware components are powered down and data is saved to non-volatile memory. The shutdown procedure should also ensure that all motors are set to a safe position before fully powering down the entire robot.

Another system feature the code should account for is an emergency stop function. The emergency stop should immediately hold all motor activity and put the system into a safe state when triggered by a specific input. The input can be a button press, a remote command, or an automatically detected condition. One condition could be the gyroscope being a specific position which would indicate the robot has fallen over. Other possible triggers for the emergency stop to activate could be a critical error being detected, overheating, motor stall, or a loss of communication with the remote control system.

For maintenance and troubleshooting purposes, there should be a dedicated section of the code to handle data logging. The system should record all variables, possible errors, connectivity issues, joint positions, response times, controller inputs, current power level, and the temperature of the system. All the data collected should be recorded and logged on an SD card or flash drive. This allows for an easier identification of when parts need to be replaced or when parts are not working.

Designated code must be included to handle and manage all incoming signals from the controller to execute the robots physical commands. Many microcontrollers and microcomputers already come with built-in libraries and additionally there are plenty of publicly available libraries for different types of remote control systems that can be used to handle controller inputs. The code needs to continuously monitor and process inputs from the remote control, such as joystick movements, button presses, and trigger presses to determine desired pre-coded actions and states.

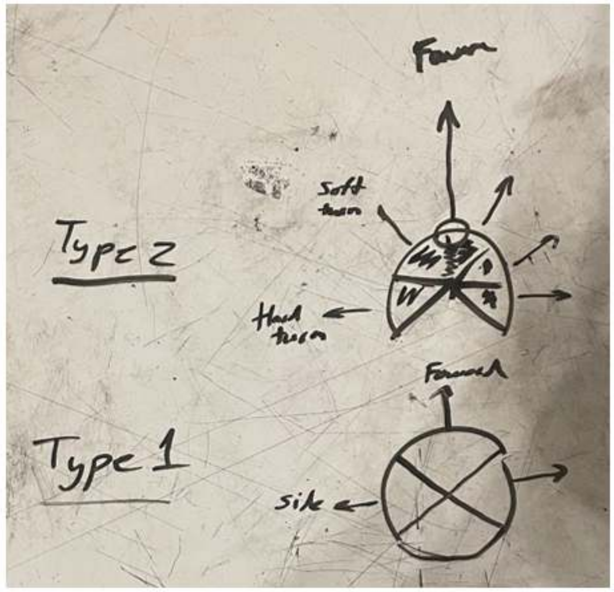


Figure 34: Joystick Inputs

In terms of audio management, the code is expected to play various sound effects or audio cues at specific events or states from start up. The sounds are used to signal when there has been an error or a successful startup. The main way the users will be able to determine if something is wrong or if all is good to proceed will be via audio cues.The audio will also be used for various animations to bring this robot to life.

In regard to power management, the code is expected to cue the user through some kind of visual or audio cue to indicate that the battery is low and a recharge is needed; This may be through an audio chime or a LED light indicator.

Included in the code, the robot is expected to have four different modes and to seamlessly transition between the states based on user input; these modes are currently defined as States 1-4.

State 1,2,3,4 Descriptions:

State 1, otherwise known as the idle state, refers to when the robot's motors are set to a pre-coded position and are now waiting for further user input. This state is mainly for start-up purposes and as a safe transition state to other states.

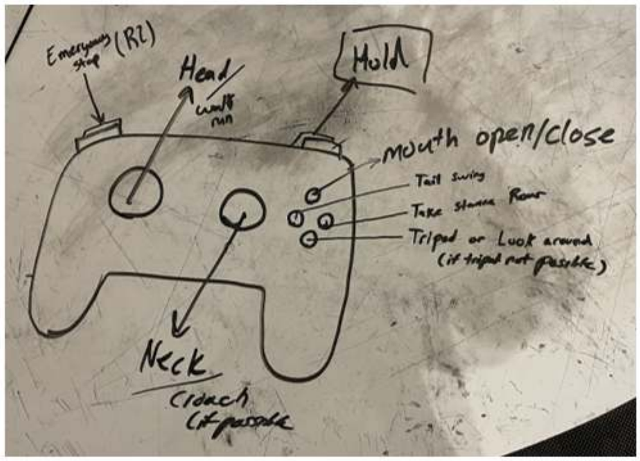


Figure 35: Theoretical Key Mapping of Inputs

State 2 refers to when the robot is in motion. This state is described as when the motors move according to registered inputs and the gait study in a predefined cycle. The stop and start of state 2 is all through user input, and shall remain in this state until a stop input is received. The controller joystick and buttons can modify behavior, speed, and direction. If a stop command is received the robot is expected to transition from state 2 to state 1 before proceeding to any other states.

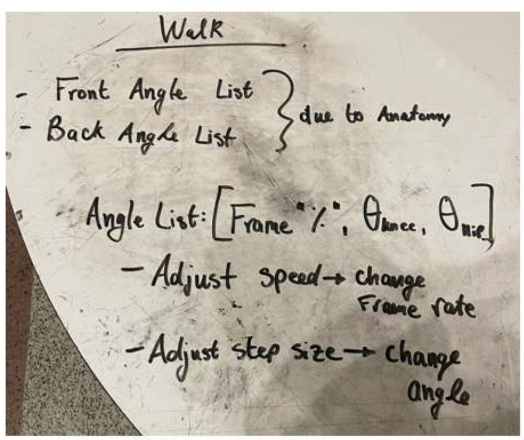


Figure 36: Concept of Gait Study paired with Control System

State 3 is defined as an animation state. The motors are expected to move according to a predetermined animation indicated by an input and executed to completion before returning back to the idle state. Through the user's input they may disrupt and stop the animation halfway through and return to idle state. The robot is projected to have three animations: a tail swing, a roar animation, and a third emotive animation which has yet to be determined. The tail swing animation is expected to have the robot look around and set up and swing its tail as if attacking. The roar animation is expected to have the robot dinosaur take a stance and open its mouth when an audio effect should sound.

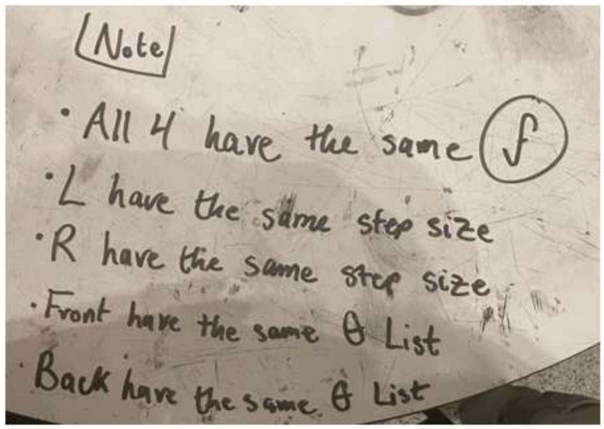


Figure 37: Concept of Data received from Gait Study

During the fourth and final state, State 4, the head, neck and tail will be active and will be controlled directly. The head, neck, and tail motors should move according to user input. The robot is expected to be able to move its head and neck left, right, up, and down. Likewise the neck and tail should be able to move left and right. The jaw should also be able to open and close. The joystick and controller buttons are expected to be able to modify motor behavior, such as speed and direction, to carry out these physical motions. Once state four has been initiated by the user, a second input must be signaled to exit this state and return back to the idle state.

The visual below represents the code structure of the project as was outlined above. It highlights the intended design, providing a clear reference point for how the dinosaur was meant to operate before constraints and adjustments were introduced and should be used as reference for future iterations.

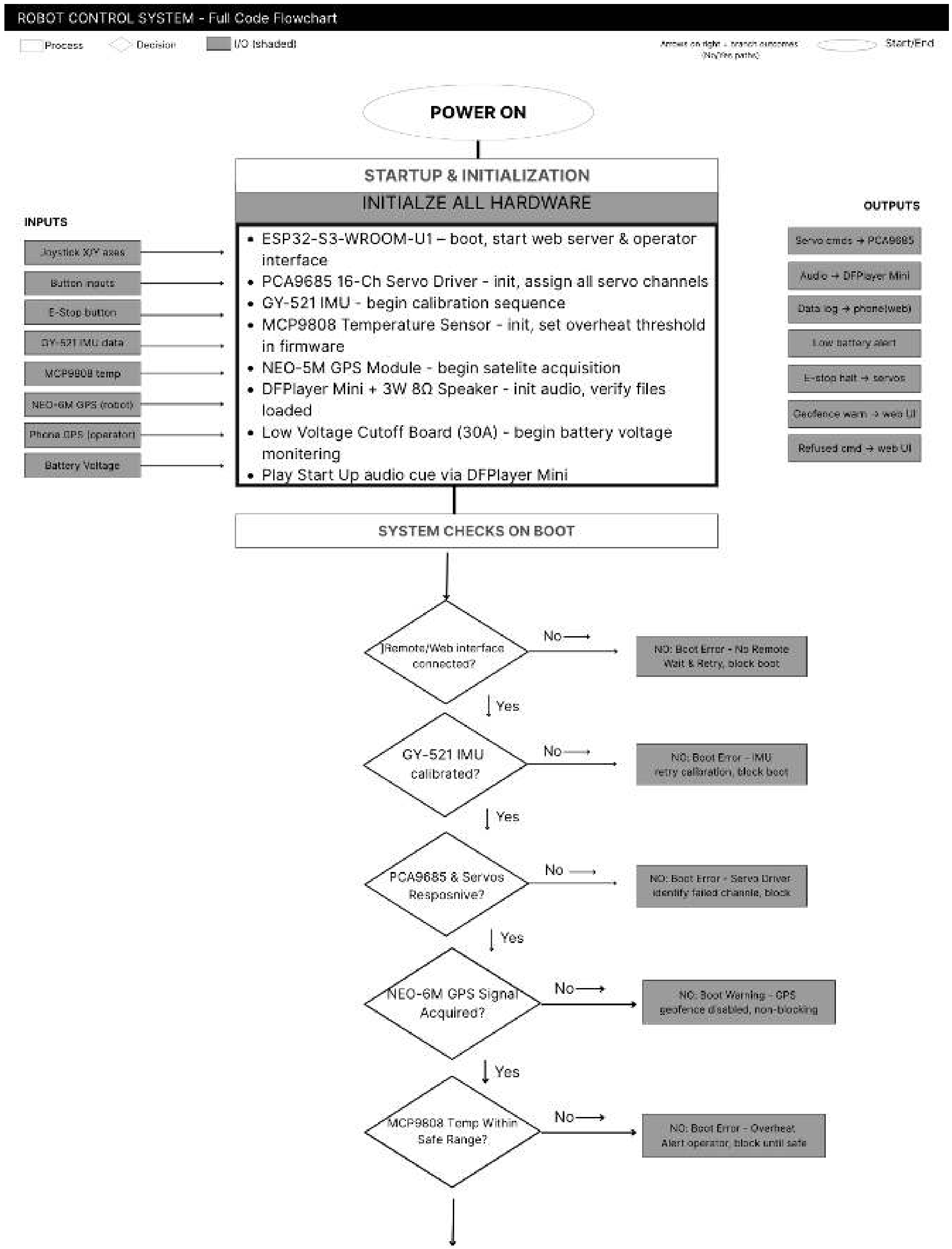
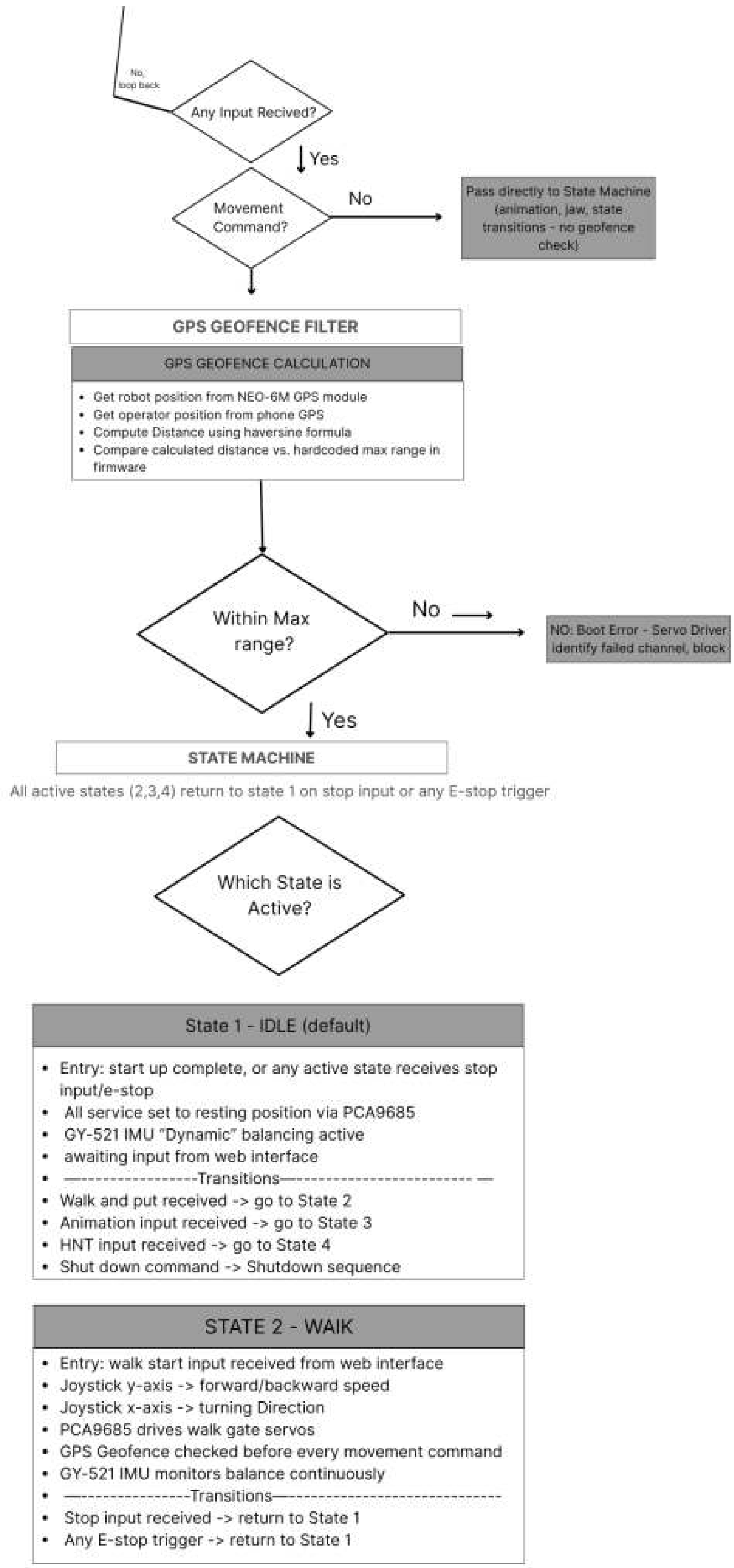
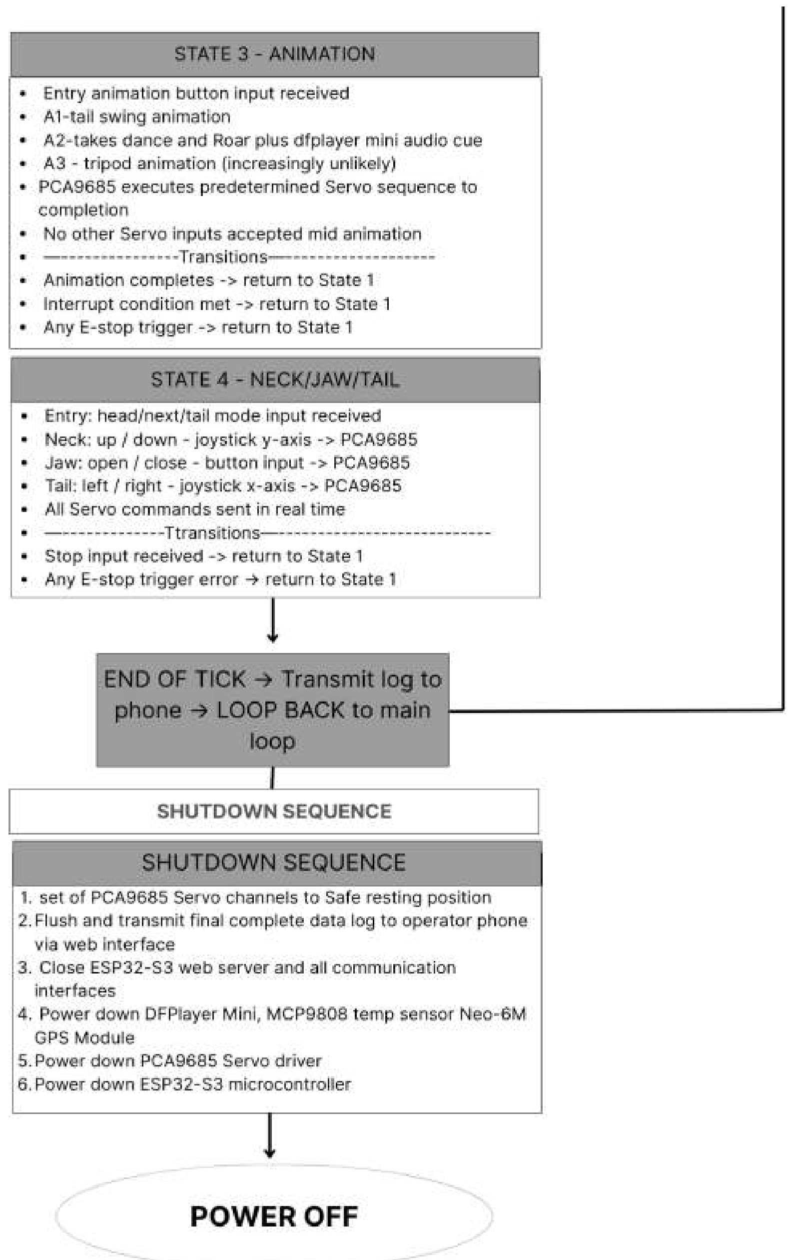


Figure 38: Code Flow Chart


No, loop back
Any Input Recived?
Yes
Movement Command?
No
Pass directly to State Machine (animation, jaw, state transitions - no geofence check)
GPS GEOFENCE FILTER
GPS GEOFENCE CALCULATION
• Get robot position from NEO-6M GPS module
• Get operator position from phone GPS
• Compute Distance using haversine formula
• Compare calculated distance vs. hardcoded max range in firmware
Within Max range?
No
NO: Boot Error - Servo Driver identify failed channel, block
Yes
STATE MACHINE
All active states (2,3,4) return to state 1 on stop input or any E-stop trigger
Which State is Active?
State 1 - IDLE (default)
• Entry: start up complete, or any active state receives stop input/e-stop
• All service set to resting position via PCA9685
• GY-521 IMU "Dynamic" balancing active
• awaiting input from web interface
• ——————-Transitions—————————— —
• Walk and put received -> go to State 2
• Animation input received -> go to State 3
• HNT input received -> go to State 4
• Shut down command -> Shutdown sequence
STATE 2 - WAIK
• Entry: walk start input received from web interface
• Joystick y-axis -> forward/backward speed
• Joystick x-axis -> turning Direction
• PCA9685 drives walk gate servos
• GPS Geofence checked before every movement command
• GY-521 IMU monitors balance continuously
• ——————-Transitions——————————
• Stop input received -> return to State 1
• Any E-stop trigger -> return to State 1


STATE 3 - ANIMATION
Entry animation button input received
A1-tail swing animation
A2-takes dance and Roar plus dfplayer mini audio cue
A3 - tripod animation (increasingly unlikely)
PCA9685 executes predetermined Servo sequence to completion
No other Servo inputs accepted mid animation
—---------------Transitions—------------------
Animation completes -> return to State 1
Interrupt condition met -> return to State 1
Any E-stop trigger -> return to State 1
STATE 4 - NECK/JAW/TAIL
Entry: head/next/tail mode input received
Neck: up / down - joystick y-axis -> PCA9685
Jaw: open / close - button input -> PCA9685
Tail: left / right - joystick x-axis -> PCA9685
All Servo commands sent in real time
—-------------Ttransitions—--------------------------
Stop input received -> return to State 1
Any E-stop trigger error → return to State 1
END OF TICK → Transmit log to phone → LOOP BACK to main loop
SHUTDOWN SEQUENCE
SHUTDOWN SEQUENCE
1. set of PCA9685 Servo channels to Safe resting position
2. Flush and transmit final complete data log to operator phone via web interface
3. Close ESP32-S3 web server and all communication interfaces
4. Power down DFPlayer Mini, MCP9808 temp sensor Neo-6M GPS Module
5. Power down PCA9685 Servo driver
6. Power down ESP32-S3 microcontroller
POWER OFF

Electrical System:

The electrical system poses a unique problem due to the power supply needed to power all components within a highly constrained space. This requires a high density battery that can be easily accessed, recharged, or replaced to support repeated operation. The battery should be able to provide enough power to all subsystems for a long duration or for repeated use. The power must be able to supply two different voltages and current requirements.The control computer, microcontroller boards, and the audio modules operate at approximately 3.3-5 V with relatively low current demands. In contrast, the motor components require between 6-8.7 Volts and can draw higher current under stall conditions. The power system needs to be able to accommodate this, the following section outlines the power subsystem in detail, including the componentment, electrical considerations and rationale behind design choices to address constraints.

Battery

Battery selection requires careful consideration of several factors, but first appreciate the class of battery for the system but be identified. The battery can fall into three categories: Primary non rechargeable batteries, secondary rechargeable batteries, and specialized industrial batteries. Common examples of primary non-rechargeable batteries are, alkaline batteries, zinc carbon batteries, silver oxide and zinc air batteries. Secondary batteries are rechargeable so these can include lithium ion batters, lithium polymer, nickel metal hydroxide, nickel cadmium, lead acid, and lithium iron phosphate batteries. Batteries like sodium ion and solid state batteries are classified as special industrial batteries, commonly defined by their advanced chemistries for high performance, long life cycle or unique conditions.

Pugh Matrix					
			Solution Alternatives		
Concept Selection Legend: Better +, Same S, Worse - Key Criteria	Importance Rating	Benchmark Option	Non-Rechargeable Batteries	Rechargeable Batteries	Industrial Batteries
Maintinence	6		-	+	-
Cost	4		s	s	-
Safety	6		s	s	-
Size	8		+	+	-
Weight	6		+	S	-
Power	8		-	+	+
Sum of Positives			2	3	1
Sum of Negatives			2	0	5
Sum of Sames			2	3	0
Weighted Sum of Positives			14	22	8
Weighted Sum of Negatives			14	0	30
TOTALS			0	22	-22

Figure 39: Battery Pugh Matrix

The selected battery needs to accommodate the following criteria: long life span, rechargeable, compact size, and it must be capable of supplying both the high-voltage, high current demands and the low voltage and low current demand without overloading or damaging the lower-voltage components of the robot.

The battery must be able to support an extended period of operation to meet the demands of museum use. If the battery were to wear down or need to be replaced, the selected battery must be easy to source from common suppliers. This will allow the museum to easily replace the battery at a more economic cost. For example in a theoretical scenario if the robot were to have its power go out, the battery should be easily bought off of major online retailers like Amazon or any other in person option. This would significantly reduce downtime in the case where the battery needs to be replaced, as the battery will be easily accessible and might even allow for stocking spare if the selected battery for quick replacement.

Due to the different voltage and current demands, a single and multiple battery system was considered as well as a parallel battery system. The parallel battery arrangement would maintain the system required voltage but increase the current allowing for higher current accommodation that the motor system requires. Ultimately, a single battery system was selected, as suitable battery models were identified to satisfy the necessary high voltage high current requirement while remaining compact enough to fit within the tight space constraints of the animatronic .

To conclude, the key factors for battery selection were runtime, output voltage, volume, capacity, and overall weight. The main goal of the selected battery was to be easily usable while maximizing energy storage and minimizing size and mass. Having a higher power density was essential to achieve the desired longer operational periods for the anatomic. The team aims for the robot to have roughly 30 minutes of operation at maximum strain and an average operation time of 1 hour and 30 minutes to 2 hours under normal operation conditions.

Batteries commonly used in drones and RC vehicles were recognized as strong candidates for the robot. The two types of Lithium-polymer (Li-Po) type of batteries below offered compact sizing, rechargeability ease, and met several other requirements, making them well suited for our defined constraints.

Figure 40: Lithium-polymer Battery

After multiple communications and discussions with the museum, it was decided that a hot swap system for the battery is going to be utilized in the final design of the prototype. A hot swap system allows a depleted battery to be easily and quickly removed and replaced with a fully charged one. This allows for no significant interruption. Using this approach allows for a smaller battery size while still achieving extended runtime, as operation can continue as long as additional charged batteries are available.

The battery selected for this system was a Milwaukee battery traditionally used in drills. An adapter was proposed to allow the battery to be easily slotted in and out, similar to how it functions in its original tool. The specific battery chosen was the Milwaukee M12 Redlithium battery, a 12-volt lithium-ion battery.

Figure 41: Milwaukee Battery

Charging Port

Because of maintenance and for the ease of the museum, a chargeable battery was determined to be the most practical choice. The rechargeable system eliminates the need for repeatedly opening or disassembling the animatronic to replace batteries. Instead the museum would operate on a plug and charge workflow, making it far more accessible. Due to this a charging port is incorporated into the robot.

After further discussions with the museum, the design was revised to better account for system down. Because of the shift in battery approach, implementing a charging port system for the dinosaur was no longer practical. Both the museum and the team felt it was more optimal to maintain a set of fully charged batteries that could be swapped out as needed as opposed to pausing operation to recharge a single battery.

A key concern and the primary reason for this change was minimizing interruption to the exhibit. If the dinosaur relied on in place charging, the robot would be out of operation for extended periods, reducing its operation time for the general public. In constant, the hot swap battery reduces downtime to seconds or minutes. This allows for continuous or near-continuous operation. We believed that this aligns better with the goal of maintaining a consistent and engaging exhibit experience. The final system is rechargeable, long-lasting, and designed for quick, efficient battery replacement.

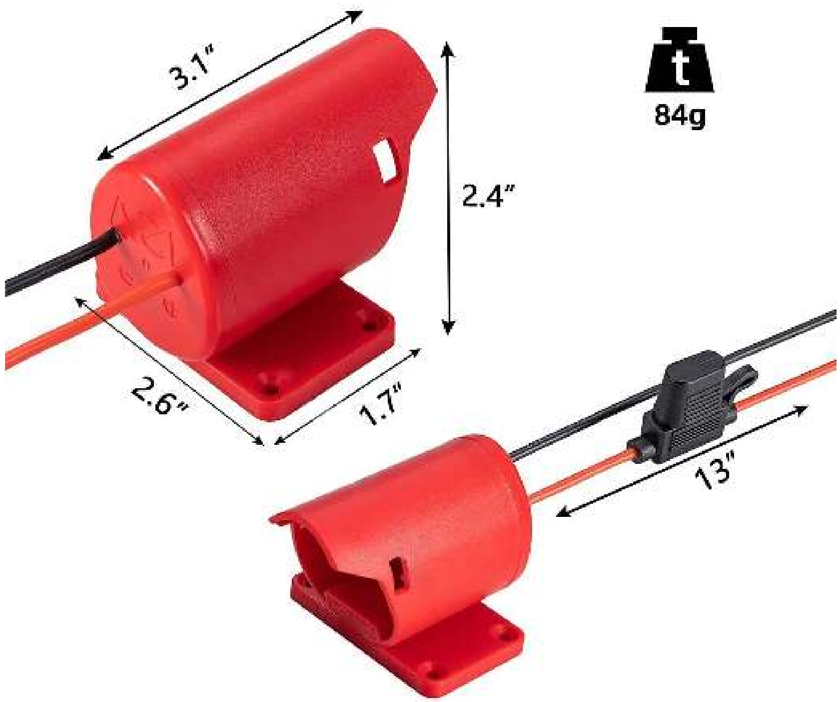


Figure 42: Battery Adapter

Power Switch

To power on and off the robot, the team decided to integrate a power switch on the robot. Multiple switch types were considered including toggle switches, slide switches, and rocker switches as potential candidates [58]. Because the system is going to have a battery with a voltage output ranging from 7.4 to 12 Volts and may experience high current draw from the motors, the switch must be able to withstand these conditions without melting or welding shut.

The design will include a manual on and off switch in so that the system is not constantly running power when the battery is plugged. The selected switch was identified as a toggle switch, due to its ability to handle high-level current and its reliability to withstand the currents when the motors are stalled. The motors are expected to require at max stall is roughly 32 Amps of current which would strain the system but the toggle switch would be able to handle high current without any major concern. It is also important to note that the probability that all motors will ever achieve max stall conditions is low and is only taken into consideration as a precaution.

The animatic switch will be located in an easily accessible area, which will allow the users to toggle the power system on and off without worry of injury. The switch is only intended to be used before the robot is in operation. It should not be used while the robot is in motion. It should be powered off once all action has halted and it is in its idle state ready to be powered off to conserve the battery [58]

Figure 43: Toggle Switch

Low volt power Cutoff

With the change to a hot swap battery system, an additional circuit board was required to prevent the battery from draining beyond a safe level that may cause damage. To address this, we used a low-voltage cutoff board. The board allows the cutoff threshold to be set to a specific voltage, ensuring that once the battery reaches that level, power is automatically disconnected and further discharge is prevented. When the cutoff is triggered, it is a sign that the battery needs to be replaced with a fully charged battery to resume operation. In the current system the cutoff is set to about 10.7 volts.

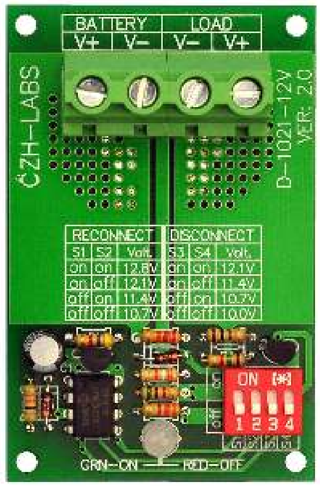


Figure 44: Low Voltage cutoff board (30A)

Fuse

For safety, a fuse will be integrated in the robot’s power system to prevent any accidents or damage to the robot. It will serve as a safeguard to prevent any electrical failure or overheating.

Figure 45: AMP Fuse

Buck converter 5V low amp

The power system design includes a low-voltage low-current buck converter, in order to meet the needs of the lower voltage components without overloading the system. The buck converter will allow us to convert the high voltage, medium to high current battery to lower voltage and lower current for the control system. This will be used to power the PCA Motor Driver and all the low volt servos.

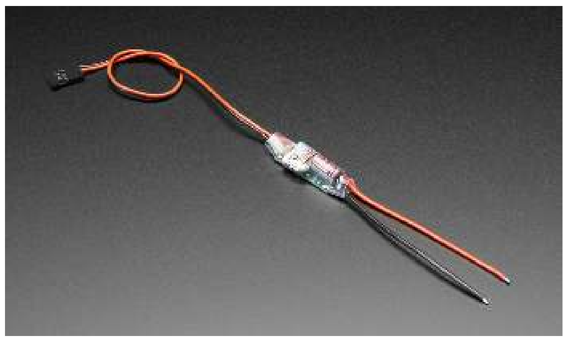


Figure 46: Low Amp Buck Converter

However, since the final design did not ultimately include the 9G servo motors or the microlinear mover for tail, neck and jaw animation, the buck converter was not required. It is important to note that if those components were to be added in the future, a buck converter would be necessary. Currently, the buck converter is physically installed and wired as it would be in a fully implemented design, but it is not actively being used in the current prototype.

Buck converter 5V USB-C Port

To power the microcontroller and its associated peripherals, a separate buck converter that connects to the microcontroller via usb-c. This ensures a stable and regulated power supply for sensitive electronic components. The power would flow from the battery into the buck converter then into the ESP32 through its USB-C input. From there, the ESP 32 distributes power and communicates with all connected peripherals, including the temperature sensor, IMU, speaker, DFPlay Mini, along with any additional modules on the system. This allows all low-power electronics to be reliably powered from a single source.

Figure 47: Buck Converter USB-C

Power Supply

Likewise, a power supply will be used to help power the motor system. The motors require a higher operating voltage and current at varying levels. Using a buck converter ensures that the motors receive stable and sufficient power while isolating it from the lower voltage control system.

This would also be used to power the serco motors responsible for the hip shoulder and knee joint. The design includes two separate power supplies to properly distribute load and allow for stable operation. One power supply is dedicated to the two hip servo motors and the other power supply is used for the shoulders and all four knee servo motors. The hip servos would be receiving 8.4V and the knees, along with the shoulders, would be receiving 6V. This allows each group of servos to operate at their required voltage level without any strain.

Figure 48: Power Supply

Control System:

Within the control system design, the team had to determine whether to develop a custom electronics solution or adopt an off-the-shelf control system. Both approaches had their own drawbacks and advantages but the final determinant was the museum's needs as the end user.

The museum requires a system that is easy to maintain, service and replace if a part fails. The museum staff needs to be able to learn how to assemble, disassemble and trouble shoot the control hardware, therefore it must be feasible to learn and understand within a short period of time. Because of this, an off the shelf would be more practical as a custom solution would be harder to learn, fix, and replace. Although a custom solution would allow for greater space efficiency, commercial microcontrollers would be readily available, more economic and familiar to a wide user base. Although the commercial control systems may have parts that will go unused in our project because they are part of the general purpose design, the benefits of maintenance and replaceability outweighs the cons.

Microcontroller/Micro Computer

For this project our team used an ESP-32-S3-WROOM1U as our microcontroller. TheESP32-S3 is the correct size for our robot, since the stegosaurus has very limited internal space. It also has a lower power usage, so the efficiency helps extend the run time and avoids any power strain. Since the robot must control all the motors, sensor, communicate wirelessly and log data all at once it needs a microcontroller that can handle all this, the ESP32 can handle it reliably due to its high performance dual core processor. It has stable real time motor control with no OS interruptions as well having built in bluetooth, an external antenna, Wifi capabilities, it is durable and stable, and has enough GPIOs, high-resolution PWM, and fast communication buses. The ESP32-S3-WROOM-1U hit all our target points but is still simple enough to not over complicate our robot, allowing us to still meet our user end needs.

Figure 49: ESP32-S3-WROOM-U1

User Input Interface

For our robotic stegosaurus, we considered several communication and control methods keeping in mind an expected operating distance of 20 to 30 meters. We evaluated Infrared, Radio Control, RF modules, Bluetooth, and Wi-Fi. Bluetooth was initially selected as our preferred method due to its low power consumption, simple pairing process, and native compatibility with commercially available controllers such as the PS4 DualShock 4.

During implementation it became apparent that the PS4 controller communicates over Bluetooth Classic rather than Bluetooth Low Energy, and that iOS devices restrict direct Bluetooth Classic connections to third-party devices without manufacturer certification. This oversight meant the controller could not communicate directly with the ESP32 on iOS as originally planned.

The solution was to use the operator's smartphone as a communication bridge. The PS4 controller pairs to the phone via Bluetooth as it normally would with any device, and the phone connects to the ESP32 over Wi-Fi through a WebSocket interface served directly from the microcontroller. This approach circumvented the platform restriction while introducing the added benefit of significantly increasing the effective operating range beyond what Bluetooth alone would have provided, bringing the system well within the 20 to 30 meter requirement.

`

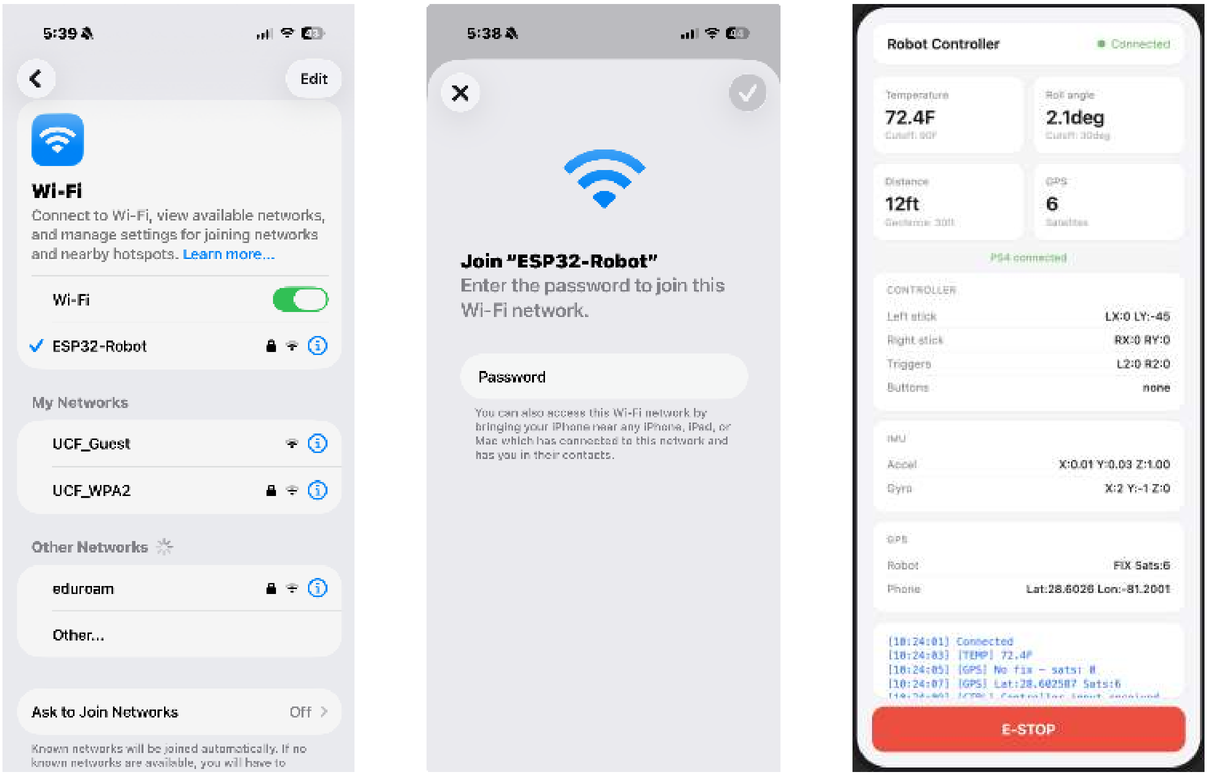


Figure 50: User Interface of Connecting to ESP-32

Controller Options:

In regards to the controller selection, we reviewed and researched multiple different types of controllers including PlayStation/Xbox controllers, Handled RC/ Drone controllers, a mobile device or computer, and FPV flight controllers. Playstation and Xbox controllers are widely commercially available and ergonomic, making them comfortable to use. Libraries for the ESP32 already exist making integration of the controller simple and straightforward. Drone controllers/Handheld RC were ruled out due to them requiring extra adaption for our robot since they are typically designed for flying drones. While a computer or mobile drive covers more flexible control it is practical for museum works and adds set up complexity. A FPV flight controller is not suitable for a land based robot due to its general design purpose being for drones, so it would add complexity and need adaptation for our robot. Finally, after taking into account user needs and accessibility and taking into account our robot design we selected a PlayStation controller.

Selecting a controller mainly took into account our selected communication method and feasibility as well as the overall complexity. Because Bluetooth is our communication method of choice, we were left with an Iphone/computer and a PlayStation. Choosing an Iphone would make the process for connecting more complex due to it probably needing a specialized app which would have to be custom programmed. A computer is also just less efficient due to it not being easily transportable and posing a similar problem as the iPhone. The PlayStation controller The Playstation controller was chosen as it is something most people are familiar with so learning to use it will not be difficult for museum operators. Between the Xbox controller and the PlayStation, the PlayStation provided available libraries to help the set compared to the Xbox controller which does not offer the libraries nor the active support for use in microcontrollers.

Figure 51: PS5 Controller

Audio

The audio plays a couple key roles within the project, we plan to use the audio as user feedback and safety cues. The audio will provide feedback to the operator when starting, stopping or encountering errors throughout the operation of the robot. We are also considering possibly having audio cues to signify battery level status conditions with regards to certain parts and connectivity with the user interface controller. The audio also plays a very key role in terms of entertainment for museum guests and making the project come to life. This enhances the visitor engagement. We could play roar sound effects, possibly footsteps sound effects as the dinosaur walks, maybe a tail swing sound effect. The overall goal is to enhance the engagement and bring the dinosaur to life as much as possible. The audio effects should enhance physical motions. The audio needs to be synchronized with the motion, this adds complexity to the dinosaur but will make a big difference when done right.

The audio Hardware we selected is DFPlayer Mini MP3 Player (FIT0502). This Hardware works natively with ESP32 platforms as well as many other MCU platforms using a simple serial command to control. It takes MP3 and wave files directly from a microSD card making it very simple to use and easy to code. It also has a very low power requirement making it suitable for battery powered robotics without affecting the run time in a negative way. As we reiterate time and time again, size is a very big factor we need to consider due to our limited space and this component is very small and compact. It does not take up much space in the stegosaurus, allowing it for ease of use and easy to fit in.

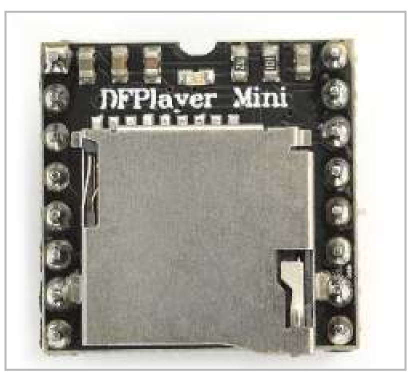


Figure 52: DFPlayer Mini

In addition to the audio processor we also have a stereo enclosed speaker to actually play the audio out loud. We plan to package the speaker into the dinosaur itself. The speaker has a 3-Watt output which is loud enough to be heard within a museum environment while also maintaining a clear and crisp audio. The stereo feature allows for directional audio which is also a consideration that we wanted to have the option for. The speakers are also very durable and economical as it only costs about $5. This makes it the perfect speaker due to its low cost, easy replaceability while maintaining good quality.

Figure 53: Speaker

While the audio in the robot does promote engagement which is a user end request, the main purpose is ut being used for safety and operational feedback. The DFPlayer Mini and the 3W stereo enclosed speaker will give a compact and easy to use system that is compatible with the ESP32. This allows us to meet our goal of simplicity while maintaining reliability and low maintenance.

Motor Controllers

In regards to motor drivers, we are incorporating one isolated driver, for the servos. This is done to guarantee stable and isolated control of each subsystem.

The board being used is the PCA9685 16-channel PWM servo Driver Board. This board is used to control 16 servos that will animate the limbs, tails and ether body parts. The board provides smooth and coordinated movements via the 16 independent PWM channels with 12-bit resolution. The selected board uses I²C to communicate with the ESP32, this minimizes complexity via wiring and preserves ESP32 pins.Srvos operate at 6V and can draw higher current than the ESP32 can supply, therefore the PCA9685 acts as a safety buffer, converting 3.3V logic signals. This simplifies power distribution across all the servos and projects the ESP32 from overcurrent and voltage mismatches.

Figure 54: PCA9685 16-Channel PWM Servo Driver Board

Using the PCA9685 for the linear motor and serves respectively, precise, safe, and reliable control of the robot is achieved. Effectively protect the ESP32 to avoid any hazards and allow for smooth movement.

Motors

For motor applications such as robotic systems that demand stable power output, precise control, and coordinated multi-axis motion, motors like servo, stepper, BLDC, and linear actuators are commonly used. These are commonly used in industrial application so they must satisfy the following criteria: energy efficiency, maintenance and long operational life span, high precision and control, high and low performance in regards to torque and speed output, rapid response, low noise and flexibility. For our project low noise is not a priority. [57]

For applications like our stegosaurus, we reviewed various types of motors and found that the most common and suitable motor types are servo motors, stepper motors, Brushless DC motors (BLDC motors) and linear motors/linear actuators. Servo Motors are able to provide precise position and speed control which would make the ideal for joint and articulated movement applications. They have a quick response and showcase relative motion. Because of this they are commonly used on robotics, 3D printers, and CNC applications. Stepper motors allow for excellent precise incremental motion and repeatability but it lacks the ability to execute dynamic and lifelike motions. This makes stepper motors less ideal for a project like ours where we want dynamic motion. BLDC motos excel in efficiency, smoothness, and speed. They are more specialized towards continuous rotational movement rather than position specific tasks. Linear motors/actuators prove linear motion, which can be used for lifting, pushing, or extending components. This would be ideal for the mouth opening/closing of our robot. Alternate motors like AC, pneumatic, piezoelectric, magnetic field motors were deemed less practical due to their size, complexity or power requirements.

After identifying the most ideal motors for our robotic stegosaurus we assigned motor types different parts based on control, size, placement, and reliability. The main motors chosen are Servo motors and linear actuators. The servo motors were chosen based on the ease of control due to the simplicity of interfacing with the ESP32 via PWM signals. This makes the programming and motion control straightforward. Another reason was the precision and responsiveness a servo motor provides. This is critical for the limbs, tail, and neck movements of the dinosaur. The Servo also has a compact size and low maintenance fitting within the limited internal space and meeting the museum’s low maintenance needs. Servo motors also handle movements like shoulder rotation, knee articulation, and tail motion without requiring complex control circuits.

The overall selection rationale was combining servos for its multi axis movements with linear actuaries for specialized linear motion to give the robot as much realism and flexibility as possible. The servos motors and linear actuators are compact, reliable, and low maintenance allowing us to meet museum needs while executing quality movement. These selected motors also allow us to avoid an overcomplicated design. The selection of motors allows for good energy efficiency, reliability, precision and torque making them the ideal motors for the project.

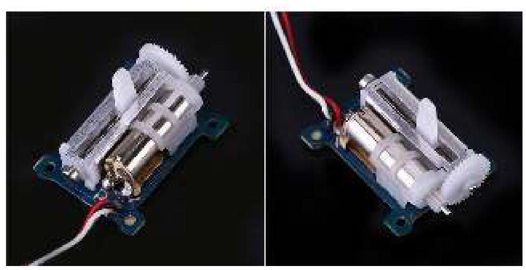


Figure 55: Micro Digital Linear Servo Actuator Servo

Figure 56: DS3230MG High Torque 30KG Digital Servo Motor

Figure 57: DS3230MG High Torque 30KG Digital Servo Motor

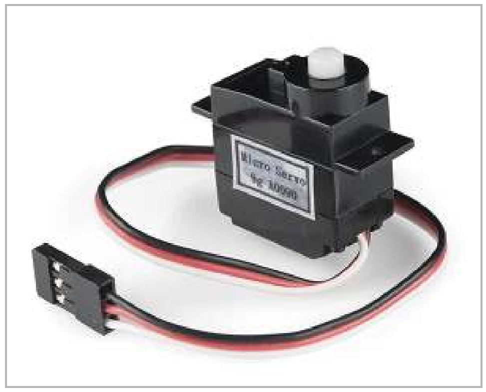


Figure 58: Micro Servo

Peripherals

A gyroscope - STEMMA QT MPU-6050 (Accelerometer & Gyroscope / 6-Axis IMU) is needed to prevent the robot from tipping over. This gyroscope will allow an audio cue to play when fallen to alert staff. If the robot falls over then all inputs from the controller will no longer be taken to prevent itself from being damaged. We are also considering adding a gyroscope and having the system correct itself from falling. This feature would take a large amount of effort, but will be implemented if possible.

The “IMU - KEAcvise 6-Pack GY-521 MPU6050 Sensor Modules” The IMU - KEAcvise 6-Pack GY-521 MPU6050 Sensor Modules are used to monitor joint motion in order to validate that the data being collected is consistent with expected ranges. By tracking accelerations and angular velocities across multiple joints, the sensors help ensure that the motion patterns are congruent with physiological norms rather than artifacts caused by noise, drift, or misalignment. Using several IMUs in tandem also allows for cross-validation between joints, improving the reliability and accuracy of the dataset. This monitoring process supports error detection and calibration, making it easier to identify anomalies such as unrealistic angles or sudden spikes in movement. Ultimately, the system provides quality assurance by confirming that the captured data is both accurate and meaningful, enabling more precise biomechanical analysis and ensuring confidence in the results.

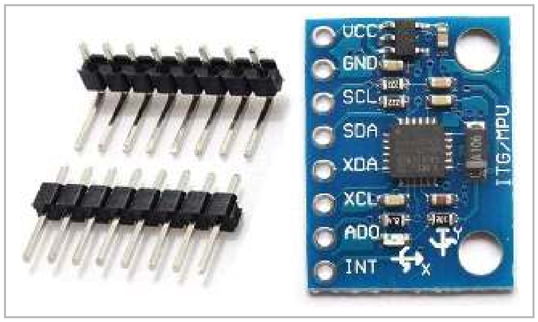


Figure 59: KEAcvise 6-Pack GY-521 MPU6050 Sensor Modules

We will have a temperature sensor - MCP9808 High-Accuracy I2C Temperature Sensor. This is a safety feature to monitor the temperature of the robot and if the temperature reaches a critical level to prevent damage the robot will alert the operator via sound cue. These sensors will alert the user if the temperatures around the motors or battery exceeds a certain temperature, making sure nothing breaks down or overheats, or even explodes.

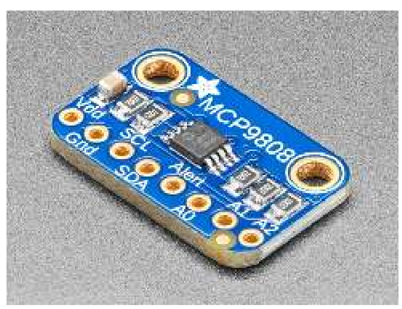


Figure 60: MCP9808 High-Accuracy I2C Temperature Sensor

The GPS - GNSS Module (SeeedStudio XIAO GNSS Add-on) allows us to keep track of distance and is meant to prevent the operator from driving the robot outside the distance of operation, again an audio cue will play to alert that the operator is getting close to being outside the operation zone. Having this system implemented in the robot is critical just in case any individual drives the robot too far and is at risk of disconnecting the robot from the controller, which would cause connection issues, and may cause malfunctioning issues.

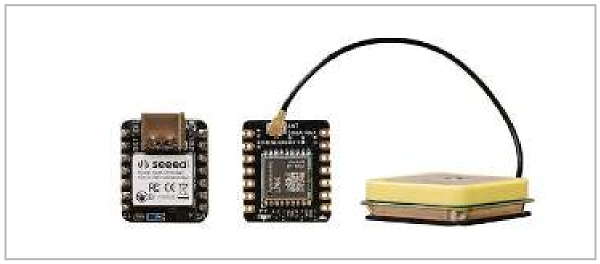


Figure 61: GNSS Module (SeeedStudio XIAO GNSS Add-on)

All the peripherals are small, low-power, I^2C compatible, and have robust ESP32/Arduino libraries, making development and maintenance.

Packaging

Fitting all the components in the robot can be tricky. All the electrical components should fit inside the ribcage. A housing will be designed for each component that will fit under the ribcage. As of now, wires will run through the bones. The bones will not be hollow to maintain structural integrity. The housing that will hold all the electrical components will be designed to specifically hold certain components. This is so no electrical component is loose or rattling inside the housing to eliminate the risk of electrical component damage.

As of now the plan is to have a hollow cylindrical cutout for the wires to run through the bones up to the housing where most of the electrical components will go. Each leg will have 3 motors, and we will have wires running up through the leg bones, and up the chassis. Wires will also be about an inch long between each hinge so that no wire is stretched over the limit to reduce risk of electrical component damage.

Gait Studies

A gait study is a biomechanical analysis of how a living creature moves. Gait studies focus on a few kinematic and dynamic points of analysis. The first point of analysis is ground reaction force, which is the force of a limb hitting the ground. The second major point of analysis is joint angles, which is the calculation of one joint with respect to the bone members. Both points are measured throughout the stride of the animal. The gait can also be analyzed in terms of the angular velocity of each joint. The stride is divided into frames, where each frame can be assigned a ground reaction, force, and a list of angles with respect to multiple joints.

Pugh Matrix

Concept Selection Legend
Better +
Same S
Worse -

Key Criteria	*Importance Rating*	Solution Alternatives: Joint Angles	Angular Velocity	Torque Calculations
Data Volume	5	-	+	+
Data Specificity	10	+	-	S
Speed Adjustments	8	+	-	-
Sum of Positives		2	1	1
Sum of Negatives		1	2	1
Sum of Sames		0	0	1
Weighted Sum of Positives		18	5	5
Weighted Sum of Negatives		5	18	8
TOTALS		**13**	**-13**	**-3**

Figure 62: Gait Study Method Concept Generation

The team considered these methods of compiling data for a gait study and used joint angles to coordinate the control system and the joint actuators. The use of angular velocity provides a more limited data package, as the animatronic will not require positional data for each point in its stride. However, this limited data will lack the specificity needed. The control system would not be able to [*Figure 62:*] account for the position of the limbs at each frame of motion, meaning if anything unexpected occurred, the animatronic would have no capacity to correct itself.

Similar problems arise when designing the gait using torque. Torque calculations would provide the angular acceleration of the joints, which would similarly reduce the data package. But, without positional data, the animatronic would not be able to handle unexpected changes to its joint angles. Additionally, both the angular velocity and torque methods would struggle to handle changes to speed.

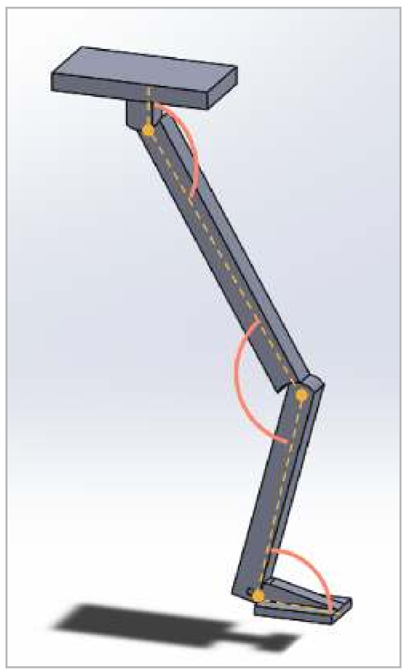


Figure 63: Hind Limb Simplified Model with Notated Joint Angles

The team ultimately decided that the use of joint angles will lead to the greatest success. A data point would be required for each frame of motion; however, the control system would have positional data for the entire stride. Further, because the data is purely positional, the control system can more easily adjust the speed of the animatronic by increasing the frequency of the frames.

The team initially intended to use human gait data to assist in the development of the joint angle lists. However, after consulting stegosaurus biomechanical expert Matthew Dempsy from the University of Liverpool, the team pivoted to use preexisting animations of stegosaurus's gaits. The

To complete the gait study, the team will need to assemble a list of tabulated data, including the angles of the joints at specific frames of motion. Then, the team will need to coordinate the motion of the four limbs and tail with respect to each other. The control system can then read this data, and relay the information to the joint actuators, because the data will be based off of frames of motion, the control system can increase the frequency of the frames to increase the speed.

VII. Design Analysis

Physical Prototyping

In order to fully develop a prototype and to see if our design will meet the project requirements, a physical model will need to be created. To accomplish this the model was finalized from M5 and then the majority of which was 3D printed.

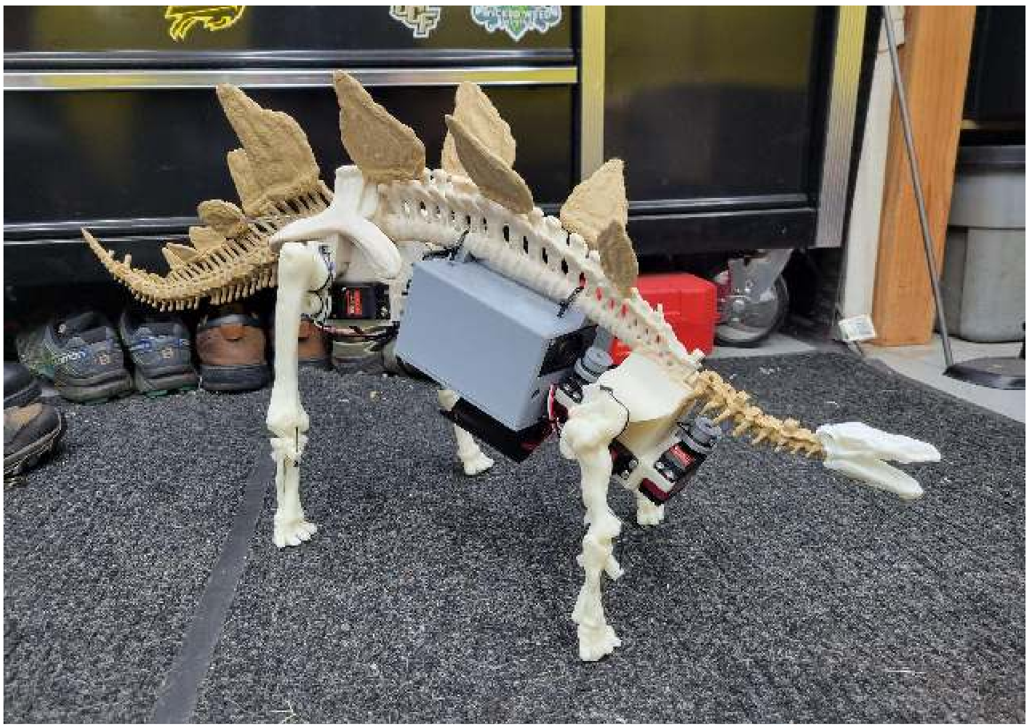


Figure 64: FPrototype Assembly.

For material selection, ASA was used for all hard parts in both a natural and gray color to duplicate bone color and to hide the electronics packaging. The brown sections as seen in Figure 63, were printed in TPU 95A to keep these sections flexible in order to meet the bending and flexibility requirements of the tail and neck.

Figure 65: Main Tail Section

Another topic not previously discussed were adhesives, as the majority of the bone structure was designed to be glued together vs bolted. This reduces the number of mechanical fasteners needed in the final prototype. And the deletion of them allows for the bone structure to remain more pronounced and true to the base 3D scan. For this task, a semi flexible epoxy design specifically for plastics was chosen. It is possible to use super glue, or other types of adhesive. But this epoxy was chosen as it is able to fill small gaps, is relatively strong, and from an aesthetics standpoint was available in a tan color to closely match the color of the bones.

The next main components that needed to be assembled were the legs. These were 3D printed like the rest of the bone structure, but included a lot of fine assembly as they are not static pieces.

Figure 66: Front Leg Assembly

For joints themselves, this remains consistent from M5, the elbow/knee joints are held together with a single #10-32 screw. Another piece that required more fabrications were the main support shafts which are made out of 3/8" steel shaft in order to interface with the bearings in the hips/shoulders. A small portion needed to be cut out of each shaft as seen in figure XX to clear the channels for the internal cable routings. For installing the main shafts, an interference fit was found to be the cleanest method of attachment, the plastic legs made this quite forgiving and a final press fit of around 0.010 was achieved.

Figure 67: Main Leg Shafts

Drive wheels were also made to attach the servo motors to the cables from the legs themselves, a simple adapter was made to wrap the cables around, and a set screw was added to provide the ability to lock the cable in place, and to adjust the leg angle as needed.

Figure 68: Drive Wheels Attached to Servo

Structural Analysis

FEA is an important tool that will allow us to use predicted loads and see if our base designs will hold up and are in line with the materials we want to use. Below is an example pulled off of the hip joint. This is going to see the highest amount of load by far due to the large motor size, and relatively small support area. Another consideration is in the 3D printing material and layer orientation, which will be further experimented with in this project. For a general idea, the maximum yield strength was set to ABS plastic. At roughly 4000 psi, below the finished simulation predicts a mad stress of 2320 psi in figure 14, which is well under the lower estimate for ABS.

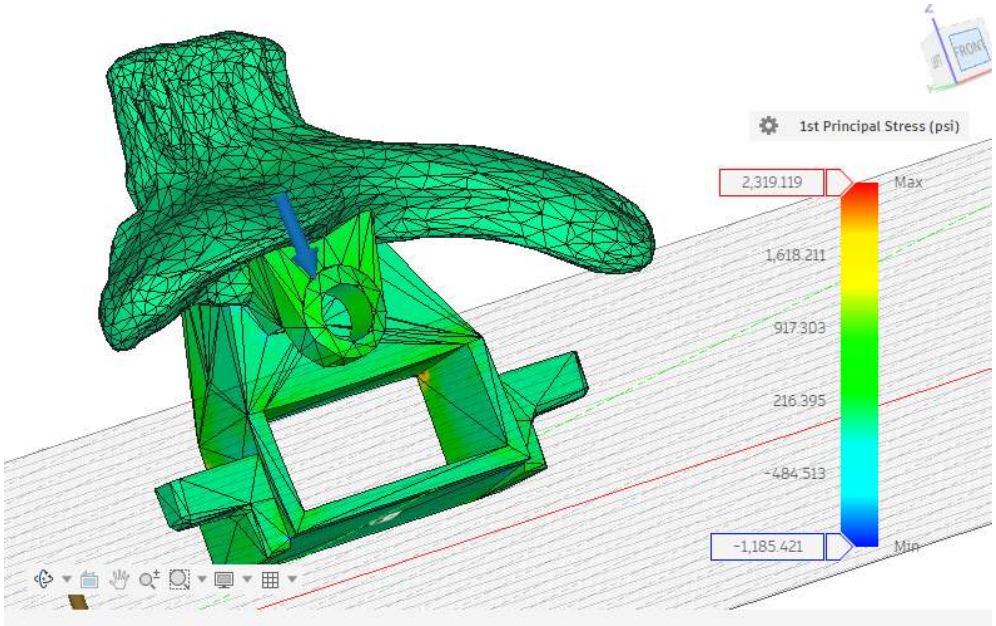


Figure 69: Hip 1st Principal Stress

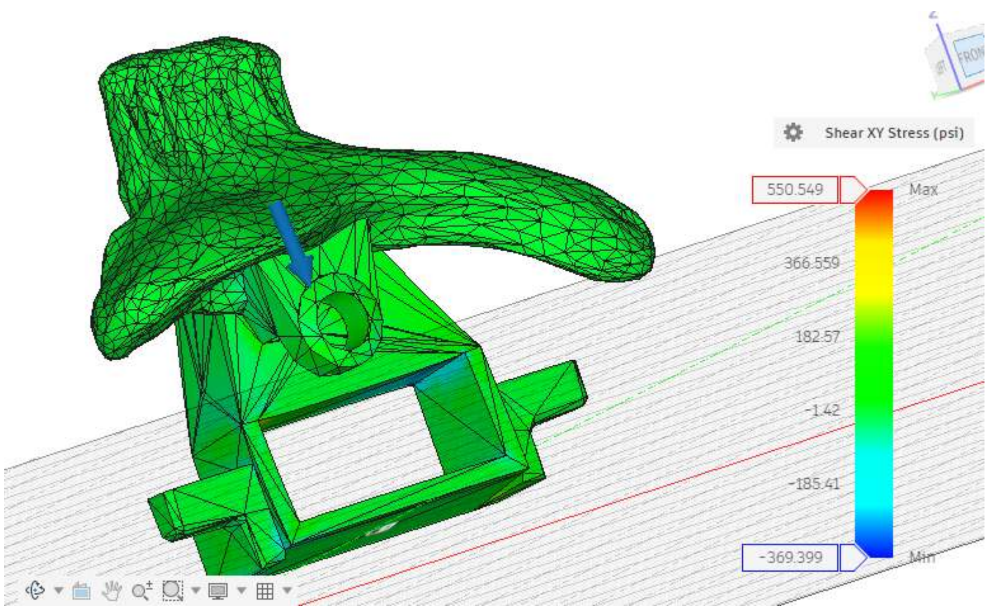


Figure 70: Hip Shear Stress

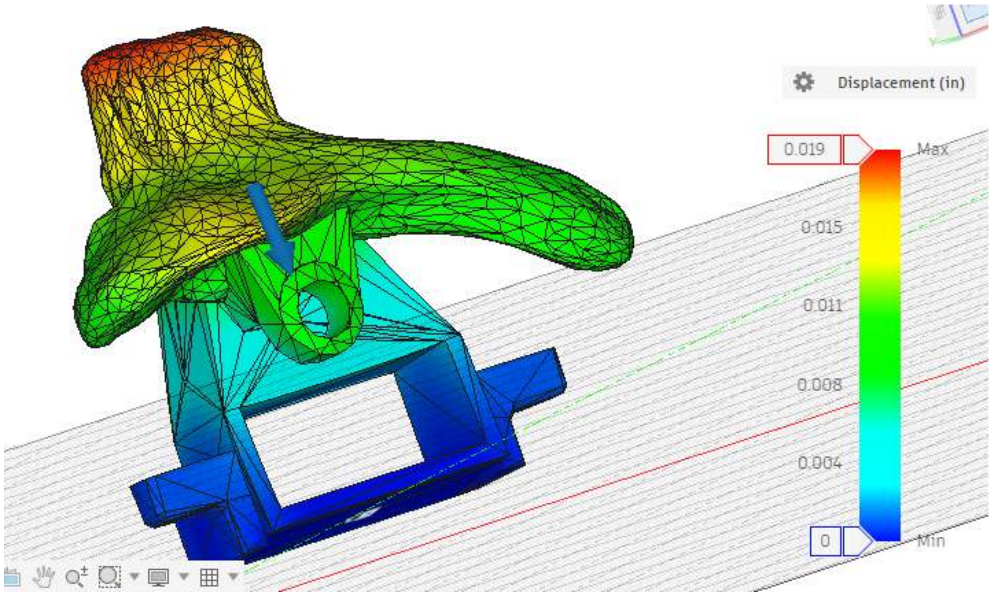


Figure 71: Hip Displacement

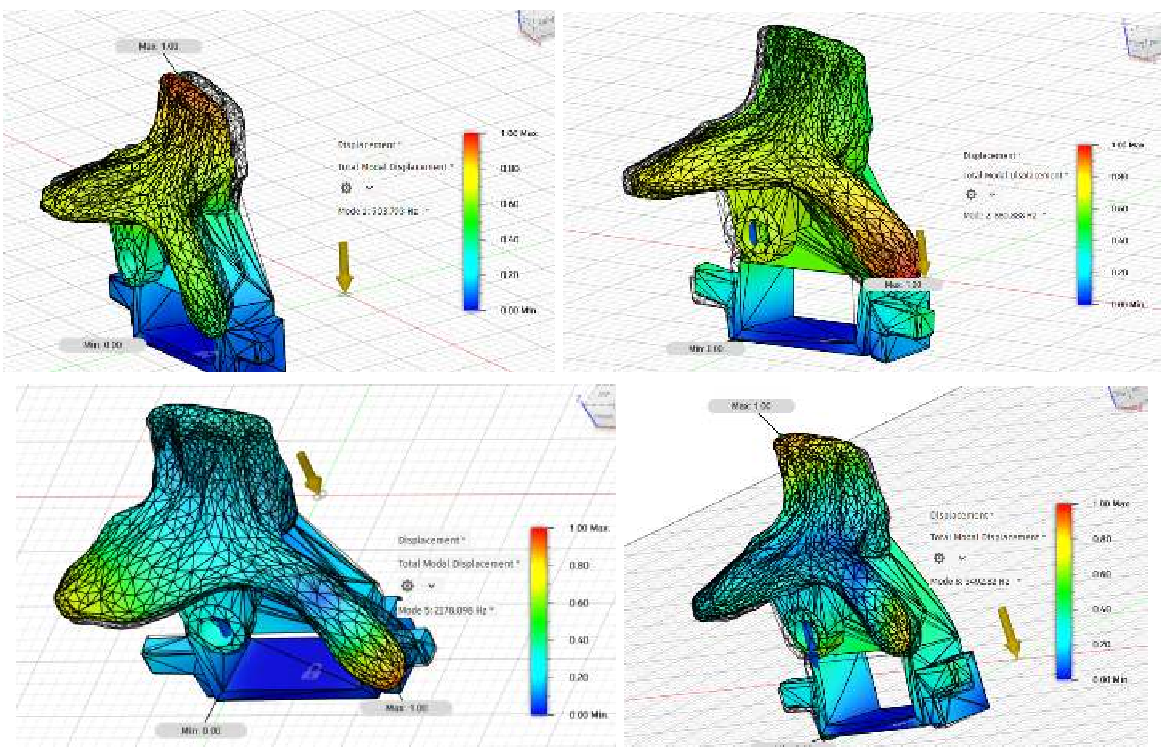


Figure 72: Mode Shapes from Vibration Analysis

Implemented Electrical Design

This entire system is powered by a Milwaukee M12 battery. This serves as the primary energy source for all subsystems. Power flows from the battery to the battery adapter, then through a main switch, enabling or disabling the entire system manually. After the current passes through a fuse for overcurrent protection, and then goes into a 30A low-voltage cutoff board that prevents excessive battery discharge. From the cutoff board, power is distributed to two bus bars that act as central distribution points for positive voltage and ground. All ground connections throughout the system return to the common ground bus, ensuring a shared electrical reference for proper signal operation.

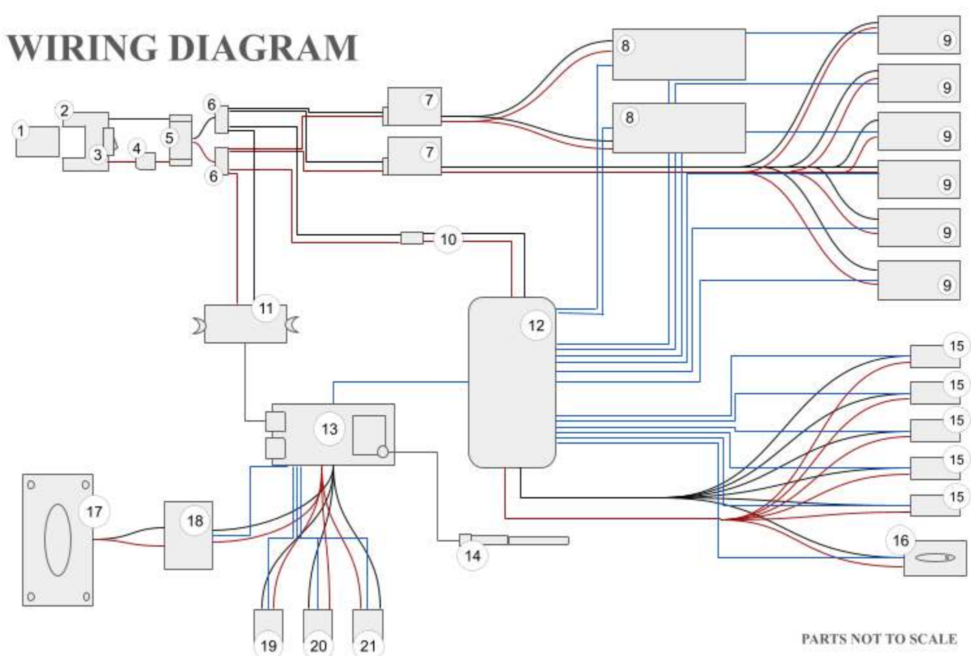


Figure 73: Wiringing Diagram

ITEM NO.	SUBSYSTEM	COMPONENT	QTY
1	**Power**	Milwaukee M12 Battery	1
2	**Power**	Battery Adapter	1
3	**Power**	Switch	1
4	**Power**	Fuse	1
5	**Power**	Low Voltage Cutoff Board (30A)	1
6	**Power**	BusBar	2
7	**Motor**	2-8S LiPo UBEC-8A For 5 Servos Motors	2
8	**Motor**	DS5180 High-Torque Servo - 80Kg servo	2
9	**Motor**	DS3230MG Servo - 30Kg servo	6
10	**Motor**	5V 3A Low Volt Buck Converter	1
11	**Motor**	5V 3A Buck Converter USB-C	1
12	**Motor**	PCA9685 16-Ch Servo Driver	1
13	**Control**	ESP32-S3-WROOM-U1	1
14	**Control**	Antenna	1
15	**Motor**	Micro Linear Servo	5
16	**Motor**	Goteck 1.5g Micro Digital Linear Servo Actuator	1
17	**Audio**	3W 8Ω Speaker	1
18	**Audio**	DFPlayer Mini MP3	1
19	**Control**	MCP9808 Temp Sensor	1
20	**Control**	GY-NEO6MV2 NEO-6M GPS Module	1
21	**Control**	Additional GY-521 IMUs	1

Figure 74: Bill of Electrical Components

From the busbars, raw battery voltage is routed to two 2–8S LiPo UBEC 8A regulators, which step the voltage down to a stable level suitable for high-current servo operation. These UBECs provide regulated power and ground to the high-torque DS5180 servos and the DS3230MG servos. Additional regulated 5V power is generated by two buck converters: a 5V 3A low-voltage buck converter and a 5V 3A USB-C buck converter. These supply stable logic-level voltage to the control electronics and peripheral modules.

The ESP32-S3-WROOM-U1 serves as the system's central controller. It receives power from a buck converter that is regulated to 5V and shares the common ground reference. The ESP32 communicates with the PCA9685 16-channel servo driver over an I²C interface. The PCA9685, powered from the logic rail, generates PWM control signals for all servo motors. While the servo driver supplies signal lines, the servos themselves receive their operating power directly from the UBEC-regulated supply. This separation guarantees that high-current motor loads do not destabilize the logic circuitry.

In addition to motor control, the ESP32 interfaces with multiple sensors and communication modules. The MCP9808 temperature sensor and the GY-521 IMU communicate with the ESP32 via I²C, sharing the same bus as the servo driver. The GY-NEO6MV2 GPS module communicates over a UART interface. All sensors are powered by the regulated logic supply and share the microcontroller's ground.

The audio subsystem consists of a DFPlayer Mini MP3 module powered from the 5V logic rail. The ESP32 communicates with the DFPlayer via UART to control any audio queues. The DFPlayer directly powers a 3W 8Ω speaker, providing amplified sound output. The ESP32 is connected to an external antenna, enabling wireless communication with the user's phone and PS4 Controller.

Figure 75: Electronics Housing

Implemented Programming Layout

The following sections outline the systems and capabilities that were successfully implemented during the development phase of the dinosaur robot. Partial achievements and limitations were noted alongside the original design as the development of the project occurred.

Communication System

The ESP32-S3 hosts its own Wifi access point enabling wireless communication between the robot and the operator without an external network. A WebSocket server running on port 82 handles live, two-way communication between the robot and the operator's phone. The PS4 DualShock 4 controller is paired via Bluetooth to the user's phone. It transmits joystick positions, button presses, and trigger inputs through the phone's browser using the Gamepad API to the ESP32, updating at around 60 frames per second. This setup removes the need for a separate receiver and allows for any modern smartphone to act as the bridge between the controller and the robot. When we tested in an open space we found the range to be approximately 100 to 120 ft of distance. Although the testing conditions were not optimal they were similar to as if it were operating within the Lawrence Hall of Science.

Operator Interface

A web based operator indeed is hosted on the ESP32 and can be accessed from any device connected to the robot's WiFi network by navigating to 192.168.4.1. The interface provides real time updates on the system through four main panels showing temperature, IMU roll angle, GPS geofence distance, and GPS satellite count. It displays live controller inputs, joystick movement, trigger values, and button states, allowing the user to see exactly how inputs are being interpreted by the system. A log panel section streams all serial output from the microcontroller directly to the phone, so system messages and events can be monitored without needing a separate computer. There is also always a visible emergency stop button at the bottom of the interface and remains active at all times. This provides an immediate way to halt the system if needed.

Safety System (Emergency Stop)

To ensure safe operation of the system four independent emergency stop mechanisms, four independent emergency stop mechanisms were implemented.

Thermal Cutoff

The MCP9808 temperature sensor monitors the system temperature and triggers an emergency stop if the temperature exceeds 90 degrees fahrenheit. Once activated it does not resume operation until the temperature drops below 85 degrees F.

Fall Detection

The MPU6050 IMU activates an emergency stop when the roll angle exceeds 30 degrees and remains there after multiple readings. To improve reliability and prevent false triggers caused by electrical noise from the servos several filtering techniques were added, including a complementary filter combining accelerometer and gyroscope data, outlier rejection, a rolling average window, and a bidirectional debounce counter.

Geofence Violation

Using data from the NEO-6M GPS module and the phone's GPS, a Haversine distance calculator activates an emergency stop if the robot moves beyond 30 feet from the operator.

Manual Trigger

The user can trigger the emergency stop at any time using the button at the bottom of the web interface. Once activated, all motor activity and usage is paused and any inputs are blocked until the triggering conditions are resolved.

State 1: Idle State

The idle state was successfully implemented as a part of the system's start up behavior. When the robot is powered on, the robot runs an initialization sequence that brings each hardware competent online one at a time. Each device , including the temperature sensor, IMU, servo driver, GPS module, and audio system, are checked to confirm it is functioning correctly before any further actions. Once the initial setup is complete, all the active servo channels are moved to the predetermined restring positions of 40 degrees, within the operating range of the shoulder and hip motors. If any part were to fail to initialize it is reported to both the serial monitor and the user's phone interview via the log panel. The system then waits in idle state until another input is received.

State 2: Locomotion (Partial)

Basic motor oscillation was achieved across four shoulder and hip servo channels (PCA9685 channels 0 through 3). When the user pushes the left joystick forward it triggers a sine-wave alternating gate batten that moves the servos between 20 and 60 degrees at a controlled rate. The right and left sides alternate in opposition, creating a walking-like motion. Because the movement is driven by a sin function rather than by step-based commands, the motion appears fluid and continuous. IF the user releases the joystick, the motion stops and all servos hold their current position.

In addition, a more advanced 200-frame gait cycle was developed from the MATLAB gait study. This sequence defined coordinated hip and knee movements, with hip angles running from 75 to 110 degrees and 130 to 180 degrees respectively across the cycle, This gait sequence was successful in simulation, but it was not implemented onto the physical prototype due to limitations in the chassis design that prevented the installation and actuation of the knee servos. Both the gait data and the supporting code structure have been preserved for future integration.

State 3: Animations (Not Achieved)

Planned animation features, including tail movement, roaring, and other expressive behaviors, were not implemented in this prototype due to physical components required that were not installed due to the incomplete chassis build. An example would jaw and neck servo assemblies. Although the physical actions were not installed, the audio system intended to support them is fully functional. The DFPlayer Mini is able to play the required sound effects in response to controller input.

State 4: Head, Neck, and Tail Control (Not Achieved)

Control of the head, neck, jaw, and tail was not included in this version of the prototype. Similarly to the animation feature, the primary reason behind this came down to the physical build and the mount required for the servos were not completed. Even though they were not implemented, the system was built with these features in mind. The PCA9685 servo driver can handle up to 16 channels, and there is still available capacity set aside for these additional components. On the software side, everything is already structured to support expansion. The WebSocket communication and joystick input system can easily be extended to include new features. Once hardware is installed, these features can be added without any significant work needed to be redone.

Audio System

The DFPlayer Mini audio module and speaker were successfully set up and are working reliably in the current prototype. Five different audio tracks are stored on a FAT32-formatted microSD card and mapped to the controller. Each button corresponds to a different audio: the X, O, Square, and Triangle buttons each play a specific sound, while the right trigger activates a fifth. In order to prevent unwanted looping, the system uses edge detection, so each sound plays once per button press instead of repeating while the button is held down. If needed the audio system is prepared to be extended with future audio features.

GPS Geofence System (Partial)

The NEO-6M GPS module was integrated and tested, and works well as long as it is used near a window or outdoors, this is due to it needing to acquire satellite signals. The ESP32 uses the Haversine formula to calculate how far the robot is from the operator's phone in real time. This allows for the geofencing feature, and in testing, the 30 foot cutoff triggered exactly as expected whenever GPS connection is stable.

GPS data transmission works on Android devices straightforwardly. Using Google Chrome, the phone was able to send GPS data to the robot without much trouble, as long as the right permissions and settings were enabled. iOS devices are a lot more limited. Because Safari requires a secure HTTPS connection to share location data, which doesn't work easily with the ESP32, iOS devices are unable to transmit GPS data. We did try to implement HTTPS, but ran into compatibility issues with the libraries we were using. A Bluetooth Low Energy solution was also explored, but that would have required building a custom app, which was outside the scope of this phase.

Data Logging

SD card logging was part of the original design specification in this project but it was not implemented in the current prototype. Instead, all system information is shown live through the serial monitor and also sent straight to the operator's phone through the WebSocket log panel. The current system works well for real time monitoring since everything can be seen without a second device. The main downside is that nothing gets saved once the system is turned off, so there's no long-term record of what happened during a run.

Controller

As for the controller layout, the majority of the intended control scheme was successfully implemented. However, the actual controller system does not fully match the ideal configuration. This is mainly because several features were not implemented. This includes neck movement, tail movement, jaw opening and closing, and the associated animation sequences but they were not implemented across layers. On the physical design side, some groundwork was partially in place. For example flexible materials were used for the tail and neck, and a hinge mechanism was included for the jaw. Although it was started, mounting points and final motor placement were not integrated.

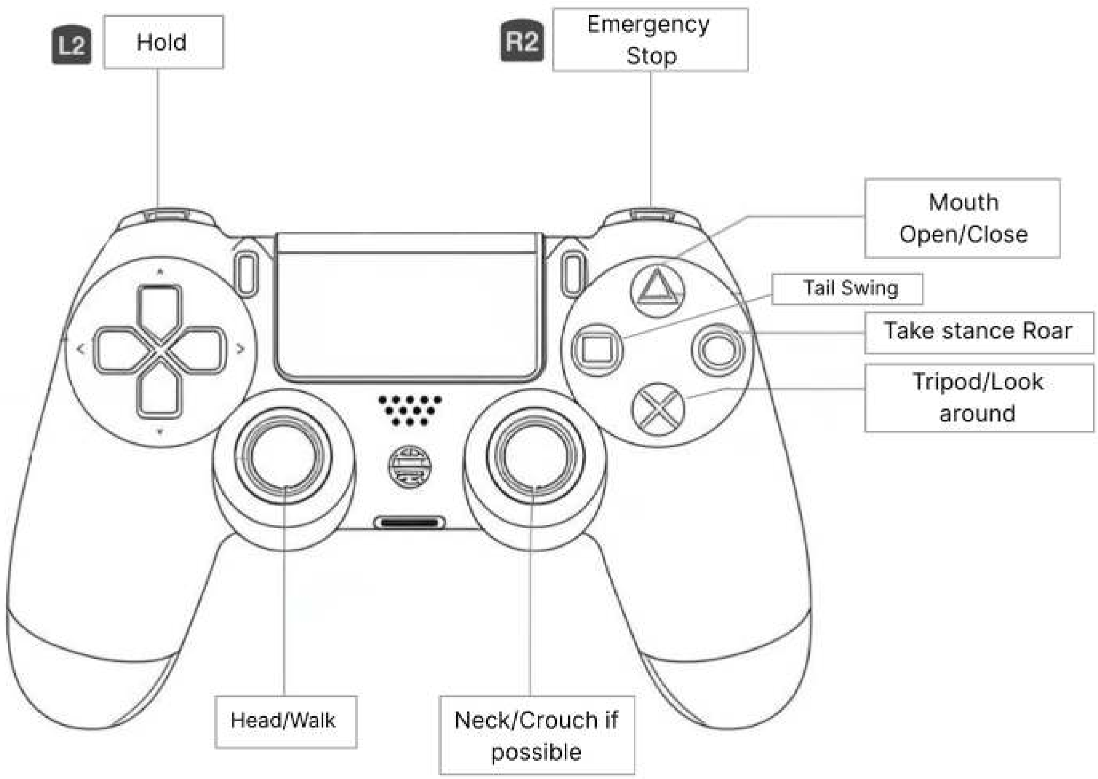


Figure 76: Proposed Controller Layout State 4

Gait Study

A gait study provides data on how a creature moves. For the purposes of this project, the gait study provided the angular position of each joint. This information was taken by the programmer and incorporated into the control system. The control system would send signals to the motors to move each joint to a specific angle at each point during a walk cycle.

Due to the extinct nature of the stegosaurus, there is no precise information on the joint angles of the animals walking pattern. The team analysed the skeletal structure of the dinosaur to determine the maximum range of motion of each joint. This gave the outer limits of the range of motion. The team then utilized tabulated information on the gaits of other animals, particularly humans, to create angle lists for each joint with the correct cadence. An animation using MATLab was created to visualize the walk of the animatronic. From there, edits could be made to the angle lists. The angle data was then handed over to the programmer to be integrated into the control system.

Observing the stegosaurus skeleton, the rear leg has a maximum range of motion of 85 degrees. The front leg does not move further than 15 degrees past the vertical, while having a much more range of motion backwards. An animation resource was used to pinpoint key joint angles throughout the strides and are captured in the tables below. The joint angles were averaged between each point in the stride to create a smooth transition.

Table 1: Front Leg Joint Angles

Stride Status	Gait Percent	Hip Angle	Knee Angle
Heel Touch	0%	135°	150°
Full Load	50%	115°	210°
Toe Off	65%	105°	150°
Leg Swing 1	75%	70°	100°
Leg Swing 2	85%	120°	95°
Heel Touch	100%	135°	150°

Table 2: Rear Leg Joint Angles

Stride Status	Gait Percent	Hip Angle	Knee Angle
Heel Touch	0%	75°	160°
Full Load	50%	90°	175°
Toe Off	65%	100°	160°
Leg Swing 1	75%	110°	150°
Leg Swing 2	85%	95°	130°
Heel Touch	100%	75°	160°

Figure 77: Joint Angles at Key Benchmarks for Stegosaurus Gait

Once the angles were averaged, the angles were passed through a function to smooth the sharp transitions further. The resulting joint angles are portrayed below. The joints of all the limbs point in towards the center of the body. This is noted in the shoulder and hip joints. The elbow and knee joints have similar paths, though the knees have a peak just after the 50% mark of the gait that is lacking in the elbow stride. The elbow has a sharper trough than the knee. The hip and the knee have mirrored paths that are nearly identical.

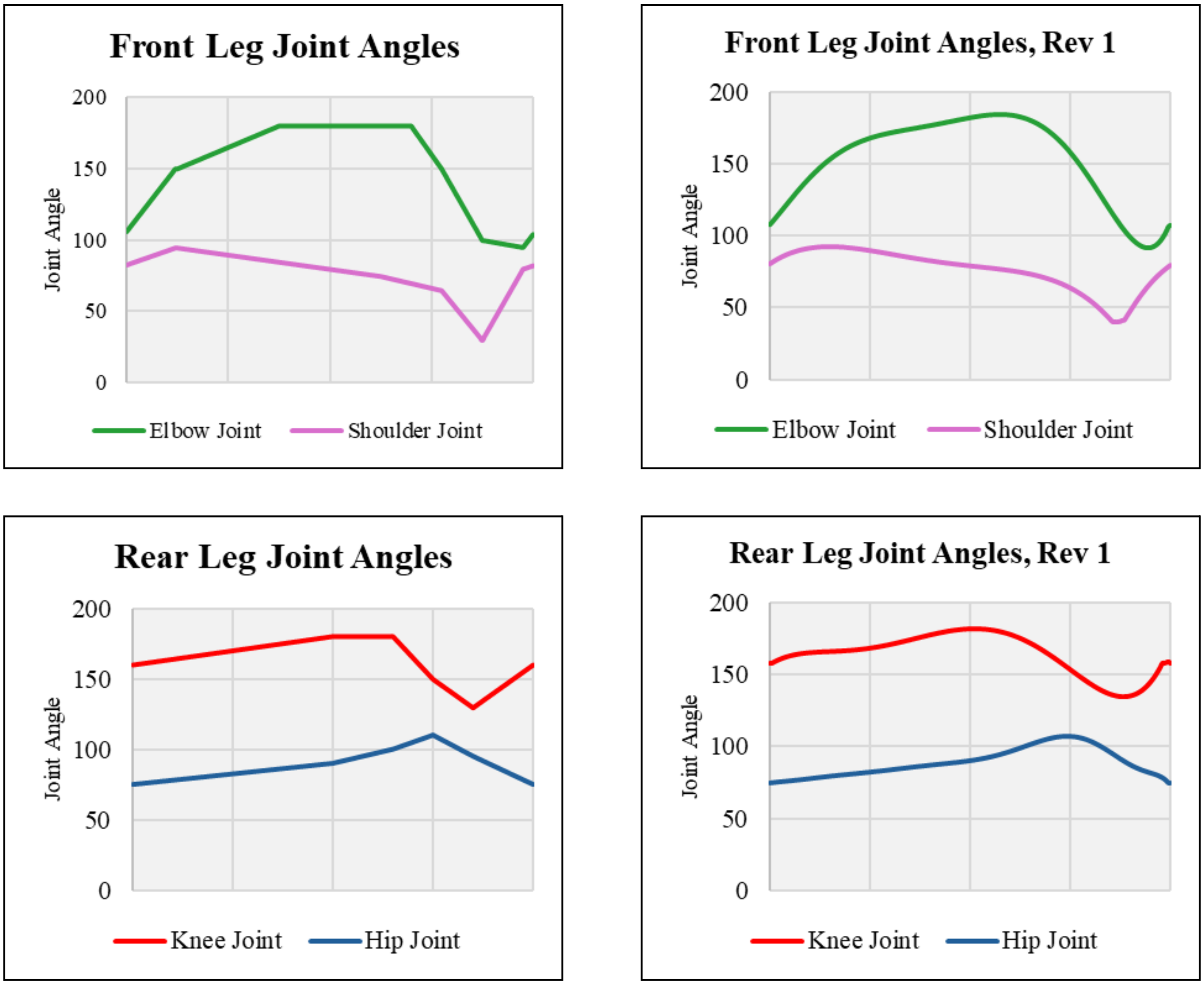


Figure 78: Joint Angle Data

The tabulated joint angles were added to MATLab to be animated. The code included two modifying variables, one to change the overall speed, and one to turn the animatronic. The speed is adjusted by the frame rate. Increasing the frames per second easily allows the animatronic to walk faster. To turn the animatronic, a multiplier was applied to the left and right side of the body. The multiplier adjusts the joint angles evenly between the two sides of the body. If one side of the body has a narrower range of motion than the other, it will allow the body to turn.

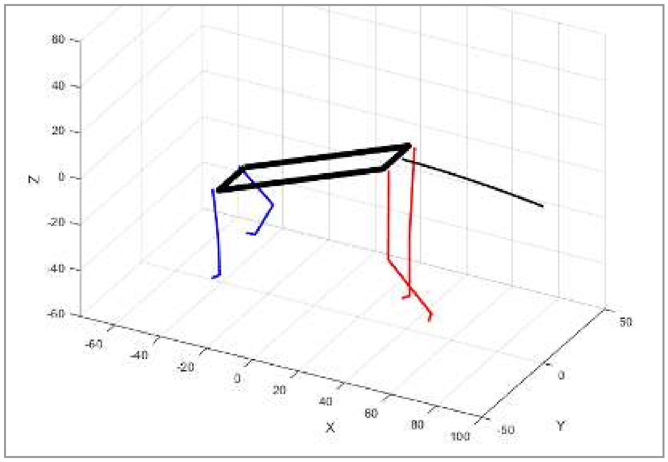


Figure 79: MATLab Animation of the Gait

An interesting point of comparison exists between a human gait and the stegosaurus's gait. The human hip has a range of 50 degrees, where the stegosaurus has a range of motion of the front shoulder of 60 degrees and a hip range of motion of 35 degrees. However, the human hip has a longer continuous change in motion, whereas the range of motion of the stegosaurus hip is brief but sharp. The human knee features two sharp peaks in joint angle. The stegosaurus knees and elbows have one strong peak with one minimal peak. In general, the path of the human knee joint is very similar to that of the stegosaurus, while the hip differs dramatically. This follows with the anatomy of the human and stegosaurus. The bipedalism allows for the human hip to be more flexible, whereas the stegosaurus has more stable hip and shoulder joints to support its four limbs. The knees and elbows, by comparison, are anatomically similar.

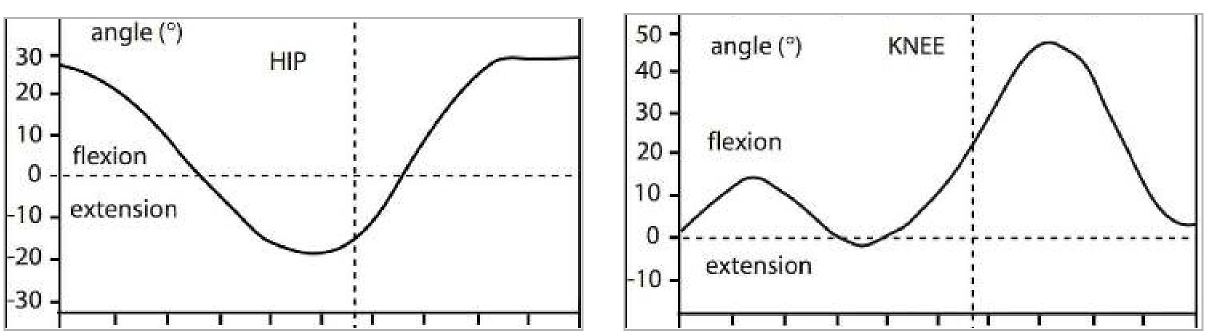


Figure 80: Human Joint Angles [71]

Torque Calculations

Below are the results of the torque calculations. When the animation was run, the angular position was integrated to provide angular acceleration. This was done at 100 frames per second, where the gait is set into 200 frames. Therefore, each leg of the stegosaurus was taking one step every two seconds.

Table 3: Torque Data				
	Front_Femur	**Front_Tibia**	**Rear_Femur**	**Rear_Tibia**
Max Absolute Angular Acceleration	8298.22	9326.6	6518.8	11216.47
Length (in)	4.0000	3.4000	7.8200	5.2800
Mass (lb)	1.19E-01	1.01E-01	2.33E-01	1.58E-01
Max Possible Torque (lb*ft)	3.06	2.11	17.954	9.51

Figure 81: Estimated Torques Due to Gait Study

The maximum absolute torque was found. Then, using the lengths of each limb and the mass-per-volume of the modeled parts, the maximum possible torques were calculated. These torques only provide data on the torques from positional data, but do not include material data. Without the printing being completed for the physical model, the team cannot yet account for torque increases due to friction. One proposed solution is currently to apply a lubricant to reduce friction and improve motor control.

$$I_{yy} = \Sigma x_i^2 \, dA_i$$

Figure 82: Formula - Moment of Inertia [71]

$$\tau = I\alpha$$

Figure 83: Formula - Torque [72]

VIII. Final Design & Engineering Specifications

Although a lot of the final design/ prototype was covered in the previous section the Engineering Specifications remain largely consistent to previous reports. Notable changes include the Museum's desire to have a swappable battery in order to reduce downtime. And the revision on the required degrees of freedom in each leg from 3 to 2.

Appendices B and C show the full requirement breakdown and an indicator as to which requirements were met and which ones weren't. General successes include the robot's final realism and similarity to the 3D scan. Overall the robot came in at 36in long, and the only modifications that were required to the scan was to straighten out the tail and spine in order to place the dinosaur in a neutral position. Other than that, overall proportions remain exact to the scan.

For programming, the user interface was a large success, as it easily provided the user with feedback on controller and motor positions, IMU and GPS data. Along with a real time display of any errors the code had in order to speed up diagnosing issues. Another success on the control side of things was range and the implementation of safety features in case the dino fell over. The IMU was very responsive in shutting down motor movement if it measured to be >45 deg from level. And the range was found to be greater than 100ft, in practice easily reaching the full length of the Health Sciences Building on campus.

One challenge the team had was in getting the dinosaur to walk. The joints themselves did meet the requirements for general degrees of freedom and articulation, however a big challenge was seen in properly tensioning the cables, and friction in the hip section for both the knees and elbows.

Another issue in the joints themselves were in the pinions that actuated the cables themselves. These did work well for the capstan drives on the legs, however the lock screw for the cable itself caused issues with slipping, as when the servos provided a high amount of torque, the cables would effectively slip. Causing the robot to lose its angular reference.

IX. System Evaluation

Gait Study

For the gate study, once a list of joint angles per frame is compiled, this data needs to be attached to the mechanical model to see if the data will generate a smooth gate on the model. From this point, iterations may be made to the list of joint angles.

Torque Calculations

Once the mechanical model and gate study are completed, torque calculations can be conducted. The mechanical model will indicate where the insertion points of the “tendons” will be placed. Knowing the angle per frame, as well as the insertion point will allow the team to calculate the torque on each joint. Torque calculations will also provide data for an effective gear ratio from each joint to calculate the desired motor torque.

Finite Element Analysis

The team will then conduct stress analysis on the model components. The torque calculations will allow the team to determine the stresses that each member will experience. It is important to understand the axial and shear stresses that will occur. This will determine the method of printing for the members.

The structure will require testing of material, infill densities, infill patterns, etc. The primary risks in the joints of the robot will be their ability to meet the final torque and speed requirements. After running FEA, a reasonable safety factor for each bone can be determined depending on the stresses each bone will undergo. The team can also reinforce each bone with supporting material to reduce risk. In the FEA, different load cases will be considered to account for different ways this robot can experience external forces, such as from a falling object or a kick from a child. Although the electrical components are extremely sensitive, the structural body will be protective to defend the components from unexpected impact.

$$\sigma = \frac{P}{A_0} \quad \epsilon = \frac{l - l_0}{l_0} \quad \sigma = E\epsilon \quad \tau_{\max} = \frac{Tr}{J}$$

Figure 84: Formula - Material Stress [68]

Electrical System

Before full assembly, the electrical system will be tested by subsystems. This will protect the components so that if a test fails, the entire system is not destroyed. Additionally a factor of safety needs to be determined with each motor so that they are properly sized and issues such as overheating and excessive wear can be mitigated.

For motor safety, the team will need to make sure the motors can handle exceptionally high torques to take into account any external forces that will be applied from testing and public use. Safety factors will need to be doubled or tripled when it comes to calculating torque required to have the robot properly move under certain loads. The animatronic will also have an indicator when it comes to battery life, as having the battery fall below a certain percentage can cause it to burn or explode.

Programming

The team intends on using multiple methods to test the code of the robot. First, each function will be tested individually. Functions include the gait and walking, as well as the animations of the head and tail movement. After each of the coded functions are tested, the total code will be assembled and tested to ensure that they work in conjunction with one another. In addition, there will be a data logging feature which will store all the data and variables of the motors, microcontrollers, power system, temperature, and any other data collected by sensors in an individual file with a time stamp. This will allow the team to know when errors occur, which improves diagnostic work.

Integration

In order to integrate all subsystems, the team will do tests on smaller components. For example, before assembling the entire animatronic, the team will do a proof of concept on one leg. The joint actuation system will be tested with the mechanical model and the control system to ensure all our coordination. This will allow the team to make edits to the design based on their findings.

X. Significant Accomplishments & Open Issues

Physical Prototype:

For prototyping, our dinosaur looks, sounds, and acts like a stegosaurus dinosaur skeleton. With assisted walking and aesthetics, our team was able to deliver when it came to those requirements. All of our prints are based off of 3D scans, and we were able to edit and additively manufactured accurate dinosaur skeletal parts.

Initially a base 3D print was made of the skeleton from Makerworld [46] and can be seen in Figure 84. Although a very small portion of this initial design will remain unmodified, it was critical to get an idea of the physical proportions of the dinosaur.

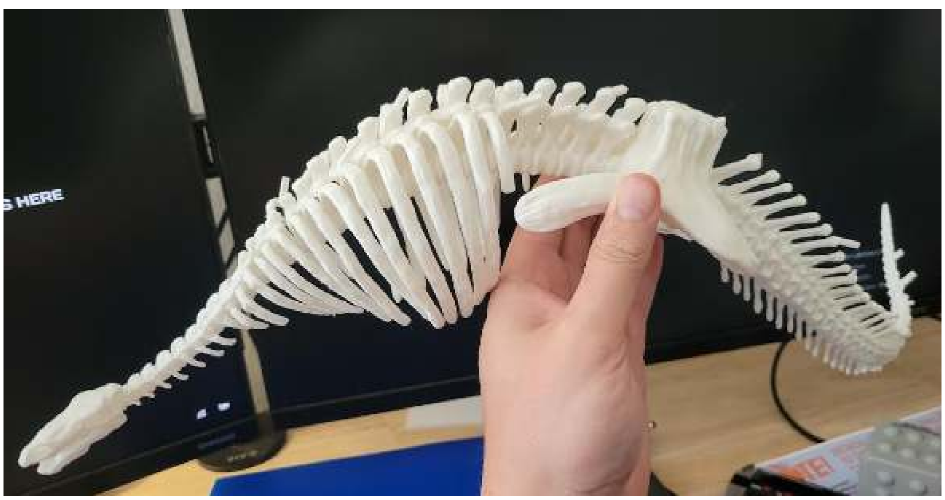


Figure 85: Initial Stegosaurus Print

From this point, basic measurements have been taken off of this model in order to get a rough idea of the amount of space that is available in the inner section of the body for packaging. Also a scale factor was found from this initial print to be 150% in order to attain the desired 3ft length from head to tail as discussed in concept generation. The final length of the prototype ended right at our goal of 36in

Smaller sections of the flexible elements have also been made to gain a greater understanding of the physical flexibility and the effect that varying infill densities will have on flexibility. There is a tradeoff between flexibility and structural integrity that needs to be analyzed as we get further towards our final design.

Figure 86: Tail Section from TPU

The rest of the physical model went together relatively smoothly, however it did require a small level of general fabrication and modifications to the printed pieces in order for everything to fit as intended.

Contact With Museum

The team got in contact with Lawrence Hall of Science UC Berkeley, and the museum offered to help us as we move forward with the project. The museum is also interested in having our robot in their indoor facility where visitors would interact with the robot. The team is in contact with paleontologists.

Open issues

Several open issues became clear during the development of the robot. One of the main concerns is the overall size of the structure. In its current form, the robot does not provide enough room for the electronics, which makes installation difficult and increases the chance of wires becoming loose or disconnected. Although the ribcage cavity offers the most practical central location for these components, the available space is still too limited for an organized and reliable setup.

A second unresolved issue involves the method used to connect the motors to the limbs. The wire- or rope-based system created repeated problems with tensioning, and keeping that tension consistent during use was difficult. Because of this, the current approach does not provide the level of reliability needed for smooth operation. A gear-based connection may be a stronger alternative, but this has not yet been fully explored or implemented.

The robot's stability also remains an important issue. Its current leg length raises the body too far off the ground, which reduces balance and makes walking trials more difficult. This suggests that the proportions of the lower body still need refinement before stable movement can be achieved.

Another open issue is the integration of the gait study into the robot's motion. Translating the intended walking pattern into a functional system proved more challenging than expected, which limited the team's ability to fully test performance. This challenge was made more noticeable by the lack of a team member with a stronger background in electrical or computer engineering, leaving some technical problems unresolved and reducing the time available for troubleshooting and improvement.

XI. Conclusions & Recommendations

This project was developed by Dr. Kurt Stresau, who wanted to see his daughter's stuffed dinosaur toy come to life. It is also inspired by the film, The Night at the Museum, where a T-Rex skeleton begins walking around. While the team has no specific stakeholder, the project is being developed with a museum application in mind. Key requirements of the project include an anatomically accurate skeleton model of a stegosaurus, and the stegosaurus model must have a realistic walk.

In order for the animatronic to walk, a gait study must be performed. The team has determined that the bulk of the gait study will be based off of a human gate study. The team will compile data from human joint angles and apply them to the stegosaurus model. From there, adjustments must be made in order to coordinate the four limbs and the tail, as well as adjust for discrepancies that arise from differences in anatomy between humans and the stegosaurus.

In order for a mechanical model to walk, adjustments to the joints have been made. The team will then use a series of servos to actuate these joints. Only 2 degrees of freedom will be required for the legs. Hips will be attached on hollow shafts, allowing for knee actuation cables to pass through. Knees have a hidden joint and are tuned so the joint appears to slide instead of rotate. The joints will be cable driven by a remotely mounted servo motor and have a single flexible section. They will be driven by a single servo using a tendon style joint to have a smooth curve. The neck design is similar to the tail, but will move in two axes. To produce side to side movement, cables are hidden into the existing bone structure using a modified mesh. Back plates and spines will help to disguise the cable pull blocks. For the skull, a simple hidden hinge will allow a range of 20 degrees of movement, and will be driven by a hidden micro servo inside of the skull.

For the code overview, there will be four states. State 1 is the idle state, where the motors are set to position and wait for command to transition to other states on input, and will be the default state on startup. State 2 is the active walk state, where the motors move according to registered inputs. During State 2, other inputs from the joystick and buttons can modify motor behavior (speed, direction, etc). State 3 is the active animation state, where the motors move according to predetermined animation initiated by an input and executed to completion, then return to state 1. During State 4, other inputs from the joystick and buttons will cycle the animatronic through predetermined animations.

When it comes to the bulk of our material, ASA will be used because of its availability and properties. ASA is strong and has deflection properties similar to ABS, and it can handle heat better than PLA and ABS due to its thermal properties. Hex, or honeycomb, is the strongest infill pattern when it comes to strength in all directions. If a part or bone is experiencing load in one direction, line and grid are the strongest option. Gyroid is also another alternative to hex. When it comes to infill density, anything above 40% is adding more weight than strength, and there is a tradeoff. The team has also taken into consideration motor strength, as having lighter bones should be considered when calculating torque. Printing orientation is extremely important when it comes to the stresses a print will experience. Errors in prints such as “u” and “v” notches can cause a 3D printed part to fail when undergoing tensile and stress forces. The initial design has been analyzed for point of failure throughout all the subsystems.

In general, the team has accomplished the majority of the preparatory work necessary for a successful project. The goals for the following semester will be to generate prototypes of the animatronic and use testing and analysis to eliminate faults. The goal of this team is to utilize fundamental engineering principles to bring ROAR to life.

After careful thought and thorough consideration, our team came up with future recommendations to whom it may concern. We have decided that the robot needs to be larger, not only so that more students can work on it in an easier manner, but so that the electronics can have more space to fit into the robot. The only reasonable place to hold the electronics is inside the cavity of the ribcage because that is where we have the most space, and because it is in the center of the robot where the wires can be branched out to all the electrical components. If the robot was for example 20-30% larger, the electrical components could easily fit into the housing underneath the ribcage, and there would be no need to shove the electrical components into the housing and have the wires disconnect because of the tight space.

Another recommendation is to have geared connections between the limbs and motors, instead of connections by wire/rope. Tensioning the rope was extremely challenging, and maintaining the tension was a struggle. While gears can have lash or "give", you can choose tight tolerance gears which have little to no lash and can be 3D printed. Another advantage to gears is that you can have a gear ratio, if the motor's torque is too weak, you can add a gear ratio to increase the torque.

Another recommendation is to lower the length of the legs. The higher the robot is relative to the ground, the more unstable the robot will be, and that gave us issues when we tried to have it walk. Finally, we had trouble with implementing the gait study into the robot. Having an electrical or computer engineering student on our team would have streamlined some of the tasks and gave us more time to test our robot.

References

[1]“Home.” https://lawrencehallofscience.org/. Point of contact: Michelle Zhu-ting Rodriguez: Public Experience Design Lead, mzrodriguez@berkeley.edu

[2]“eTools : Solutions for Electrical Contractors - Materials Handling - Heavy Lifting Occupational Safety and Health Administration,” *www.osha.gov*. https://www.osha.gov/etools/electrical-contractors/materials-handling/heavy

[3] C. A. Brassey, S. C. R. Maidment, and P. M. Barrett, “Body mass estimates of an exceptionally complete Stegosaurus (Ornithischia: Thyreophora): comparing volumetric and linear bivariate mass estimation methods,” Biology Letters, vol. 11, no. 3, p. 20140984, Mar. 2015, doi: 10.1098/rsbl.2014.0984.

[4] R. Redelstorff and P. M. Sander, “Long and girdle bone histology ofStegosaurus: implications for growth and life history,” Journal of Vertebrate Paleontology, vol. 29, no. 4, pp. 1087–1099, Dec. 2009, doi: 10.1671/039.029.0420.

[5] P. P. Skutschas et al., “Wear patterns and dental functioning in an Early Cretaceous stegosaur from Yakutia, Eastern Russia,” PLoS ONE, vol. 16, no. 3, p. e0248163, Mar. 2021, doi: 10.1371/jthenal.pone.0248163.

[6] D. B Weishampel, P. Dodson, and H. Osmólska, The Dinosauria, 2nd ed. Berkeley: University of California Press, 2003. doi: 10.1525/9780520941434.

[7] R. Grandia *et al.*, “Design and Control of a Bipedal Robotic Character.” Available: https://la.disneyresearch.com/wp-content/uploads/BD_X_paper.pdf

[8] Walt Disney Imagineering, “Droids in Training: Imagineers Conduct Playtest at Star Wars: Galaxy’s Edge,” *www.youtube.com*, Nov. 17, 2023. https://www.youtube.com/watch?v=qNfRgZMWyCI

[9] Walt Disney Imagineering, “The team Call It Imagineering | Audio-Animatronics®,” *YouTube*, Apr.04, 2024. https://www.youtube.com/watch?v=1e1R2vUORGI

[10] Molly, “What Is Disney Audio-Animatronics Technology?,” *Disney Experiences*, Jan. 19, 2024. https://disneyexperiences.com/disney-animatronics/

[11]A. Mazumdar *et al.*, “Synthetic Fiber Capstan Drives for Highly Efficient, Torque Controlled, Robotic Applications,” *IEEE Robotics and Automation Letters*, vol. 2, no. 2, pp. 554–561, Apr. 2017, doi: https://doi.org/10.1109/lra.2016.2646259.

[12]S. Huber, R. Poranne, and S. Coros, “Designing actuation systems for animatronic figures via globally optimal discrete search,” *ACM Transactions on Graphics*, vol. 40, no. 4, pp. 1–10, Jul. 2021, doi: https://doi.org/10.1145/3450626.3459867.

[13]W. Roozing and G. Roozing, “3D-printable low-reduction cycloidal gearing for robotics,” Oct. 2022, doi: https://doi.org/10.1109/iros47612.2022.9982006.

[14]M. Ruderman, F. Hoffmann, and T. Bertram, “Modeling and Identification of Elastic Robot Joints With Hysteresis and Backlash,” *IEEE Transactions on Industrial Electronics*, vol. 56, no. 10, pp. 3840–3847, Feb. 2009, doi: https://doi.org/10.1109/tie.2009.2015752.

[15]L. U. Odhner and A. M. Dollar, “The Smooth Curvature Model: An Efficient Representation of Euler–Bernoulli Flexures as Robot Joints,” *IEEE transactions on robotics*, vol. 28, no. 4, pp. 761–772, Aug. 2012, doi: https://doi.org/10.1109/tro.2012.2193232.

[16] Veritasium. (2024, December 6). 34 Years Of Strandbeest Evolution [Video]. YouTube. https://www.youtube.com/watch?v=IFaAjR_RRJs&t=280s

[17] Zhao, L., Luo, Z., Han, Y., Zhang, J., Chen, Y., Liu, Y., & Lu, P. (2025). Learning aggressive animal locomotion skills for quadrupedal robots solely from monocular videos. *Npj Robotics*, *3*(1). https://doi.org/10.1038/s44182-025-00048-x

[18] Biswal, P., & Mohanty, P. K. (2020). Development of quadruped walking robots: A review. *Ain Shams Engineering Jthenal*, *12*(2). https://doi.org/10.1016/j.asej.2020.11.005

[19] A. M. El-Dalatony, T. Attia, H. Ragheb, and A. M. Sharaf, “Cascaded PID Trajectory Tracking Control for Quadruped Robotic Leg,” *International Journal of Mechanical Engineering and Robotics Research*, vol. 12, pp. 40–47, Jan. 2023, doi: https://doi.org/10.18178/ijmerr.12.1.40-47.

[20] MATLAB, “Everything You Need to Know About Control Theory,” *YouTube*, Oct. 27, 2022.https://www.youtube.com/watch?v=lBC1nEq0_nk&list=PLPajznaQQtnUmnhak610IQSNVDX4dGfp (accessed Sep. 20, 2025).

[21] Y. Güven, E. Coşgun, S. Kocaoğlu, H. Gezici, and E. Yılmazlar, “Understanding the concept of microcontroller based systems to choose the best hardware for applications.,” *123.216.36*, Jul. 2019, doi: https://hdl.handle.net/20.500.11857/1024.

[22] G. Biggs and B. Macdonald, “A Survey of Robot Programming Systems.” Accessed: Apr. 28, 2024. [Online]. Available: https://www.societyofrobots.com/robottheory/Survey_of_Robot_Programming_Systems.pdf

[23]“Documentation - MATLAB & Simulink,” *www.mathworks.com*. https://www.mathworks.com/help/?s_tid=mlh_sn_help

[24] A. Koubaa, Ed., *Robot Operating System (ROS)*. Cham: Springer International Publishing, 2016. doi: https://doi.org/10.1007/978-3-319-26054-9.

[25] MicroPython, “MicroPython - Python for microcontrollers,” *micropython.org*. https://micropython.org/

[26] R. Ierusalimschy, H. Luiz, De Figueiredo, and W. Celes, “The Evolution of Lua.” Available: https://www.lua.org/doc/hopl.pdf

[27] Y. Ding, Abhishek Pandala, C. Li, Y.-H. Shin, and Hae Won Park, “Representation-Free Model Predictive Control for Dynamic Motions in Quadrupeds,” *IEEE Transactions on Robotics*, vol. 37, no. 4, pp. 1154–1171, Aug. 2021, doi: https://doi.org/10.1109/tro.2020.3046415.

[28] H. Meng, S. Zhang, W. Zhang, and Y. Ren, “Optimizing actual PID control for walking quadruped soft robots using genetic algorithms,” *Scientific Reports*, vol. 14, no. 1, Oct. 2024, doi: https://doi.org/10.1038/s41598-024-77100-7.

[29] G. Xin *et al.*, “Robust Footstep Planning and LQR Control for Dynamic Quadrupedal Locomotion,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4488–4495, Mar. 2021, doi: https://doi.org/10.1109/lra.2021.3068695.

[30] S. Thrun, W. Burgard, and D. Fox, *Probabilistic robotics*. Cambridge, Mass.: Mit Press, 2010.

[31] J. Elfring, E. Torta, and R. van de Molengraft, “Particle Filters: A Hands-On Tutorial,” *Sensors*, vol. 21, no. 2, p. 438, Jan. 2021, doi: https://doi.org/10.3390/s21020438.

[32] Steven Michael Lavalle, *Planning algorithms*. New York (Ny): Cambridge University Press, Cop, 2014.

[33] S. Karaman and E. Frazzoli, “Sampling-based algorithms for optimal motion planning,” *The International Journal of Robotics Research*, vol. 30, no. 7, pp. 846–894, Jun. 2011, doi: https://doi.org/10.1177/0278364911406761.

[34] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning quadrupedal locomotion over challenging terrain," *Science Robotics*, vol. 5, no. 47, Oct. 2020, doi: https://doi.org/10.1126/scirobotics.abc5986.

[35] A. Villarreal *et al.*, "Fast and Continuous Foothold Adaptation for Dynamic Locomotion Through CNNs," vol. 4, no. 2, pp. 2140–2147, Feb. 2019, doi: https://doi.org/10.1109/lra.2019.2899434.

[36] J. Hwangbo *et al.*, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, Jan. 2019, doi: https://doi.org/10.1126/scirobotics.aau5872.

[37] C. Cadena *et al.*, "Past, Present, and Future of Simultaneous Localization And Mapping: Towards the Robust-Perception Age," *IEEE Transactions on Robotics*, vol. 32, no. 6, pp. 1309–1332, Dec. 2016, doi: https://doi.org/10.1109/TRO.2016.2624754.

[38]G. Duarte *et al.*, "Quadruped Robot Control: An Approach Using Body Planar Motion Control, Legs Impedance Control and Bézier Curves," *Sensors*, vol. 24, no. 12, pp. 3825–3825, Jun. 2024, doi: https://doi.org/10.3390/s24123825.

[39] I. T. Jolliffe and J. Cadima, "Principal component analysis: a review and recent developments," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 374, no. 2065, p. 20150202, Apr. 2016, doi: https://doi.org/10.1098/rsta.2015.0202.

[40] M. Zucker *et al.*, "CHOMP: Covariant Hamiltonian optimization for motion planning," *The International Journal of Robotics Research*, vol. 32, no. 9–10, pp. 1164–1193, Aug. 2013, doi: https://doi.org/10.1177/0278364913488805.

[41] M. Kalakrishnan, S. Chitta, E. Theodorou, P. Pastor, and S. Schaal, “STOMP: Stochastic trajectory optimization for motion planning,” *2011 IEEE International Conference on Robotics and Automation*, May 2011, doi: https://doi.org/10.1109/icra.2011.5980280.

[42] A. J. Ijspeert, “Central pattern generators for locomotion control in animals and robots: A review,” *Neural Networks*, vol. 21, no. 4, pp. 642–653, May 2008, doi: https://doi.org/10.1016/j.neunet.2008.03.014.

[43] M. Bloesch *et al.*, “State Estimation for Legged Robots - Consistent Fusion of Leg Kinematics and,” Jul. 2012, doi: https://doi.org/10.15607/rss.2012.viii.003.

[44] F. Sygulla *et al.*, “An EtherCAT-Based Real-Time Control System Architecture for Humanoid Robots,” *Conference on Automation Science and Engineering*, Aug. 2018, doi: https://doi.org/10.1109/coase.2018.8560532.

[45] J. J. Craig, *Introduction to robotics : mechanics and control*. Upper Saddle Hall: Pearson Educación Internacional, 2005.

[46]“Makerbot Stegosaurus Skeleton Model (All Parts and Presupported) Remixed by bonfiggy MakerWorld: Download Free 3D Models,” *Makerworld.com*, 2025. https://makerworld.com/en/models/65592-makerbot-stegosaurus-skeleton-model-all-parts-and?from=search#profileId-196876 (accessed Nov. 21, 2025).

[47] American Museum of Natural History, “Tyrannosaurus rex Fossil Exhibit | American Museum of Natural History,” American Museum of Natural History, 2010. https://www.amnh.org/exhibitions/permanent/saurischian-dinosaurs/tyrannosaurus-rex

[48] Kurt Stresau, “The ROAR Project”, Unpublished, Aug 2025, https://The teambcourses.ucf.edu/courses/1492674/files/folder/Project%20Descriptions?preview=116985755

[49] MikroElectronika, “Li-Polymer battery datasheet,” Mar. 21AD. Available: https://www.farnell.com/datasheets/2369105.pdf

[50] A. Wood and S. Force, “What is a Lithium Polymer Battery? | LiPo,” Ossila, 2021. https://www.ossila.com/pages/lithium-polymer-battery#advantages (accessed Oct. 18, 2025).

[51] Boston Dynamics, “Spot,” Boston Dynamics, 2023. https://bostondynamics.com/products/spot/

[52] Vertoy, “Vertoy Aosgadino Instructions,” Amazon. https://m.media-amazon.com/images/I/A1aXZ7h3sHL.pdf

[53] Sketchfab, “Sketchfab,” Sketchfab, Jul. 05, 2018. https://sketchfab.com/3d-models/stegosaurus-skeleton-dc6e1c748484449587b81426d41dA6cb

[54] Simplify3D Software. (2022a, November 22). Properties table. Simplify3D Software. https://www.simplify3d.com/resources/materials-guide/properties-table/

[55] Akhondi, Shima, et al. “A Study on Mechanical Behavior of 3D Printed Elastomers with Various Infills and Densities.” Manufacturing Letters, vol. 35, 1 Aug. 2023, pp. 592–602, www.sciencedirect.com/science/article/pii/S2213846323000925, https://doi.org/10.1016/j.mfglet.2023.08.035.

[56] (PDF) effect of infill percentage and pattern on compressive behavior of FDM-printed GF-cf pa6 composites. (n.d.). https://www.researchgate.net/publication/380597376_Effect_of_infill_percentage_and_pattern_on_compressive_behavior_of_FDM-printed_GF-CF_PA6_composites

[57] (PDF) effect of infill percentage and pattern on compressive behavior of FDM-printed GF-cf pa6 composites. (n.d.-b). https://www.researchgate.net/publication/380597376_Effect_of_infill_percentage_and_pattern_on_compressive_behavior_of_FDM-printed_GF-CF_PA6_composites

[58] "Electronics Club - Switches - SPST, SPDT, DPST, DPDT, reversing, ON-ON, ON-OFF-ON, momentary ON-(ON), push, rotary, multiway, tilt, reed," *electronicsclub.info*. https://electronicsclub.info/switches.htm

[59] "Top 10 Popular Microcontrollers Among Makers," *Electronics-Lab.com*, Jun. 19, 2020. https://www.electronics-lab.com/top-10-popular-microcontrollers-among-makers/

[60] "Arduino Hardware," *www.arduino.cc*. https://www.arduino.cc/en/hardware/

[61] "Hardware," *Arduino.cc*, 2025. https://www.arduino.cc/pro/platform-hardware/

[62] R. P. (Trading) Ltd, "Buy a Raspberry Pi," *Raspberry Pi*. https://www.raspberrypi.com/products/

[63] "MSP430 microcontrollers | TI.com," *Ti.com*, 2025. https://www.ti.com/product-category/microcontrollers-processors/msp430-mcus/overview.html

[64] "Home - STMicroelectronics," *St.com*, 2019. https://www.st.com/content/st_com/en.html

[65] Espressif, "ESP32 Overview | Espressif Systems," *www.espressif.com*, 2023. https://www.espressif.com/en/products/socs/esp32

[66] "8 Different Types of Robot Motors: What They Are, Why Use Them, and How to Choose Motor," *Mosrac.com*, 2018. https://www.mosrac.com/resources/blog/robot-motors.html

[67] "AGV & AMR Motors including Servo, Frameless, and More," *Linengineering.com*, 2023. https://www.linengineering.com/industries/autonomous-guided-vehicles-agv?utm_source

=google&utm_medium=paid&utm_campaign=10276975858&utm_content=175716740570&utm_term=robot%20motors&gadid= (accessed Nov. 25, 2025).

[68] (PDF) effect of infill percentage and pattern on compressive behavior of FDM-printed GF-cf pa6 composites. (n.d.-c). https://www.researchgate.net/publication/380597376_Effect_of_infill_percentage_and_pattern_on_compressive_behavior_of_FDM-printed_GF-CF_PA6_composites

[69] Marsh Othniel C. "Restoration of Stegosaurus in: American Journal of Science, series 3, vol. 42 (1891), pl.9..

[1] P. Berkelman, P. Rossi, T. Lu, and J. Ma, "Passive Orthosis Linkage for Locomotor Rehabilitation," 2007 IEEE 10th International Conference on Rehabilitation Robotics, Jun. 2007, doi:.

[71] S. S. Bhavikatti, Mechanics of Solids. Newage International Publishers, 2000.

[72] R. Richards, Principles of Solid Mechanics. CRC Press, 2000.

Appendices

I. Appendix A: Engineering Design Competence Evaluation

Project Title:	Bring ROAR to Life Team 1				**Semester:**	Fall 2025	
MECHANICAL	**Critical / Main contributor**	**Strong contributor**	**Necessary but not a primary contributor**	**Necessary but only a minor contributor**	**Only a passing reference**	**Not Included in this Design Project**	**Report Section Pg.**
Thermal-Fluid Energy Systems				X			74
Machines & Mechanical Systems	X						34-49
Controls & Mechatronics	X						50-78
Materials Selection		X					85-89
Modeling & Measurement Systems		X					34-49, 77-78
Manufacturing	X						77-78 85-89

II. Appendix B: Functional Project Requirements

FUNCTIONAL REQUIREMENTS				
Req. #	**REQUIREMENT**	**TARGET**	**VALIDATION METHOD**	**Met?**
S	**STRUCTURAL REQUIREMENTS**			
F.S1	The animatronic shall balance statically.	The animatronic must be able to remain upright on its 4 limbs on a smooth surface within a range of +/- 10° in any cardinal direction.	Test: Stand animatronic on a surface, changing its degree of incline in the cardinal directions.	
F.S2	The animatronic shall balance while walking.	Barring any external disturbances, animatronic must retain balance when cycling through all its gaits.	Test: The animatronic will cycle through all gaits on a smooth, flat surface.	

F.S3	Joints and structure should support the weight and movement of the robot.	The animatronic shall not weigh more than 50 pounds, including all systems involved. The animatronic shall be able to support its entire weight.	Analysis: G force calculations. FEA.	
M	**MATERIAL REQUIREMENTS**			
F.M1	Shall be durable.	Robot shall be able to withstand falls, small drops, and regular handling.	Testing.	
F.M2	Shall use additive manufacturing.	3D printing shall be considered for rapid prototyping and custom parts.	Material Analysis and Selection.	
F.M3	Shall be economical	Material shall be economical to produce and use standard parts when possible.	Analysis on material costs.	
F.M4	Shall be reliable.	Material can consistently withstand the loads and forces applied during operation.	Analysis.	
F.M5	Shall be easily repairable.	Shall utilize sub-assemblies that are easily replaceable.	Analysis.	
P	**POWER REQUIREMENTS**			
F.P1	Power source shall have a long run time.	Robot runs for at least 30 mins before signaling low battery.	Test.	>2hrs
F.P2	Power systems shall comply with all relevant FCC and Industry safety standards.	Refer to relevant standards.	Analysis, testing.	

F.P4	Power distribution infrastructure will not interfere with mechanical articulation.	Joints will be able to have full range of motion without being impeded or damaging the power system.	Test.	
F.P5	Power system shall be light weight.	The power system and battery combined must weigh less than 5 lbs.	Analysis.	2lbs
F.P6	Robot will visibly and audibly communicate diagnostic data.	When the robot reaches 20% battery left, the robot will lay down and its eyes will flash red.	Test.	
F.P7	Robot shall be rechargeable.	There will be a USB C port on the back of the robot.	Visual.	
F.P8	Software will monitor power source health and fuel consumption.	When the robot reaches below 5% power, it will automatically shut off.	Test.	
F.P9	The power system shall use an internal power source.	the whole robot will function off the same single battery, subsystems will not have independent power systems.	Visual.	
F.P10	The power system shall support all current requirements.	The main power source shall be able to handle the max current loads from all internal electronics.	Test.	
A	**ARTICULATION REQUIREMENTS**			
F.A1	Shall have a high resolution.	An individual joint should be able to stop and hold within 2 degrees of the desired angle.	Testing.	
F.A2	Shall be flexible.	Each leg shalt have a minimum of 2 degrees of freedom.	Analysis.	

F.A3	Shall emulate realistic movement.	General walking gate and structure will appear to move in a biological way.	Analysis on Gate.	
F.A4	Shall have adequate range of motion.	Each joint and limb will move XX degrees to allow for full range of motion.	Analysis.	
F.A5	Shall be able to ROAR.	The stegosaurus mouth will open at a 20° angle with respect to the top jaw with a tolerance of +/- 2°.	Test.	
F.A6	Shall be able to turn to avoid obstacles.	The animatronic shall be able to turn at a 1:1 body length to turn radius ratio.	Test.	
F.A7	Shall be able to move its tail.	Can wag tail at minimum of 45 degrees from the center axis.	Test.	
F.A8	Shall be compact.	Wiring and circuitry shall be hidden in the main body of the robot.	Analysis.	
C	**CONTROL SYSTEM REQUIREMENTS**			
F.C1	Shall have a control system.	Shall be controlled by a single person.	Test.	
F.C2	Shall respond to user input.	Control system shall have a low latency and update rate, with a command latency <= 100ms (Input to Response).	Test.	
F.C3	Shall provide and store continuous diagnostics.	Stores Joint positions, IMU, Errors, Response Time, Controller Inputs, Power, and Temperature.	Demonstration.	

F.C4	Shall have pre-recorded movement.	The interaction shall fully execute within 10 to 20 seconds then return to its idle state. The animation can be canceled mid execution.	Test.	
F.C5	User shall be able to control individual motion.	The controller will provide of head and neck motors in the yaw and pitch directions and hold the position within Plus of minus 2 degrees of the commanded orientation.	Test.	
F.C6	Shall make noise/sound.	ROAR volume shall be at minimum 70 decibels.	Test.	
F.C7	Movement and walking shall be easily controlled.	The controller shall control direction of gait and speed.	Test.	
F.C8	Shall have an operators instruction manual.	Easy to operate and learn control mapping, with an included instruction manual.	Visual.	
F.C9	Shall have a balance feedback system.	Stop animation and return to idle if animation is interrupted or loses balance.	Test.	
F.C10	Shall be connected wirelessly.	Shall have at least a 50ft range.	Test.	>100ft
E	**SAFETY AND ENVIRONMENT REQUIREMENTS**			
F.E1	Shall operate in dry conditions.	Robot shall not be submerged in water or exposed to rain.	Test.	
F.E2	Shall operate in standard atmospheric conditions.	Standard atmospheric conditions are defined as a temperature of 15°C (59°F) and a pressure of 101,325 pascals (14.7	Test.	

		psi).		
F.E3	Shall have an integrated shut down system.	Program shall be able to sense if the robot falls over and turns off.	Test.	
F.E4	Shall have a mechanical shut down system.	Will be able to turn off in less than 2 sec if a switch is pressed.	Test.	
F.E5	Shall shut down if remote contact is lost.	Shall shut down within 5 seconds of loss of contact.	Demonstration.	
F.E6	Shall have low emissions.	Shall not damage its surrounding environment or objects.	Analysis.	

III. Appendix C: Component Project Requirements

COMPONENT REQUIREMENTS								
Req. #	**System**	**Component**	**Stretch Goal**	**Requirement**	**Validation Method**	**Units**	**Range**	**Met ?**
C.C1	Control System	On/Off switch		Shall be easily accessible	Inspection	N/A	N/A	
C.C2	Control System	On/Off switch		Shall immediately cut all power to the robot.	Inspection	N/A	N/A	
C.C3	Control System	Remote Transmitter/ Controller		Shall have a minimum range of [50] ft.	Testing	Ft	>= 50	>100 ft
C.C4	Control System	Remote Transmitter/ Controller		Shall stop all movement if the robot loses range or remote contact.	Testing	N/A	N/A	

C.C5	Control System	Controller		Will contain a computer that is able to be programmed allowing for precise control of joints and motion.	Testing	N/A	N/A	
C.C6	Control System	Motor Controller		Will interface between the primary computer and the motors, combining positioning requests and handling motor electrical requirements.	Testing	N/A	N/A	
C.C7	Control System	IMU	X	Will be able to keep track of inertia when the robot moves.	Testing	N/A	N/A	
C.C8	Control System	Camera	X	Will be able to take in visual data which will affect how the robot moves.	Testing	N/A	N/A	
C.C9	Control System	Gyroscope	X	Determine whether the position of the robot is upright or if it has fallen over.	Testing	N/A	N/A	
C.C10	Control System	GPS	X	Determine where the robot is located in space.	Testing	N/A	N/A	
C.C11	Control System	I/O		Input/Output connections to support everything and there are enough connections.	Analysis	N/A	N/A	
C.E	Electrical System							
C.E1	Electrical System	Battery Pack		Shall have a run time of at least [30] min.	Test	min	>=30	>2hr
C.E2	Electrical System	Battery Pack		Shall charge in under [180] min.	Analysis	min	<=180	1hr
C.E3	Electrical System	Drive Motors		Shall have adequate torque in order to move each limb.	Analysis	N/M	N/A	

C.E4	Electrical System	Drive Motors		Shall have adequate speed in order to replicate a realistic gate.	Analysis	Rad /Se c	N/A	
C.E5	Electrical System	Animation Motors		Shall have [XX] torque in order to move non-structural elements.	Analysis	N/ M	N/A	
C.E6	Electrical System	Speaker		Will be able to communicate audio at a volume of at least [XX] decibels.	Testing	dB	>=7 0	
C.E7	Electrical System	Eyes		Will be able to display at least 2 colors.	Analysis	Cou nt	>=2	
C.S	MAIN STRUCTU RE							
C.S1	Main Structure	Power Module		Shall not exceed the physical dimensions of the robot.	Analysis	N/ A	N/A	
C.S2	Main Structure	Power Module		Shall be able to store and protect all main electrical components with the exception of motors.	inspecti on	N/ A	N/A	
C.S3	Main Structure	Power Module		Shall be impact resistant	Inspecti on	N/ A	N/A	
C.S4	Main Structure	Hardware/ Fasteners		Shall use off the shelf, metric Nuts & Bolts unless a specific part requires otherwise.	Inspecti on	N/ A	N/A	
C.S5	Main Structure	Hardware/ Fasteners		Shall use off the shelf standard size Bearings, Pins, & other hardware.	Inspecti on	N/ A	N/A	
C.S6	Main Structure	Bones		Main bone elements and geometry will be within [XX] percent of measurements taken from an existing scan.	Analysis	%	<= 20	0% from scan

C.S7	Main Structure	Back Bone		The main structure, spine & rib cage, shall be able to support [XX] weight.	Analysis	LBS	TBD	30lbs
C.S11	Main Structure	Fins		Shall have fins along its spine, as the fossil record shows.	Analysis	Count	11	
C.S12	Main Structure	Tail Spikes		Shall have tail spikes in accordance with the fossil.	Analysis	Count	4	
C.S13	Main Structure	Feet		Interface material shall provide adequate friction to be able to have proper traction.	Analysis	N/A	N/A	
C.J	JOINTS							
C.J1	Joints	Leg Joints		Each leg will have at least 2 degrees of freedom.	Analysis	N/A	N/A	
C.J2	Joints	Leg Joints		Each joint shall be able to carry the weight of the robot.	Analysis	FOS	>=2	
C.J3	Joints	Leg Joints		Each joint shall be able move [XX] degrees in order to emulate movement.	Analysis	Deg	TBD	
C.J4	Joints	Neck		Neck shall be made from a single flexible section. And articulate >45 deg	Analysis	Deg	>45	
C.J5	Joints	Tail		Neck shall be made from a single flexible section. And articulate >90 deg	Analysis	Deg	>90	
C.J6	Joints	Mouth		Mouth shall be able to open at least [20] degrees.	Analysis	Degrees	>=20	

IV. Appendix D: Failure Modes and Effects Analysis

Structure								
Function	**Failure Mode**	**Effect**	**Severity**	**Causes**	**Probability**	**Controls**	**Detection**	**RPN**
Balance statically	Loses balance/doesn't balance	Robot falls over	3	weight imbalance, external strike, uneven ground	3	Internal gyroscope	1	9
Balance dynamically	Fails to balance during motion	Robot falls during motion	3	incorrect data input, gyroscope malfunction, joint malfunction	7	Microcomputer system check	3	63

Materials								
Function	**Failure Mode**	**Effect**	**Severity**	**Causes**	**Probability**	**Controls**	**Detection**	**RPN**
Joints and structures support weight	Robot too heavy	joint breaks, weight imbalance, energy inefficient, slow movement speed	5	internal parts too heavy, load placed on top of robot	3	scale	1	15
Parts are durable	Part breaks from material fatigue	Robot cannot walk, head or tail no longer move properly	7	overuse, external strain on parts	3	Factor of Safety	5	105
Use of 3D printing	Part breaks along layer line fault	legs/body stop functioning, Robot can no longer support weight	7	Shear force along printing layer line	3	infill density, Factor of Safety	2	42
Economical	Parts/manufacturing over budget	decreases potential customer base	1	design flaw	1	project budget, design choices	3	3
Economical	Product outside of consumer target range	decreases potential customer base	2	design flaw	1	project budget, design choices	3	6
Reliable	Part or parts of robot consistently break and/or require replacement	Maintenance cost increase, consumer interest decrease	5	Part weakness, shear forces, environmental damage	5	infill density, Factor of Safety, structural supports	3	75
Repairable	Robot maintenance excessively time	Maintenance cost increase, consumer interest	3	design flaws	2	design choices	3	18

	consuming	decrease						
Reparable/reliable	Robot maintenance excessively expensive	Maintenance cost increase, consumer interest decrease	5	design flaws, part choices	2	material/design choices, testing	2	20
Parts are durable	Parts experience severe deflection without breaking	Robot cannot walk, head or tail no longer move properly	7	overuse, external strain on parts	5	Factor of Safety	5	175
Replaceable	Robot maintenance excessively complex or consistently requires expert assistance	Maintenance cost increase, consumer interest decrease	7	part choice, design flaws	2	documentation, training, design choices	2	28

Power								
Function	**Failure Mode**	**Effect**	**Severity**	**Causes**	**Probability**	**Controls**	**Detection**	**RPN**
Runtime	Robot has low runtime	Use in exhibits is limited	5	battery capacity, excess weight, motor inefficiencies	6	battery choice, testing	3	90
Standards	Robot fails to meet a relevant FCC/Engineering standard	Lawsuit :(	9	design flaw, law oversight	1	research	5	45
Power distribution infrastructure	Wires break	Part or whole robot stops working until problem is fixed	9	Material fatigue, part interaction	3	design choices, testing	4	108
Power distribution infrastructure	Wires short	Potential fire hazard, risk of electrocution	9	wire breakage, water damage	3	design choices, testing	3	81
Overall function	Critical infrastructure not integrated properly	robot fails to work correctly	7	human error, broken/damaged parts	3	testing, visual verification	5	105
Weight	Power system overburdens robot/causes excessive strain on joints	Increased power usage, possible joint breakage	5	decreased runtime, battery strain	2	design choices, testing	2	20

Visible and audible communication of diagnostic data	Sound levels too loud	annoying to consumer	2	voltage to speaker too high, code problem	3	testing	1	6
Visible and audible communication of diagnostic data	Sound levels too soft	unusable to consumer, audio cue misinterpreted or missed	6	voltage to speaker too low, code problem	3	testing	1	18
Interference with mechanical articulation	something interferes with joint movement	Part or whole robot stops working, potential wire short	7	foreign objects, wires, internal defects	4	shielding, design choices	4	112
Rechargeable	Overcharging	potential battery explosion	9	code bug, microcontroller defect, voltage regulator defect	1	testing, voltage regulator	5	45
Rechargeable	Undercharging	potential battery explosion, decreased runtime	8	code bug, microcontroller defect, voltage regulator defect	1	testing, voltage regulator	5	40
Safety hazard	Battery puncture	Fire :(	10	external damage to battery	1	shielding	3	30
Software monitored	Software crashes	Robot stops working until problem is rectified	7	code bug, microcontroller defect	2	testing	6	84
Software monitored	Software receives bad sensor information	robot behaves erratically or not at all	6	external interference with sensor, damage to sensor, microcontroller problem, code bug	3	testing, calibration	5	90
Internal power source	Exceeds weight requirement	Increased power usage	4	human error	2	scale	2	16
Design error	Exceeds internal spatial dimensions	Less space for critical infrastructure, more vulnerable if not properly protected	4	human error	2	product description	3	24

Improper safety hazard	Malfunction damages other interior components	robot stops working or causes further damage to itself or environment	8	manufacturing defect, microcontroller problem, voltage regulator problem	2	testing of parts	1	16

Articulation								
Function	**Failure Mode**	**Effect**	**Severity**	**Causes**	**Probability**	**Controls**	**Detection**	**RPN**
High resolution movement	Irregular or erratic movement	unpredictable or dangerous behavior	5	code error, joint problem, motor problem	5	testing, torque and ROM limitations	7	175
Degrees of freedom	Failure to move in a prescribed degree of freedom	partial or complete failure in locomotion	9	blocked joints, code error, mechanical failure, foreign object	4	testing, material and design choices	7	252
Realistic movement	Jerky or robotic movements	consumer immersion lost, interest decreases, potential loss of balance	4	blocked joints, code error, mechanical failure, foreign object	3	visual verification, gait study	7	84
Range of motion	Failure to move smoothly through entire range of motion	potential loss of balance/ ability to move	1	blocked joints, code error, mechanical failure, foreign object	3	testing, gait study	3	9
Range of motion	Range of motion blocked by something	robot loses balance/ ability to move	7	foreign debris	4	visual verification, shielding	3	84
Roar	Fails to produce sound	customer expectations not met	3	speaker defect, code error, microcontroller problem	2	test	1	6
Communication	Produces sound without movement	customer expectations not met, no movement	7	speaker defect, code error, microcontroller problem	2	test	1	14
Avoid obstacles	Hits obstacle	loss of balance, damage to robot or environment	8	operator failure, detection failure	3	test, training	1	24
Tail movement	Tail only partially moves	fails to meet standards	3	foreign object, material fatigue, broken part	3	test	2	18

Jaw movement	Jaw doesn't move or too much load	fails to meet standards	2	material experiences stresses past the yield point	3	tests, FEA	1	0

Control System								
Function	**Failure Mode**	**Effect**	**Severity**	**Causes**	**Probability**	**Controls**	**Detection**	**RPN**
Has a control system	loses control system	impossible to control, feedback loop or complete inaction	8	damage to control system, control system defect	4	shielding, integration	3	96
Responds to user input	User input misinterpreted by CS	erratic behavior	5	control system defect	2	Testing, component choice	3	30
Responds to user input	No response to user input/ blocked user input	erratic or no behavior	4	RC signal blockage, code issue, controller defect	4	testing, environment	3	48
Provides and stores diagnostic data	Reports inaccurate diagnostic data	erratic or no behavior	4	memory corruption, diagnostic tool damage/defect	3	testing, code debugging	2	24
Provides and stores diagnostic data	Fails to store diagnostic data	control system accuracy compromised	3	memory corruption, diagnostic tool damage/defect	3	testing, debugging	2	18
Pre-recorded movement	pre recorded movement does not play	breaks immersion, fails design standards	3	memory issue, code issue, microcomputer defect	2	testing, debugging, memory integrity validation	2	12
User-controlled individual motion	User error, robot piloted to dangerous area	robot breaks either structurally or electronically	8	operator	5	training, documentation	3	120
Makes noise/sound	audio files do not work	no sound, breaks immersion, fails design specifications	3	code problem, file corruption	2	testing	5	30
Makes no sound	audio devices broken or compromised	no sound, breaks immersion, fails design specifications	3	external damage, water damage, manufacturing defect	4	protection, testing	2	24

Easy to control	controls exceedingly complex	operator fails to properly operate robot	8	design flaw, human error	4	testing	3	96
Comes with instruction manual	instructions unclear	owner does not know how to use robot, potential unintended damage or failure to operate normally	8	design flaw, human error	4	Documentation, feedback	1	32

Safety and Environment								
Function	**Failure Mode**	**Effect**	**Severity**	**Causes**	**Probability**	**Controls**	**Detection**	**RPN**
Standard atmospheric conditions	Excessive heat	electronics/material damage	8	environment	5	operating environment and safety constraints	1	40
Standard atmospheric conditions	Excessive cold	material damage	5	environment	5	operating environment and safety constraints	2	50
Standard atmospheric conditions	High humidity	electronics damage	8	environment	5	operating environment and safety constraints	5	200
Standard atmospheric conditions	Condensation	electronics damage	8	environment, rapid temperature change from hot to cold	5	operating environment and safety constraints	5	200
Automatic shut down	Signal failure	robot continues to operate outside of user definition	9	code error, microcontroller defect	2	test	2	36
Shut down on loss of communication	Signal failure	robot continues to operate outside of user definition	9	code error, microcontroller defect	1	test	1	9
Low emissions	Non-recyclable materials used	Environmental impact from end of product life cycle	4	material design choice	1	material design choices	1	4
Human interaction	Digits/body parts pinched in joints	medical injury, lawsuit	10	human error, insufficient safety protocol	6	operating environment and safety	1	60

						constraints		
Human interaction	Jaw bite force causes soft tissue damage	medical injury, lawsuit	10	loss of control, human error (sticking hand into jaw), no detection system	2	safe distance from people, controlled by trained operator	1	20
Human interaction	Robot damaged or hit by human	robot functions compromised, possible need for repair.	7	operator error, environment, other human error	5	operating environment and safety constraints	1	35

V. Appendix E: Cost Analysis

		TOTAL (with Tax):	$483.62	
Item ID	**Name**	**Quantity**	**Total Cost**	**Sourced From**
001	Black ASA Filament	1kg	$24.00	Personal
002	Off-White ASA Filament	1kg	$24.00	Personal
003	White PETG Filament	1kg	$15.00	Personal
004	1.3mm UHMWPE Cord (Black)	100ft	$12.00	Purchase
005	black, white, grey	4kg	$80.00	Personal
006	M12 Battery Adapter	1	$18.00	Purchase
007	Milwaukee M12, 2.0, 4.0, and 6.0 battery	3	$0.00	Personal
008	Mini Speakers	1	$11.99	Purchase
009	ESP32-S3 Microcontroller	1	$19.99	Purchase
010	DS5180 80kg servo	2	$69.98	Purchase
011	DS3230 30kg servo (2-pack)	3	$90.87	Purchase
012	1.5g Micro Linear servo actuator	1	$12.53	Purchase
013	16 channel PWM Servo Driver	1	$12.99	Purchase
014	5V 3A Buck Converter USB-C	1	$11.99	Purchase
015	2-8S LiPo servo power supply	2	$37.98	Purchase
016	Low Voltage Cutoff Board (30A)	1	$19.99	Purchase
017	MCP9808 Temperature Sensor	1	$4.41	Purchase
018	MPU 6050 Gyro./Accel. Module	1	$6.99	Purchase
019	GY-NEO6MV2 GPS Module	1	$19.99	Purchase
020	5V 3A Low Volt Buck Converter	1	$9.95	Purchase
021	GNSS GPS module	1	$19.90	Purchase
022	Mini MP3 Player	1	$9.90	Purchase
023	Threaded Instert Bolts	1	$9.97	Purchase
024	Set screw shaft collar	6	$21.60	Purchase
025	Needle roller bearing	6	$42.36	Purchase
026	Alloy Steel Round tube	1	$20.24	Purchase

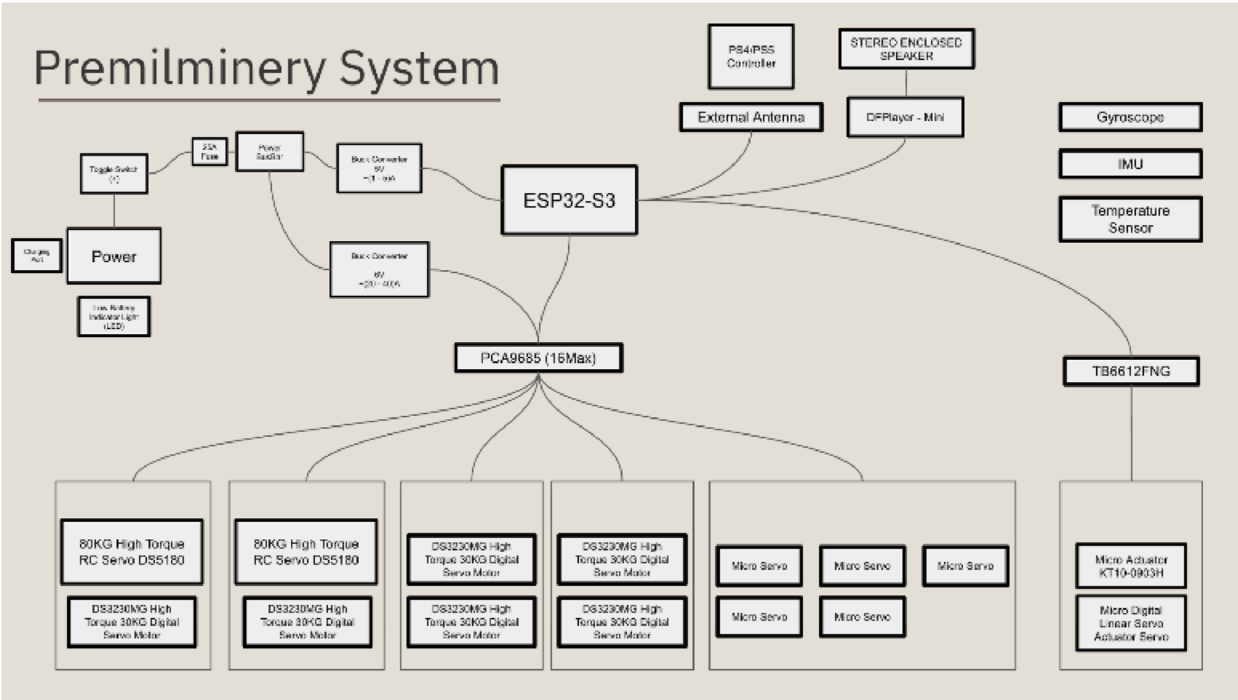
VI. Appendix F: Programming Language Selection Table

Factor	Description
Execution speed and real-time capabilities	Critical for time-sensitive control tasks where delays can compromise performance.
Hardware compatibility and low-level access	Ability to interact directly with hardware components such as sensors, actuators, and controllers.
Library and API support (e.g., for sensors, motors, vision)	Availability of pre-built modules and frameworks to accelerate development in areas like sensing, motion, and perception.
Community support and documentation	Access to tutorials, forums, and technical references that reduce development time and troubleshooting difficulty.
Integration with existing robotics platforms	Ease of interoperability with frameworks like ROS and embedded systems, ensuring scalability and modularity.

VII. Appendix G: Techniques Used in Robotics

Techniques In Robotics			
Technique	**Purpose**	**Application Area**	**Source / Reference**
PID Controller	Cascaded PID for accurate foot-tip trajectory tracking on a quadruped leg	Quadruped leg servo control / trajectory tracking	[1]
LQR (Linear Quadratic Regulator)	Optimal feedback (Riccati/LQR) integrated with quadruped planning/control to improve robust motion tracking	Dynamic quadruped locomotion (trotting/pacing)	[11]
Kalman Filter (KF)	State estimation under Gaussian noise	General robotics state estimation	[12]
Extended Kalman Filter (EKF)	Classical Gaussian state estimation (derives and analyzes the KF)	General robotics state estimation	[12]
Particle Filter	Sequential Monte Carlo methods; practical tutorial and use for localization	General estimation + (see below PF legged)	[13]
Model Predictive Control (MPC)	Representation-Free MPC (RF-MPC): represents orientation with a rotation matrix, uses variation-based linearization and QP transcription; demonstrated real-time 250 Hz control and dynamic motions including a controlled backflip.	Quadruped robot dynamic locomotion and acrobatics (Panther platform).	[9]
A Algorithm*	Foundational text on graph/discrete planning incl. A* properties and use	General motion/discrete path planning	[14]
RRT / RRT*	Sampling-based motion planning	General motion planning	[15]
Deep Neural Networks (DNNs)	Perception, classification, control policies	Vision-based Navigation, Object Recognition	[18]
CNNs (Convolutional Neural Nets)	Feature extraction from images	Robotic Vision, SLAM	[17]
Reinforcement Learning (RL)	Learning control policies via rewards	Autonomous Navigation, Manipulation	[16]
Inverse Kinematics	Convert end-effector position to joint angles	Humanoid Robots, Manipulators	[27]
Forward Kinematics	Compute end-effector position from joint states	Robot Motion Modeling	[27]
SLAM (Graph-based, EKF, etc.)	Simultaneous localization and mapping	Mobile Robots, Autonomous Vehicles	[19]
Bezier and B-Spline Curves (Parametric Animation Curves)	Smooth trajectory generation through parametric curves	Motion planning for arms, mobile robots	[20]
Principal Component Analysis (PCA)	Dimensionality reduction and feature extraction	Sensor fusion, gesture recognition, object tracking	[21]
Trajectory Optimization (CHOMP)	Gradient-based trajectory optimization	Motion planning for manipulators and legged robots	[22]
Trajectory Optimization (STOMP)	Stochastic trajectory optimization with constraints	High-dimensional motion planning	[23]
Central Pattern Generators (CPGs)	Generate rhythmic signals for periodic movement (e.g., walking gaits)	Quadruped locomotion, bio-inspired robots	[24]
IMU Fusion (e.g., EKF)	Combine IMU data with other sensors (e.g., encoders, GPS) to estimate pose	Balance control, walking on uneven terrain	[25]
ROS Teleop / Joystick Input	Manual control of robot motion using joystick or keyboard commands	Manual operation, remote teleoperation	[6]
Direct Control Loops	Send low-level velocity or position commands directly to actuators	Low-latency control, gait tuning	[26]

VIII. Appendix H: Preliminary System



IX. Appendix I: Table of Necessary Motors

Robot Part	Movement / Action	Motor Type / Example	Notes
Mouth	Open / close	1 × Micro Linear Servo Actuator	Lightweight, precise linear motion
Neck	Up, down, left, right	2 × Servo Motors	Medium torque servos for multi-axis head control
Fins	Cam movement / wave	1 × Linear Motor Servo	Smooth oscillatory motion, Lightweight, precise linear motion
Tail	Up, down, left, right	2 × Servo Motors	Requires torque and precise multi-axis control
Shoulders (2 joints)	Y-axis rotation	2 × High-torque Digital Servo per joint (DSServo DS5180, 80kg)	Strong enough to handle limb weight
Knees and shoulders (2 joints)	Y-axis rotation	2 × DS3230MG Metal Gear Digital Servo per joint (30kg)	High precision for leg articulation

Knees (4 joints)	Y-axis rotation	4 × DS3230MG Metal Gear Digital Servo per joint (30kg)	High precision for leg articulation
Ankles (4 joints)	Spring mechanism	No motor	Passive movement using springs

IX. Appendix J, Additional Pictures